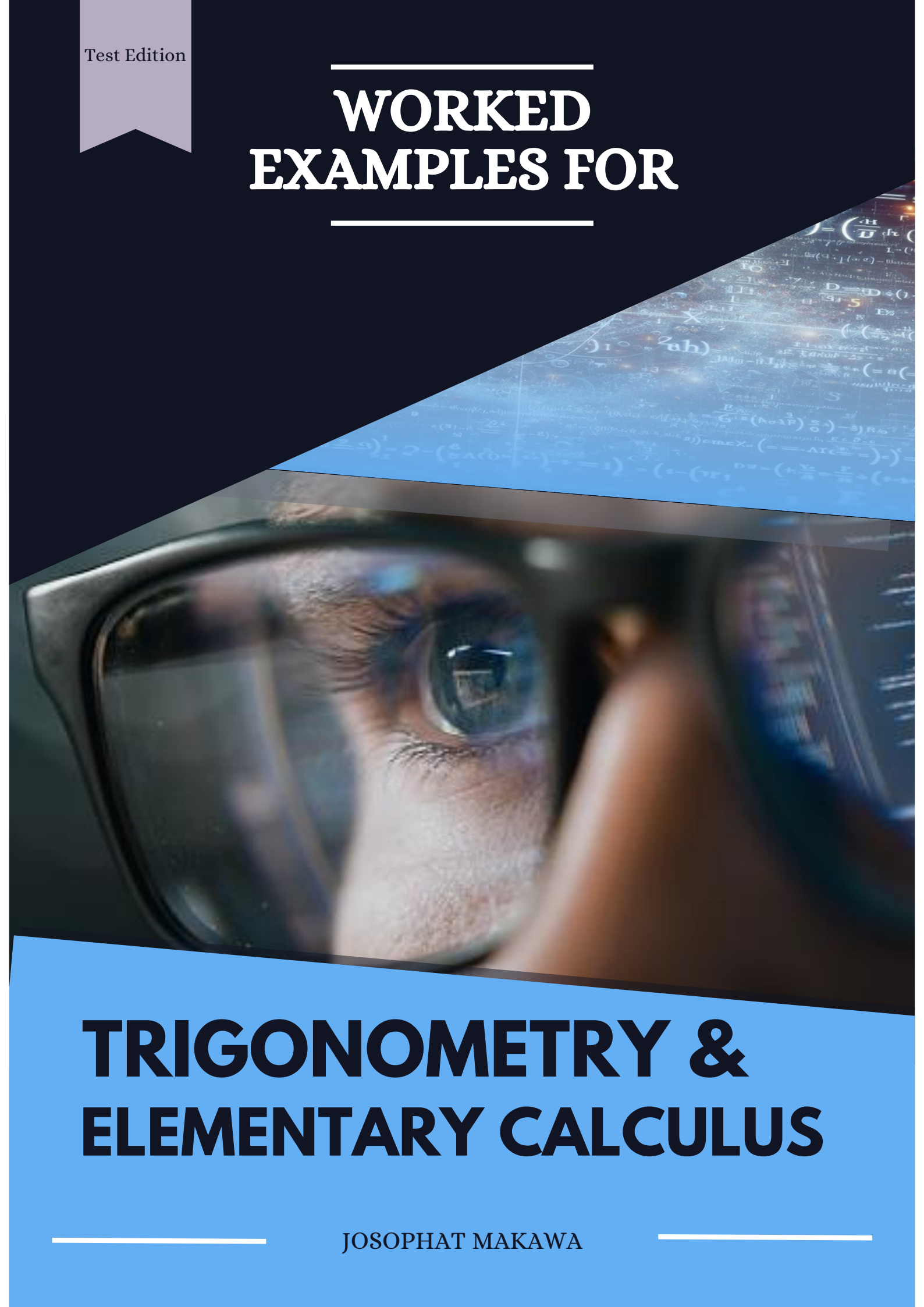


Test Edition

WORKED EXAMPLES FOR



TRIGONOMETRY & ELEMENTARY CALCULUS

JOSOPHAT MAKAWA

Josophat Makawa

Worked Examples For

TRIGONOMETRY AND ELEMENTARY CALCULUS

With Limits of Trigonometric Functions

Josophat Makawa*

January 28, 2025

Reviewed by Penjani Chawinga

Josophat Makawa

Disclaimer

This book has been compiled by a third year student of Mathematics at the University of Malawi. It was mainly as a practice for LaTeX typesetting software but grew into a passion to produce something that might help some friends. Make sure use this document together with all other documents given to you by your lecturer/teacher. Any problem should be reported to the [Author](#)

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Colophon

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(You are welcome to contribute!)

Mathematics is not about numbers, equations, computations or algorithms:
it is about understanding.

– William Paul Thurston

Preface

Mathematics has always been the language of the universe, forming the foundation for countless discoveries, innovations, and applications. Among its diverse branches, trigonometry and calculus stand out as essential pillars, bridging the gap between abstract theory and real-world problem-solving. Whether predicting the motion of celestial bodies, designing cutting-edge technology, or analyzing financial trends, the principles of trigonometry and calculus permeate our lives in profound and often unseen ways.

This book, *Worked Examples for Trigonometry and Elementary Calculus*, was born out of the need for a clear, accessible, and practical resource to aid students, educators, and self-learners alike. While many mathematical texts emphasize rigorous theory or inundate readers with endless exercises, this work takes a balanced approach by coupling foundational concepts with meticulously solved examples. It is our belief that understanding blossoms when principles are not only explained but also demonstrated step by step in a variety of contexts.

The structure of this book reflects philosophy. Each sub-topic begins with a concise yet comprehensive explanation of the relevant concepts, ensuring that even readers with minimal prior exposure can grasp the material. Following this, worked examples are presented in increasing levels of complexity to build confidence and deepen understanding. These examples are designed not only to solve problems but also to illustrate why certain methods are chosen and how they can be adapted to new situations.

Recognizing the diverse needs of learners, the book also includes practice problems to reinforce understanding, as well as tips and tricks to simplify challenging topics. Additionally, visual aids such as graphs and diagrams are incorporated to clarify abstract ideas, while historical notes and practical applications provide context and inspiration.

While this book is primarily aimed at students beginning their journey into trigonometry and calculus, it will also serve as a valuable reference for educators seeking to enrich their teaching materials and for professionals revisiting these subjects for practical purposes.

In writing this book, great care has been taken to ensure clarity, precision, and relevance. Yet, mathematics is an ever-evolving field, and there is always room for growth and improvement. As such, I welcome feedback from readers and educators alike to further refine this work in future editions.

It is my hope that this book will not only demystify the complexities of trigonometry and calculus but also ignite a deeper appreciation for the beauty and power of mathematics. May it serve as a stepping stone for your academic and professional aspirations, and may it inspire you to explore the boundless horizons of mathematical discovery.

December 2024
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TRIGONOMETRY



1 Trigonometry

Introduction

Trigonometry is a branch of mathematics that deals with the relationships between the angles and sides of triangles, particularly right-angled triangles. Its applications extend far beyond geometry, finding relevance in fields such as physics, engineering, astronomy, computer graphics, and even music. By understanding the principles of trigonometry, we gain the ability to model periodic phenomena, solve spatial problems, and analyse data in innovative ways.

At its core, trigonometry revolves around six primary trigonometric ratios: sine (sin), cosine (cos), tangent (tan), cosecant (csc), secant (sec), and cotangent (cot). These ratios describe specific relationships between the lengths of a triangle's sides, offering powerful tools for solving a variety of problems involving angles and distances.

Historical Context

The word "trigonometry" comes from the Greek words **trigonon** (triangle) and **metron** (measure). The **origins** by **Joseph Hunt** of trigonometry can be traced back to ancient civilizations, where it was first developed to aid in astronomy and land surveying. Over centuries, mathematicians like Hipparchus, Ptolemy, and al-Khwarizmi refined trigonometric principles, laying the groundwork for advanced applications we see today

Success Criteria

By the end of this section, you should be to:

- ▶ understand how angles are measured in degrees and radians
- ▶ calculate lengths of arcs and areas of sectors
- ▶ manipulate trigonometric functions,
- ▶ solve trigonometric equations,
- ▶ prove trigonometric identities,
- ▶ sketch graphs of trigonometric functions.

Let us embark on this journey of discovery, unlocking the power of angles and ratios to solve problems and uncover new insights.

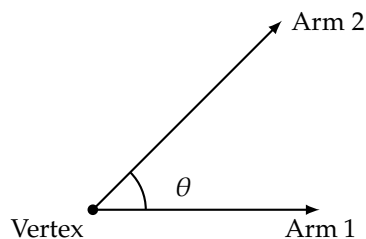
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1.1 Angular Measure

1.1.1 Core Insights

Definition 1.1.1 An *angle* is a geometric figure formed by two rays (called the *arms* or *sides* of the angle) sharing a common endpoint, known as the *vertex* of the angle. Angles measure the amount of rotation between the two arms, and this rotation can be described in degrees ($^{\circ}$) or radians.

Figure 1.1: An Angle Formed by Two Rays Sharing a Common Vertex



An angle that is measured in an anticlockwise direction is a **POSITIVE** and angle measured in a clockwise direction is **Negative**. To visualise these said directions, look at the figure below:

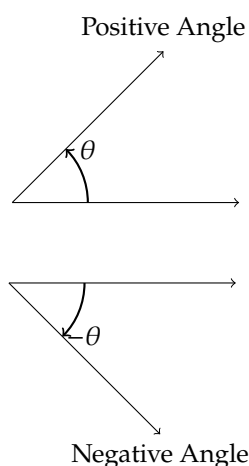


Figure 1.2: Positive and negative angles

Standard Position of an Angle

An angle is said to be in **standard position** if:

- ▶ Its vertex is located at the origin of the Cartesian plane.
- ▶ One arm (the initial side) lies along the positive x -axis.
- ▶ The other arm (the terminal side) is rotated from the initial side either counterclockwise (positive angle) or clockwise (negative angle).

A special case of the angles in standard position arise in situations where the terminal side aligns with either x – axis or y – axis. These angles include 0° , 90° , 180° , 270° and 360° . In addition, these angles have a special name, **QUADRANTAL ANGLES**¹.

1: These angles are called **Quadrantal angles** because they divide the circle into four equal parts called **Quadrants**. We will use them in solving equations later in this section.

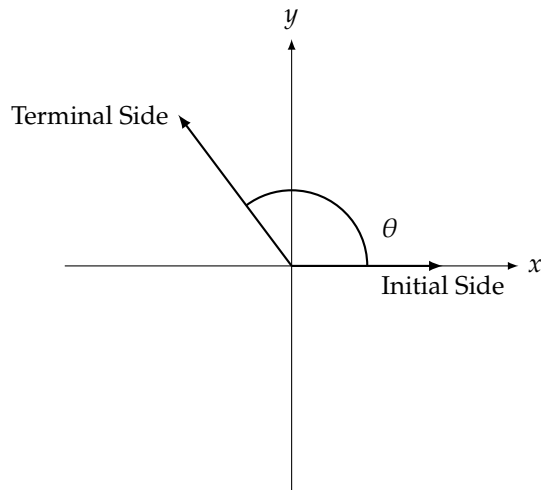


Figure 1.3: Angle in Standard Position

The figure below shows these quadrantal angles in their standard positions.

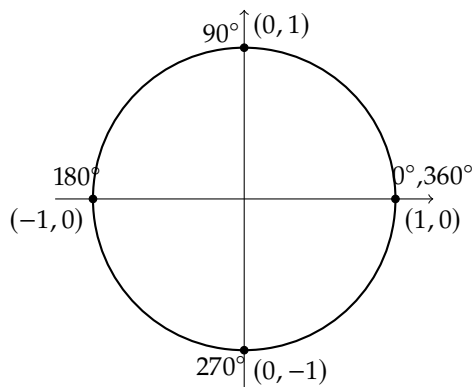


Figure 1.4: Quadrantal Angles

Coterminal Angles

Theorem 1.1.1 Two angles are *coterminal* if they share the same initial side and terminal side, but differ by a full rotation of 360° . For any given angle θ , the coterminal angles can be expressed as:

$$\theta + 360^\circ k \quad \text{where } k \in \mathbb{Z}.$$

For visualisation, below are two angles that are sharing the same initial and terminal side in figure 1.5 Another simple method that can be used to find coterminal angles is continuously adding or subtracting 360° to the given angle. Every time you add or subtract 360° to an angle, you find a coterminal angle of the given angle.

Degree Measure

A complete revolution, i.e. when the initial and terminal sides are in the same position after rotating clockwise or anticlockwise, is divided into 360 units called degrees. So, if the rotation from the initial side to the

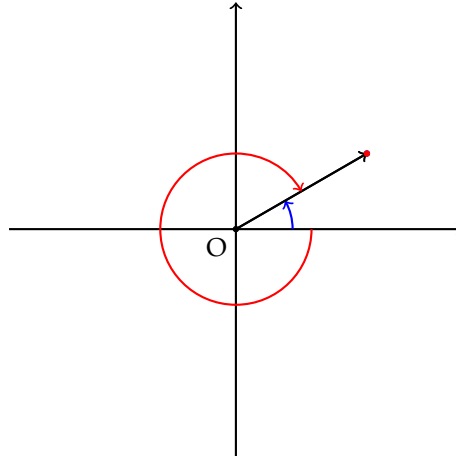


Figure 1.5: Coterminal angles

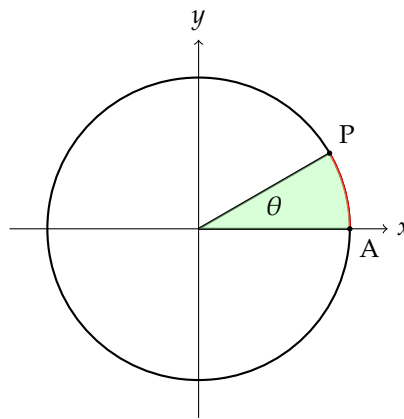


Figure 1.6: Unit Circle - Degree Measure

terminal side is $\left(\frac{1}{360}\right)^{\text{TH}}$ of a revolution, then the angle is said to have a measure of one degree. It is denoted as 1° .

Mathematically, consider a unit circle on Figure 1.6 below.

As $\theta = \angle AOP$ increases, the length of arc AP increases as well. Thus, AP is proportional to θ . Then, we shall say that $\angle AOP$ has degree measure if and only if

$$\frac{\text{Length of arc } AP}{\text{Circumference}} = \frac{\theta}{360^\circ}$$

Making θ the subject of the formula yields

$$\theta = \frac{360^\circ \times \text{Length of arc } AP}{2\pi}$$

Note that Circumference = $2\pi r$ where $r = 1$. Also, the angle is measured in an anticlockwise direction from A .

Radian Measure

Definition 1.1.2 *Radian Measure* is the standard unit of angular measurement which is based on the ratio of the arc's length to the radius of the circle it is part of. The unit used in this method is the "**radian**" (commonly denoted by ("**rad**"))

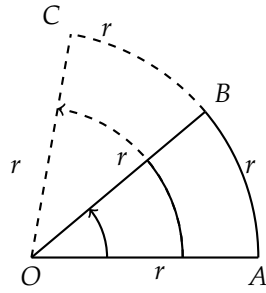


Figure 1.7: Radian Measure

In a circle of radius r , center O , we take an arc AB also of length r (Fig. 1.7). Then the angle $\angle AOB$ is the unit of radian measurement, called **one radian**.

- For example, in Fig. 1.7, if arc $AC = 2r$, then $\angle AOC = 2$ radians, and so on.
- If the arc is kr , the angle subtended is k radians.
- **Key Note:** The size of 1 radian does not depend on the length of r or any arbitrary number.

Theorem 1.1.2 We define 1 radian as:

1 radian = Angle subtended at the center by an arc of length equal to the radius.

Since an angle in radians is based on the multiple of the radius, and circumference of a circle is given by Circumference = $2\pi r$, the angle subtended by the circumference at the centre of a circle is equal to 2π . And from degree measure, the complete revolutions 360° , hence, the complete revolution in radians is 2π .

Radian-Degree Conversions

Converting angles from degree to radians or vice versa is based on the fact that

$$2\pi \text{ rad} = 360^\circ$$

or in its simpler form

$$\pi \text{ rad} = 180^\circ$$

Given an angle x , you can use if-then relations or use the following

1. If you are given an angle in degree to be converted to radians, multiply the angle by a conversion factor $\frac{\pi}{180^\circ}$
2. And if the angle is in radians to be converted to degrees, multiply by $\frac{180^\circ}{\pi}$

1.1.2 Worked Examples

Example 1.1.1 (Find the negative coterminal angle of 30°)

Solution:

To find a negative coterminal angle, we subtract multiples of 360° from the given angle until the result is negative. Starting with 30° :

Start with the formula for coterminal angles:

$$\theta + 360^\circ k.$$

$$\begin{aligned}\text{Negative coterminal angle} &= 30^\circ + 360^\circ(-1) \\ &= 30^\circ - 360^\circ = -330^\circ\end{aligned}$$

Thus, the negative coterminal angle of 30° is -330° and the representation is shown in the figure 1.8.

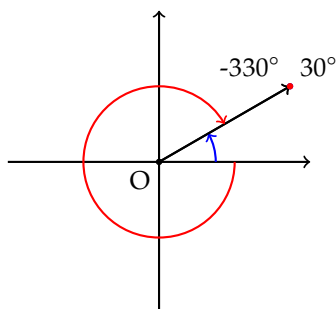


Figure 1.8: Coterminal Angles for 30°

Example 1.1.2 (Finding Coterminal Angles) Find one positive and one negative coterminal angle for $\theta = 45^\circ$.

Solution:

Start with the formula for coterminal angles:

$$\theta + 360^\circ k.$$

For one positive coterminal angle:

$$45^\circ + 360^\circ(1) = 405^\circ.$$

For one negative coterminal angle:

$$45^\circ + 360^\circ(-1) = -315^\circ.$$

Therefore, the coterminal angles are 405° and -315° as shown in Figure 1.9:

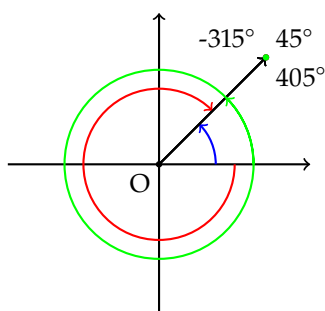


Figure 1.9: Coterminal angles for 45°

Example 1.1.3 (Find the least possible measure coterminal with each angle)

Solution:

We recall that coterminal angles differ by integer multiples of 360° . That is, to find a coterminal angle, we add or subtract $360^\circ k$ (where $k \in \mathbb{Z}$) until the angle lies in the range $0^\circ \leq \text{angle} < 360^\circ$ (for the least positive coterminal angle) or $-360^\circ < \text{angle} \leq 0^\circ$ (for the least negative coterminal angle).

(a) 908° :

Start with the given angle:

$$908^\circ.$$

Subtract multiples of 360° until the angle falls within $0^\circ \leq \text{angle} < 360^\circ$:

$$908^\circ - 360^\circ \cdot 2 = 908^\circ - 720^\circ = 188^\circ.$$

Thus, the least positive coterminal angle is:

$$\boxed{188^\circ}.$$

Another Method

If the angle is positive, then divide the angle by 360° . The remainder will be a coterminal angle between 0° and 360° . Now, if the given angle is negative, then divide the angle by 360° . The remainder gives a negative coterminal angle between -360° and 0° . Finally, add 360° to this angle to get a coterminal angle between 0° and 360° .

(b) For -75° :

Start with the given angle:

$$-75^\circ.$$

Add multiples of 360° until the angle falls within $0^\circ \leq \text{angle} < 360^\circ$:

$$-75^\circ + 360^\circ = 285^\circ.$$

Thus, the least positive coterminal angle is:

$$\boxed{285^\circ}.$$

If we need the least negative coterminal angle, it is the angle itself since -75° is already within the range $-360^\circ < \text{angle} \leq 0^\circ$:

$$\boxed{-75^\circ}.$$

Example 1.1.4 Check if the following pairs of angles are coterminal or not.

NOTE:

Two angles are coterminal if they differ by a multiple of 360° (or 2π radians).

(a) 30° and 330°

$$330^\circ - 30^\circ = 300^\circ$$

Since 300° is not a multiple of 360° , these angles are *not* coterminal.

(b) 125° and 845°

$$845^\circ - 125^\circ = 720^\circ$$

Since $720^\circ = 2 \times 360^\circ$, these angles *are* coterminal.

(c) $\frac{\pi}{3}$ and $\frac{7\pi}{3}$

$$\frac{7\pi}{3} - \frac{\pi}{3} = \frac{6\pi}{3} = 2\pi$$

Since the difference is 2π , these angles *are* coterminal.

(d) -60° and 250°

$$250^\circ - (-60^\circ) = 250^\circ + 60^\circ = 310^\circ$$

Since 310° is not a multiple of 360° , these angles are *not* coterminal.

Example 1.1.5 (Convert the following radians to degree measure)

Solution

(a) $\frac{\pi}{3}$:

$$\text{Degrees} = \frac{\pi}{3} \times \frac{180^\circ}{\pi} = \frac{180^\circ}{3} = 60^\circ.$$

Answer: 60° (b) $\frac{\pi}{6}$:

$$\text{Degrees} = \frac{\pi}{6} \times \frac{180^\circ}{\pi} = \frac{180^\circ}{6} = 30^\circ.$$

Answer: 30° (c) $\frac{3\pi}{4}$:

$$\text{Degrees} = \frac{3\pi}{4} \times \frac{180^\circ}{\pi} = \frac{3 \cdot 180^\circ}{4} = 135^\circ.$$

Answer: 135°

(d) 2:

$$\text{Degrees} = 2 \times \frac{180^\circ}{\pi} \approx \frac{360^\circ}{3.14159} \approx 114.59^\circ.$$

Answer: 114.59° (e) $\frac{\pi}{8}$:

$$\text{Degrees} = \frac{\pi}{8} \times \frac{180^\circ}{\pi} = \frac{180^\circ}{8} = 22.5^\circ.$$

Answer: 22.5° (f) $\frac{3\pi}{2}$ rads

$$\frac{3\pi}{2} \times \frac{180^\circ}{\pi} = 270^\circ$$

Answer: 270°

(g) 3 rads

$$3 \times \frac{180^\circ}{\pi} = 3 \times 57.296 = 171.9^\circ$$

Answer: 171.9° (h) $\frac{\pi}{4}$ rads

$$\frac{\pi}{4} \times \frac{180^\circ}{\pi} = 45^\circ$$

Answer: 45°

Example 1.1.6 Convert the following to radian measure as a multiple of π

Solution(a) 30° :

$$\text{Radians} = 30^\circ \times \frac{\pi}{180^\circ} = \frac{\pi}{6}.$$

Answer: $\frac{\pi}{6}$ (b) 135° :

$$\text{Radians} = 135^\circ \times \frac{\pi}{180^\circ} = \frac{3\pi}{4}.$$

Answer: $\frac{3\pi}{4}$ (c) 540° :

$$\text{Radians} = 540^\circ \times \frac{\pi}{180^\circ} = 3\pi.$$

Answer: 3π (d) 105° :

$$\text{Radians} = 105^\circ \times \frac{\pi}{180^\circ} = \frac{7\pi}{12}.$$

Answer: $\frac{7\pi}{12}$ (e) 75° :

$$\text{Radians} = 75^\circ \times \frac{\pi}{180^\circ} = \frac{5\pi}{12}.$$

Answer: $\frac{5\pi}{12}$ (f) 135° If $180^\circ = \pi$ radians, then 135° is less.

Thus, we have:

$$135^\circ \times \frac{\pi}{180^\circ} = \frac{3\pi}{4} \text{ rads}$$

Answer: $\frac{3\pi}{4}$ (g) 210° If $180^\circ = \pi$ radians, then 210° is more.

Hence, we have:

$$210^\circ \times \frac{\pi}{180^\circ} = \frac{7\pi}{6} \text{ rads}$$

Answer: $\frac{7\pi}{6}$

1.2 Arcs and Sectors

1.2.1 Core Insights

Definition 1.2.1 Before we start anything, let's the following terms:

- **Arc:** An arc is a portion of the circumference of a circle. It is defined by two points on the circle and the continuous portion of the circle between those two points. An arc can either be a minor arc (less than half the circle) or a major arc (more than half the circle).
- **Sector:** A sector is a region of a circle enclosed by two radii and the arc between them. It resembles a "slice of pie" or "pizza slice."

Below are diagrams to illustrate an arc and a sector:

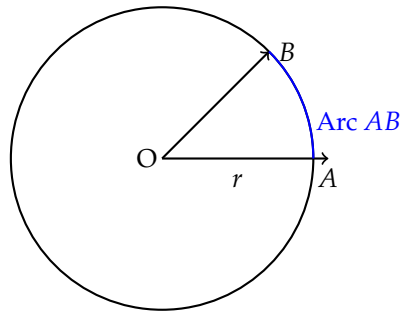


Diagram of an Arc

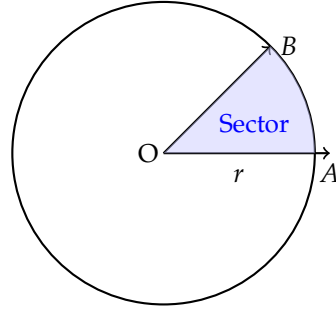


Diagram of a Sector

Figure 1.10: Illustrations of an Arc and a Sector.

From the definition of degree measure (Fig. 1.6), we looked at the relationship between the length of arc and the circumference as follows.

$$\frac{\text{Length of arc AP}}{\text{Circumference}} = \frac{\theta}{360^\circ} = \frac{\theta}{2\pi}$$

Thus,

$$\text{Length of arc AP} = \frac{\theta \times \text{Circumference}}{2\pi}$$

$$= \frac{\theta \times 2\pi r}{2\pi} \quad \text{Since Circumference} = 2\pi r$$

$$= r\theta$$

Therefore, the Length of arc AP = $r\theta$, where θ is in radians.

Theorem 1.2.1 In general, length is commonly denoted as s and is given by

$$s = r\theta$$

for any θ in radian measure.

In the figure 1.10, AOB is a sector of angle θ radians in a circle with centre O and radius r . From the same concept of similarities, the area of the sector is proportional to θ .

$$\frac{\text{Area of sector AOB}}{\text{Area of circle}} = \frac{\theta}{2\pi}$$

Hence,

$$\text{Area of sector AOB} = \frac{\theta}{2\pi} \times \pi r^2 = \frac{1}{2} r^2 \theta$$

Theorem 1.2.2 In general, area of a sector is given by

$$\text{Area of sector of angle } \theta = \frac{1}{2}r^2\theta$$

for any θ in radian measure.

1.2.2 Worked Examples

On Solving Questions in this Section

The following guidelines should be followed to get accurate answers and better explanations of your work.

1. You must make a diagrammatic representation of the information given in the question where there isn't any given in advance. This helps in having a better understanding of what the questions really want.
2. Convert any angle given in degrees to radians since our formulas in practice only allow angles that are in radians.
3. Keep your solved values that will be used on the next stages of the question in fraction or surd forms. **Do not round them off** as this might lead to changes in the values as compared to the originals. Although those changes might not affect the final answer at all, we shouldn't take chances.

Example 1.2.1 Find the length of an arc that subtends the central angle of 30° in a circle of radius 4 cm.

Solution

First of all you have to draw the diagrammatic representation of the given question. You have to do this for each and every question. Consider the figure 1.11 for this question.

Since the given angle is in degree measure, it has to be converted to radians to be useful in our formula. As such we will multiply 30° by a factor and we will get

$$30^\circ \cdot \frac{\pi}{180^\circ} = \frac{\pi}{6}$$

Then we will use the theorem 1.2.1 to find the length of the arc. Thus'

$$AB = (4) \left(\frac{\pi}{6} \right) = \frac{2\pi}{3}$$

Therefore, the length of the arc is 2.094 cm to 3 decimal places.

Example 1.2.2 The length of an arc in a circle of radius 5 cm is 6 cm. Find the angle subtended by the arc at the centre of the circle.

solution

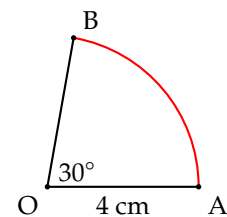


Figure 1.11: Illustration for Example 1.2.1

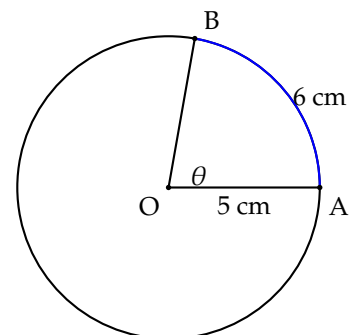


Figure 1.12: Illustration for Example 1.2.2

We have been given a radius of a circle and an arc length. Using theorem 1.2.1, $s = r\theta$. Thus,

$$6 = 5\theta$$

$$\theta = \frac{6}{5} = 1.2 \text{ rad}$$

Example 1.2.3 A sector of a circle has radius r and angle θ . Find the value of θ correct to 3 significant figures if the perimeter of the sector equals half the circumference of the circle.

solution

The perimeter of the sector $= r + r + r\theta = r(2 + \theta)$.

Then $r(2 + \theta) = \frac{1}{2} \times 2\pi r$ and

$$2 + \theta = \pi$$

giving $\theta = \pi - 2 \approx 1.14 \text{ rad}$.

Example 1.2.4 A sector of angle θ in a circle of radius r cm has an area of 5 cm^2 and its perimeter is 9 cm . Find the values of r and θ .

Solution

$$\text{Area} = \frac{1}{2}r^2\theta = 5 \text{ so } r^2\theta = 10 \quad (\text{i})$$

$$\text{Perimeter} = r + r + r\theta = 2r + r\theta = 9 \quad (\text{ii})$$

We solve these two equations.

From (i), $\theta = \frac{10}{r^2}$.

Substitute in (ii), $2r + r\frac{10}{r^2} = 9$ which gives $2r^2 + 10 = 9r$ or $2r^2 - 9r + 10 = 0$.

Hence $(2r - 5)(r - 2) = 0$ and $r = 2\frac{1}{2}$ or 2 cm .

From (i), the corresponding values of θ are 1.6 or 2.5 rad .

Example 1.2.5 In a circle with centre O , AOB and COD are two concentric sectors as shown in Figure 1.13. The lengths of the arcs AB and DC are 2.8 cm and 2 cm respectively, and $AD = 2 \text{ cm}$. Calculate:

- (a) the length of OC ,
- (b) $\angle AOB$ in radians,

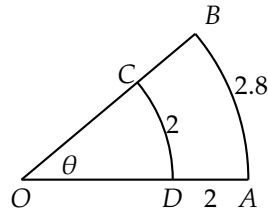


Figure 1.13: Illustration for Example 1.2.5

(c) the area of ABCD.

Solution

We have seen from Theorem 1.2.1 that $s = r\theta$. Hence, if we let OC to be r ; we will have $OA = r + 2$. From here we will define two simultaneous equations.

From the inner sector;

$$2 = r\theta \quad (i)$$

and from the outer sector;

$$2.8 = (r + 2)\theta \quad (ii)$$

If we simplify equation 2, we will get, $2.8 = r\theta + 2\theta$. Let's use substitution method to solve this system of equations. We will there go ahead to substitute 2 from (i) for $r\theta$ above..

$$2.8 = 2 + 2\theta \text{ and } 2.8 - 2 = 0.8 = 2\theta$$

That means

$$\theta = \frac{0.8}{2} = 0.4 \text{ rad}$$

(a) OC is the radius to the inner sector. As such, we will use $s = r\theta$ to solve this part.

$$2 = r(0.4) \text{ and } r = \frac{2}{0.4} = 5 \text{ cm}$$

(b) $\angle AOB = \theta$. Therefore, $\angle AOB = 0.4 \text{ rad} \approx 22.92^\circ$.

(c) Area of $ABCD$ is a bordered area. That is, the area is simply the difference of the inner sector from the outer sector. Using $A = \frac{1}{2}r^2\theta$;

$$ABCD = \left(\frac{1}{2} \times (5 + 2)^2 \times 0.4 \right) - \left(\frac{1}{2} \times 5^2 \times 0.4 \right)$$

$$ABCD = \frac{49}{5} - 5 = \frac{24}{5} = 4.8 \text{ cm}^2$$

Example 1.2.6 A piece of wire, 10 cm long, is formed into the shape of a sector of a circle of radius r cm and angle θ radians.

(a) Show that $\theta = \frac{10-2r}{r}$.

(b) Show also that $A \text{ cm}^2$ of the sector is given by $a = 5r - r^2$

Solution

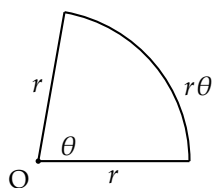


Figure 1.14: Illustration for Example 1.2.6

- (a) The length equals perimeter of the sector.

The length of the wire = $r\theta + 2r = 10$.

$$r\theta + 2r = 10.$$

$$r(\theta + 2) = 10.$$

$$\theta + 2 = \frac{10}{r}.$$

$$\theta = \frac{10}{r} - 2.$$

Therefore

$$\theta = \frac{10 - 2r}{r}.$$

- (b) Using θ from (a) above,

$$A = \frac{1}{2}r^2\theta.$$

$$A = \frac{1}{2}r^2\left(\frac{10 - 2r}{r}\right).$$

$$A = \frac{1}{2}r(10 - 2r).$$

$$A = \frac{1}{2} \times (10r - 2r^2).$$

$$A = 5r - r^2.$$

Example 1.2.7 Figure 1.15 shows a semicircle $AOBC$ with centre O . CD is an arc of another circle with centre A . If the length of AO is 4 cm, calculate the area of the shaded part.

Solution

In this figure, you might notice that $AO = OC = OB = 4$ cm since those are just radii of the semicircle. For the same reason, $AC =$

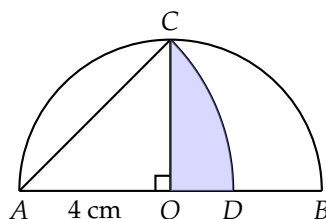


Figure 1.15: Illustration for example 1.2.7

AD (radii of the sector ADC).

Using Pythagoras theorem, $AC^2 = AO^2 + OC^2$, Thus,

$$AC = \sqrt{4^2 + 4^2} = \sqrt{32} = 4\sqrt{2} \text{ cm}$$

² Now, for us to be able to find the area of the sector ADC , we need to know the value of $\angle CAD$, since $\triangle AOC$ is right angled, then $\tan AOC = \frac{OC}{AO} = \frac{4}{4} = 1$.
Thus,

$$\angle A = \tan^{-1} 1 = 45^\circ = \frac{\pi}{4} \text{ rad}$$

If you look at the picture carefully, you will notice that the area of the shaded part is simply a difference of the area of triangle AOC from the area of the sector ACD .

a. Area of $\triangle AOC$

Recall, area of triangle = $\frac{1}{2} \times \text{base} \times \text{height}$. Thus,

$$\text{Area of } \triangle AOC = \frac{1}{2} \cdot 4 \cdot 4 = 8 \text{ cm}^2$$

b. Area of sector ACD

Using theorem 1.2.2, area of sector = $\frac{1}{2}r^2\theta$ in this case,

$$\text{Area of } ACD = \frac{1}{2} \cdot (\sqrt{32})^2 \cdot \left(\frac{\pi}{4}\right) = \frac{1}{2} \cdot 32 \cdot \left(\frac{\pi}{4}\right)$$

This means that

$$\text{Area of } ACD = 4\pi \text{ cm}^2$$

Lastly,

Area of the shaded part = area of sector ACD – Area of triangle AOC

$$\text{Area of the shaded part} = (4\pi - 8) \text{ cm}^2$$

Therefore, the area of the shaded part is 4.566 cm^2 to 3 decimal places.

Example 1.2.8 The area of the sector OAB (Fig. 1.16) is 150 cm^2 . Calculate

- θ (in radians),
- The length of arc AB ,
- If this sector is folded up to form a cone, what is the radius of the cone?

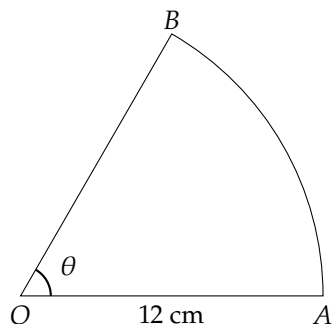


Figure 1.16: Illustration for Example 1.2.8

Solution

- (a) We've been given the area of a sector and its radius. Using Theorem 1.2.2, $\text{Area} = \frac{1}{2}r^2\theta$. Thus;

$$150 \text{ cm}^2 = \frac{1}{2}(12 \text{ cm})^2\theta = \frac{1}{2}(144 \text{ cm}^2)\theta = 72\theta \text{ cm}^2$$

If we make θ subject of the formula,

$$\theta = \frac{150 \cancel{\text{cm}^2}}{72 \cancel{\text{cm}^2}} = 2.083 \text{ rad}$$

- (b) Since we have the radius given, and angle found in (a) above, using theorem 1.2.1,

$$AB = 12 \times \frac{150}{72} = 25 \text{ cm.}$$

3

- (c) Once a sector is transformed into a cone, the arc length of the sector is eventually the circumference of the cone. From **Figure 1.17** AB will be a circle of radius R . Thus;

$$2\pi R = 25$$

And

$$R = \frac{25}{2\pi} = 3.98 \text{ cm}$$

3: It is advisable to use the fraction form of θ . Using rounded-off values has a lot of setbacks. For example, in this case $12 \times 2.083 = 24.996$ and not exactly 25 as it is in the final answer although technically, both answers might be considered as true.

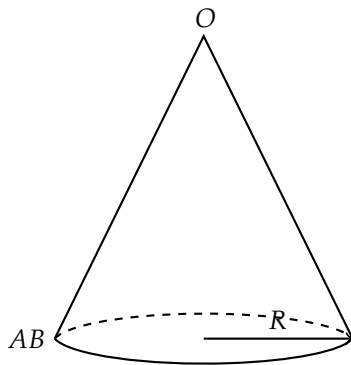


Figure 1.17: Illustration for Example 1.2.8 part (c)

1.3 Trigonometric Functions

1.3.1 Core Insights

Definitions of Trigonometric Functions

For any acute angle θ we can construct a right-angled triangle with one of the angles being θ . The three sides of this triangle are called the **adjacent** (denoted by **adj** which is the leg of the right-angled triangle that forms one side of the angle θ), the **opposite** (denoted by **opp** which is the leg of the right-angled triangle that is opposite the angle θ , i.e. does not form part of the angle), and the **hypotenuse** (denoted by **hyp** which is the longest side of the right-angled triangle).

Definition 1.3.1 (Quick review) *In secondary mathematics, we looked at three primary trig functions namely,*

1. Sine (denoted as $\sin$)
2. Cosine (denoted as $\cos$)
3. Tangent (denoted as $\tan$)

From **Figure 1.18**, these functions were defined as follows.

$$\text{Sine} = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

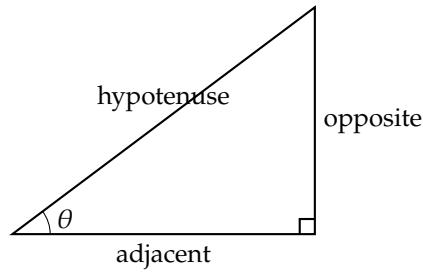


Figure 1.18: Right-angled triangle with names of its three sides

$$\text{Cosine} = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

$$\text{Tangent} = \frac{\text{Opposite}}{\text{Adjacent}}$$

Point to Note

If you look carefully, you will notice that the first letter of every word has been highlighted with a bold colour. If you take out these letters in their order from Sine through Tangent, they spell out **SOHCAHTOA** which is the acronym we used for easy recalling of these trig definitions.

Definition 1.3.2 (Reciprocal Functions) *In this book, we will also look at other three functions that are expressed as reciprocals of the primary ones. These include;*

1. Secant (denoted as sec)
2. Cosecant (denoted as csc)
3. Cotangent (denoted as cot)

As reciprocals, these trigonometric functions are defined as follows.

$$\text{Secant} = \frac{1}{\text{Cosine}} = \frac{\text{Hypotenuse}}{\text{Adjacent}}$$

$$\text{Cosecant} = \frac{1}{\text{Sine}} = \frac{\text{Hypotenuse}}{\text{Opposite}}$$

$$\text{Cotangent} = \frac{1}{\text{Tangent}} = \frac{\text{Adjacent}}{\text{Opposite}}$$

That means, in total, we have six trigonometric functions to be studied in this document.

Trigonometric Identities

Definition 1.3.3 *In mathematics, an **identity** is an equation that holds true for all values of the variables within its domain of definition. In trigonometry, identities are fundamental equations that express relationships between trigonometric functions and are valid for all angles where the functions are defined.*

From secondary school mathematics, we deduced the relationship between the three basic trigonometric ratios namely $\sin$, $\cos$ and $\tan$. For that matter, we saw that the tangent of an angle is equal to the ratio of its sine to cosine. Mathematically;

$$\tan x = \frac{\sin x}{\cos x} \quad (1.1)$$

4: Here, we are trying to highlight the fact that the angle used **MUST** be in the same measure. That is, it must be in degree or radian measure exclusively and not mixed.

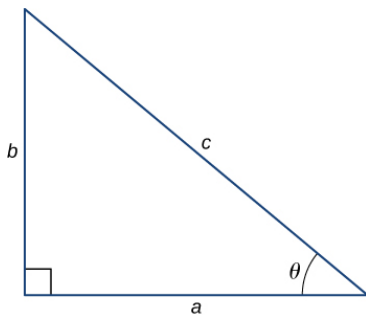


Figure 1.19: An example of a Right-Angled triangle

https://pressbooks.online.ucf.edu/app/uploads/sites/372/2021/04/CNX_UPhysics_00_EE_Triangle2_img.jpg

5: Using **SOHCAHTOA**, we know that from the diagram, $\tan \theta = \frac{b}{a}$, $\sin \theta = \frac{b}{c}$ and lastly, $\cos \theta = \frac{a}{c}$. if checked properly, $\frac{\sin x}{\cos x} = \frac{\frac{b}{c}}{\frac{a}{c}} = \frac{b}{c} \cdot \frac{c}{a} = \frac{b}{a} = \tan \theta$

for any given angle x in the same measure⁴.

From **Figure 1.19**, we used this particular diagram to deduce the said relationship⁵.

Earlier in this topic, we dealt with trigonometric ratios. And we discussed that some are reciprocals of others. We consider those as to fall under the identities too as they help in manipulating complicated identities under Proving Identities after this part. This means that the next three identities shall be,

$$\sec x = \frac{1}{\cos x} \quad (1.2)$$

$$\csc x = \frac{1}{\sin x} \quad (1.3)$$

$$\cot x = \frac{1}{\tan x} \quad (1.4)$$

But from identity 1.1, we have a definition of $\tan x$. As such we can redefine $\cot x$. That is, given,

$$\cot x = \frac{1}{\tan x}$$

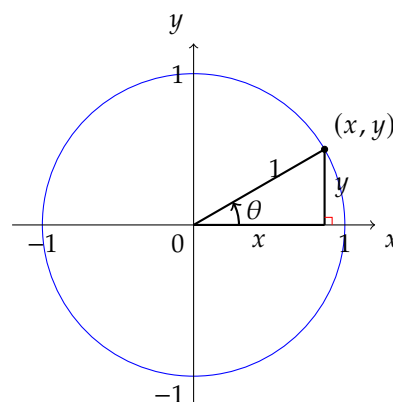
we will substitute.

$$\cot x = \frac{1}{\frac{\sin x}{\cos x}}$$

. And after a few algebraic manipulations, you will find the following subsequent identity.

$$\cot x = \frac{\cos x}{\sin x} \quad (1.5)$$

Now consider diagram of a right-angled triangle inside the unit circle.



Using **SOHCAHTOA**, $\sin x = \frac{y}{1}$ and $\cos x = \frac{x}{1}$ which means that, $\sin x = y$ and $\cos x = x$ respectively. And from the Pythagorean theorem, $x^2 + y^2 = r^2$ which means that $x^2 + y^2 = 1$.

If we substitute $\sin x$ and $\cos x$ for y and x respectively, we get,

$$(\cos x)^2 + (\sin x)^2 = 1$$

Therefore, the next trig identity and the most useful is;

$$\cos^2 x + \sin^2 x = 1 \quad (1.6)$$

From this identity, we will derive additional trig identities. We can generate other identities which fall under the same Pythagorean ideology. From $\cos^2 x + \sin^2 x = 1$, we will divide by $\cos^2 x$ which will give us

$$\frac{\cos^2 x}{\cos^2 x} + \frac{\sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x}$$

. At this point we know that $\frac{\sin^2 x}{\cos^2 x} = \tan^2 x$ and $\frac{1}{\cos^2 x} = \sec^2 x$. This means that the next identity is

$$1 + \tan^2 x = \sec^2 x \quad (1.7)$$

What if we divide by $\sin^2 x$? Here, we will get

$$\frac{\cos^2 x}{\sin^2 x} + \frac{\sin^2 x}{\sin^2 x} = \frac{1}{\sin^2 x}$$

. At this point we know that $\frac{\cos^2 x}{\sin^2 x} = \cot^2 x$ and $\frac{1}{\sin^2 x} = \csc^2 x$. If we substitute;

$$\cot^2 x + 1 = \csc^2 x \quad (1.8)$$

These are the eight trigonometric identities that will be used in this book. But memorising these identities might not be enough in some occasions. We are, therefore, required to understand the simple derivation used here. These identities will be used in many ways like solving trig equations and proving trigonometric identities.

Points to Note

Trigonometric functions raised to custom powers do not behave like most of algebraic functions. For example, $(\cos \theta)^2$ is not equal to $\cos \theta^2$. As such,

- The general option for the powers is as follows.

$$(\cos x)^n = \cos^n x$$

for all n .

- This rule applies for all the trigonometric ratios.
- All identities from the first one through the reciprocals remain true throughout all powers except the pythagorean identities which only allow power 2 and nothing else. That means, $\tan^2 x = \frac{\sin^2 x}{\cos^2 x}$ is true but $\cos x + \sin x = 1$ is not true.

ON PROVING TRIGONOMETRIC IDENTITIES

Proving trigonometric identities involves verifying that two sides of a given equation are equivalent for all valid values of the variable.

Below are the general steps:

1. Start with the more complex side of the equation.
2. Simplify using basic trigonometric identities:
 - Reciprocal identities: $\sec x = \frac{1}{\cos x}$, $\csc x = \frac{1}{\sin x}$, $\cot x = \frac{1}{\tan x}$.
 - Pythagorean identities: $\sin^2 x + \cos^2 x = 1$, etc.
3. Convert all trigonometric functions to $\sin x$ and $\cos x$, if necessary, to simplify expressions.
4. Combine fractions, factorize, or use common denominators when appropriate.
5. Compare the simplified expression with the other side to ensure equivalence.

1.3.2 Worked Examples

Example 1.3.1 (Prove that $1 + \tan^2 x = \sec^2 x$) **Solution:**

1. Start with the left-hand side (LHS): $1 + \tan^2 x$.
2. Substitute $\tan^2 x$ using its definition in terms of $\sin x$ and $\cos x$:

$$\tan^2 x = \frac{\sin^2 x}{\cos^2 x}.$$

Thus,

$$1 + \tan^2 x = 1 + \frac{\sin^2 x}{\cos^2 x}.$$

3. Combine terms using a common denominator:

$$1 + \frac{\sin^2 x}{\cos^2 x} = \frac{\cos^2 x}{\cos^2 x} + \frac{\sin^2 x}{\cos^2 x}.$$

4. Add the fractions:

$$\frac{\cos^2 x + \sin^2 x}{\cos^2 x}.$$

5. Apply the Pythagorean identity 1.6 $\cos^2 x + \sin^2 x = 1$:

$$\frac{1}{\cos^2 x}.$$

6. Recognize that $\frac{1}{\cos^2 x} = \sec^2 x$, which is the right-hand side (RHS).

Therefore, $1 + \tan^2 x = \sec^2 x$ is proven.

Tips for Proving Identities

- ▶ Always work on one side of the equation unless instructed otherwise.
- ▶ Look for opportunities to factorize or use conjugates.
- ▶ Check your steps carefully to avoid algebraic errors.
- ▶ The side you choose doesn't matter provided you support your manipulations. From personal experience, choosing a side that looks a little bit complicated than the other helps because I think it's harder to expand than to simplify (not always)

Example 1.3.2 (Prove that $\sin \theta \tan \theta = \sec \theta - \cos \theta$)

Solution:

We will prove the identity by working on the Left-Hand Side (LHS).

LHS: $\sin \theta \tan \theta$.

We know that $\tan \theta = \frac{\sin \theta}{\cos \theta}$. If we substitute this, we will get;

$$\sin \theta \tan \theta = \sin \theta \cdot \frac{\sin \theta}{\cos \theta}.$$

which then simplifies to:

$$\frac{\sin^2 \theta}{\cos \theta}.$$

But, if we look at $\cos^2 \theta + \sin^2 \theta = 1$, we can make $\sin^2 \theta$ subject of the formula and we will get, $\sin^2 \theta = 1 - \cos^2 \theta$. Thus,

$$\frac{\sin^2 \theta}{\cos \theta} = \frac{1 - \cos^2 \theta}{\cos \theta}$$

. And if we distribute the denominator, we will get

$$\frac{1}{\cos \theta} - \frac{\cos^2 \theta}{\cos \theta} = \frac{1}{\cos \theta} - \frac{\cancel{\cos \theta} \cdot \cos \theta}{\cancel{\cos \theta}}$$

Since we all know that from identity 1.2 that $\frac{1}{\cos \theta} = \sec \theta$, we will finalise this but plugging this into the main identity. Therefore,

$$\frac{1}{\cos \theta} - \cos \theta = \sec \theta - \cos \theta$$

and this is equal to the **Right Hand Side (RHS)** and it's now proved.

What if you decide to prove this from the **Right Hand Side (RHS)**?
No problem!!!

$$\text{RHS: } \sec \theta - \cos \theta$$

We know that we can eliminate $\sec \theta$ by substituting its equivalence. Hence,

$$\frac{1}{\cos \theta} - \cos \theta$$

Use $\cos \theta$ as a common denominator, combine the two fractions. Here, we will get

$$\frac{1 - \cos^2 \theta}{\cos \theta}$$

From the Pythagorean identity $\sin^2 \theta + \cos^2 \theta = 1$, we saw that, $1 - \cos^2 \theta = \sin^2 \theta$. If we substitute, we will get;

$$\frac{\sin^2 \theta}{\cos \theta}.$$

. Next, we will distribute the numerator.

$$\frac{\sin \theta \cdot \sin \theta}{\cos \theta} = \sin \theta \cdot \frac{\sin \theta}{\cos \theta}$$

From identity 1.1, we saw that $\frac{\sin \theta}{\cos \theta} = \tan \theta$. Therefore,

$$\sin \theta \cdot \frac{\sin \theta}{\cos \theta} = \sin \theta \tan \theta = \text{LHS}$$

Hence this has been proved from RHS.⁶

Example 1.3.3 (Prove that $(\cot \theta - \csc \theta)^2 = \frac{1 - \cos \theta}{1 + \cos \theta}$)

6: Despite the fact that I have solved this from both sides, you must only solve this kind of questions from one side. *Always choose one side only and apply any suitable mathematical manipulation to make it match with the other side.*

Solution:

We will simplify the Left-Hand Side (LHS) to match the Right-Hand Side (RHS).

Start with LHS

$$\text{LHS: } (\cot \theta - \csc \theta)^2.$$

At this point we already know the following equivalences: $\cot \theta = \frac{\cos \theta}{\sin \theta}$ and $\csc \theta = \frac{1}{\sin \theta}$. If we substitute, we will get:

$$(\cot \theta - \csc \theta)^2 = \left(\frac{\cos \theta}{\sin \theta} - \frac{1}{\sin \theta} \right)^2.$$

Using $\sin \theta$ as the common denominator, Combine terms:

$$\left(\frac{\cos \theta - 1}{\sin \theta} \right)^2.$$

Now we will Simplify:

$$\frac{(\cos \theta - 1)^2}{\sin^2 \theta}.$$

At this point we start with highlighting that $(\cos \theta - 1)^2 = (1 - \cos \theta)^2$.⁷ As such, we can factorise $(\cos \theta - 1)^2$ into $(1 - \cos \theta)(1 - \cos \theta)$.

$$\frac{(1 - \cos \theta)(1 - \cos \theta)}{\sin^2 \theta}.$$

Since $\sin^2 \theta$ can be written as $1 - \cos^2 \theta$ which is a difference of squares⁸, we are going to write it as $(1 + \cos \theta)(1 - \cos \theta)$. Thus,

$$\frac{(1 - \cos \theta)(1 - \cos \theta)}{\sin^2 \theta} = \frac{(1 - \cos \theta)(\cancel{1 - \cos \theta})}{(1 + \cos \theta)(\cancel{1 - \cos \theta})} = \frac{1 - \cos \theta}{1 + \cos \theta} = \text{RHS}$$

Hence this identity is proved from the Left Hand Side.

If we prove it from the Right Hand Side (RHS),

Difference of Squares

A difference of squares is a special case of quadratic expansions. In this case $a^2 - b^2 = (a + b)(a - b)$.

On Rationalising Denominators

We used this concept in secondary school mathematics when handling irrational numbers (Surds). Given a binomial denominator, a *Conjugate* is an expression similar to the original one but with an opposite sign. For example, the conjugate of $(a + b)$ is $(a - b)$. In surds, this is how the math looked like;

$$\frac{a}{\sqrt{b} + \sqrt{c}} = \frac{a \cdot (\sqrt{b} - \sqrt{c})}{(\sqrt{b} + \sqrt{c})(\sqrt{b} - \sqrt{c})}$$

And this finally leaves us with a difference of squares on the denominator.

$$\text{RHS: } \frac{1 - \cos \theta}{1 + \cos \theta}$$

Do you remember the concept of Rationalising a denominator⁹ in Surds? Here we will multiply the numerator and denominator by the conjugate of $(1 + \cos \theta)$ which is $(1 - \cos \theta)$.

$$\frac{1 - \cos \theta}{1 + \cos \theta} \cdot \frac{1 - \cos \theta}{1 - \cos \theta}$$

Knowing that the denominator is a difference of squares, we will simplify it as follows;

$$\frac{(1 - \cos \theta)^2}{1 - \cos^2 \theta}$$

At this point we know that $1 - \cos^2 \theta = \sin^2 \theta$, we will simplify as follows.

$$\frac{(1 - \cos \theta)^2}{\sin^2 \theta} = \frac{(1 - \cos \theta)^2}{(\sin \theta)^2} = \left(\frac{1 - \cos \theta}{\sin \theta} \right)^2 = \left(\frac{1}{\sin \theta} - \frac{\cos \theta}{\sin \theta} \right)^2$$

Since $\frac{1}{\sin \theta} = \csc \theta$ and $\frac{\cos \theta}{\sin \theta} = \cot \theta$;

$$\boxed{(\csc \theta - \cot \theta)^2 = (\cot \theta - \csc \theta)^2} = \text{LHS}$$

.

This is also proved from the Right Hand Side (RHS)

Example 1.3.4 (Prove that $\tan^2 \theta - \sin^2 \theta = \tan^2 \theta \sin^2 \theta$.)

Solution:

Start with the left-hand side (LHS):

$$\text{LHS: } \tan^2 \theta - \sin^2 \theta.$$

Substitute $\tan^2 \theta = \frac{\sin^2 \theta}{\cos^2 \theta}$:

$$\frac{\sin^2 \theta}{\cos^2 \theta} - \sin^2 \theta.$$

Write $\sin^2 \theta$ with the denominator $\cos^2 \theta$:

$$\frac{\sin^2 \theta}{\cos^2 \theta} - \frac{\sin^2 \theta \cos^2 \theta}{\cos^2 \theta}.$$

Combine terms over the common denominator:

$$\frac{\sin^2 \theta - \sin^2 \theta \cos^2 \theta}{\cos^2 \theta}.$$

Factor out $\sin^2 \theta$:

$$\frac{\sin^2 \theta (1 - \cos^2 \theta)}{\cos^2 \theta}.$$

From $1 - \cos^2 \theta = \sin^2 \theta$, substitute:

$$\frac{\sin^2 \theta \cdot \sin^2 \theta}{\cos^2 \theta}.$$

Simplify:

$$\frac{\sin^4 \theta}{\cos^2 \theta}.$$

Rewrite as:

$$\tan^2 \theta \sin^2 \theta.$$

Thus, the LHS is equal to the RHS:

$$\tan^2 \theta - \sin^2 \theta = \tan^2 \theta \sin^2 \theta.$$

Example 1.3.5 (Prove that $\frac{\sin x}{\csc x} + \frac{\cos x}{\sec x} = 1$.)

Solution:

Start with the left-hand side (LHS):

$$\text{LHS: } \frac{\sin x}{\csc x} + \frac{\cos x}{\sec x}.$$

Substitute $\csc x = \frac{1}{\sin x}$ and $\sec x = \frac{1}{\cos x}$:

$$\frac{\sin x}{\frac{1}{\sin x}} + \frac{\cos x}{\frac{1}{\cos x}}.$$

Simplify each term:

$$\sin x \cdot \sin x + \cos x \cdot \cos x.$$

This becomes:

$$\sin^2 x + \cos^2 x.$$

From the Pythagorean identity:

$$\sin^2 x + \cos^2 x = 1.$$

Thus:

$$\frac{\sin x}{\csc x} + \frac{\cos x}{\sec x} = 1.$$

Example 1.3.6 (Prove that $\frac{\csc^2 \theta}{1 + \tan^2 \theta} = \cot^2 \theta$)

Solution:

We will simplify the Left-Hand Side (LHS) to match the Right-Hand Side (RHS).

Start with LHS:

$$\text{LHS: } \frac{\csc^2 \theta}{1 + \tan^2 \theta}.$$

Recall the identities:

$$\csc^2 \theta = \frac{1}{\sin^2 \theta} \quad \text{and} \quad 1 + \tan^2 \theta = \sec^2 \theta.$$

Substitute these into the expression:

$$\frac{\csc^2 \theta}{1 + \tan^2 \theta} = \frac{\frac{1}{\sin^2 \theta}}{\sec^2 \theta}.$$

Using $\sec^2 \theta = \frac{1}{\cos^2 \theta}$, the expression becomes:

$$\frac{\frac{1}{\sin^2 \theta}}{\frac{1}{\cos^2 \theta}} = \frac{\cos^2 \theta}{\sin^2 \theta}.$$

This simplifies to:

$$\cot^2 \theta.$$

Hence:

$$\boxed{\frac{\csc^2 \theta}{1 + \tan^2 \theta} = \cot^2 \theta}.$$

Example 1.3.7 (Prove that $\cot x + \tan x = \csc x \sec x$)

Solution:

We will simplify the Left-Hand Side (LHS) to match the Right-Hand Side (RHS).

Start with LHS:

$$\text{LHS: } \cot x + \tan x.$$

Recall the definitions:

$$\cot x = \frac{\cos x}{\sin x}, \quad \tan x = \frac{\sin x}{\cos x}.$$

Substitute these into the expression:

$$\cot x + \tan x = \frac{\cos x}{\sin x} + \frac{\sin x}{\cos x}.$$

Combine terms using a common denominator:

$$\frac{\cos^2 x + \sin^2 x}{\sin x \cos x}.$$

Using $\cos^2 x + \sin^2 x = 1$, the numerator becomes:

$$\frac{1}{\sin x \cos x}.$$

Finally:

$$\frac{1}{\sin x \cos x} = \csc x \sec x.$$

Hence:

$$\boxed{\cot x + \tan x = \csc x \sec x}.$$

Example 1.3.8 (Prove that $(\cos \phi + \sin \phi)^2 + (\cos \phi - \sin \phi)^2 = 2$)

Solution:

We will simplify the Left-Hand Side (LHS) to match the Right-Hand Side (RHS).

Start with LHS:

$$\text{LHS: } (\cos \phi + \sin \phi)^2 + (\cos \phi - \sin \phi)^2.$$

Expand each term using the binomial expansion:

$$(\cos \phi + \sin \phi)^2 = \cos^2 \phi + 2 \cos \phi \sin \phi + \sin^2 \phi,$$

$$(\cos \phi - \sin \phi)^2 = \cos^2 \phi - 2 \cos \phi \sin \phi + \sin^2 \phi.$$

Add the two results:

$$\cos^2 \phi + 2 \cos \phi \sin \phi + \sin^2 \phi + \cos^2 \phi - 2 \cos \phi \sin \phi + \sin^2 \phi.$$

Combine like terms:

$$2 \cos^2 \phi + 2 \sin^2 \phi.$$

Using $\cos^2 \phi + \sin^2 \phi = 1$, this simplifies to:

$$2(1) = 2.$$

Hence:

$$\boxed{(\cos \phi + \sin \phi)^2 + (\cos \phi - \sin \phi)^2 = 2}.$$

Example 1.3.9 (Prove that $\frac{1}{1-\sin \theta} + \frac{1}{1+\sin \theta} = 2 \sec^2 \theta$)

Solution:

We will simplify the Left-Hand Side (LHS) to match the Right-Hand Side (RHS).

Start with LHS:

$$\text{LHS: } \frac{1}{1-\sin \theta} + \frac{1}{1+\sin \theta}.$$

Combine terms using a common denominator:

$$\frac{(1+\sin \theta) + (1-\sin \theta)}{(1-\sin \theta)(1+\sin \theta)}.$$

The numerator simplifies to:

$$2,$$

and the denominator becomes:

$$1 - \sin^2 \theta = \cos^2 \theta.$$

Thus:

$$\frac{2}{\cos^2 \theta}.$$

Using $\sec^2 \theta = \frac{1}{\cos^2 \theta}$, this simplifies to:

$$2 \sec^2 \theta.$$

Hence:

$$\boxed{\frac{1}{1 - \sin \theta} + \frac{1}{1 + \sin \theta} = 2 \sec^2 \theta}.$$

1.4 Trigonometric Equations

1.4.1 Core Insights

Quadrants and Their Signs

As we measure angles from 0° to 360° , we form a plane that can be divided into four equal quadrants using the x – and y – axis. In these quadrants, the behaviour of trig functions differ. For example if you enter, $\sin 30^\circ$ or $\sin 150^\circ$ into a calculator you will get $+\frac{1}{2}$. But if you do the same with 210° or 330° , you will get $-\frac{1}{2}$. You can notice that the values are positive in some angles and negative in the other. But it's the same **sine**, isn't it?

Consider figure 1.20 of a unit circle (radius = 1);

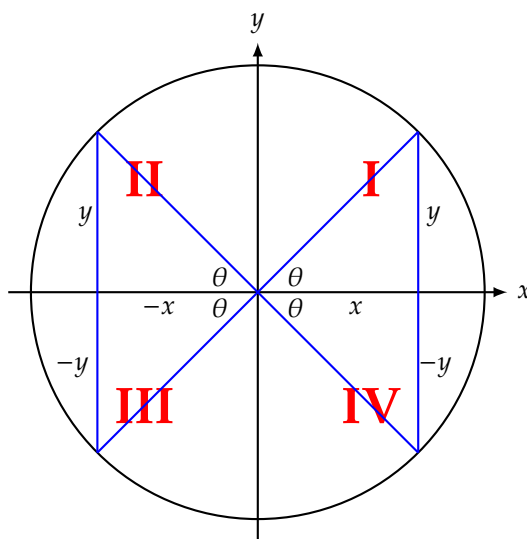


Figure 1.20: Quadrants and their Signs

Using **SOHCAHTOA**, you can define sine, cosine and tangent of θ across the quadrants as follows.

QUADRANT I: For all angles between 0° and 90° :

$$\left. \begin{aligned} \sin \theta &= \frac{y}{r} = \frac{y}{1} = y, \\ \cos \theta &= \frac{x}{r} = \frac{x}{1} = x, \\ \tan \theta &= \frac{y}{x}. \end{aligned} \right\} \text{All Trig Functions are Positive in this quadrant.}$$

QUADRANT II: For all angles between 90° and 180° :

$$\left. \begin{aligned} \sin \theta &= \frac{y}{r} = \frac{y}{1} = y, \\ \cos \theta &= \frac{-x}{r} = \frac{-x}{1} = -x, \\ \tan \theta &= \frac{y}{-x}. \end{aligned} \right\} \text{Only Sine is Positive; Cosine and Tangent are Negative.}$$

QUADRANT III: For all angles between 180° and 270° :

$$\left. \begin{aligned} \sin \theta &= \frac{-y}{r} = \frac{-y}{1} = -y, \\ \cos \theta &= \frac{-x}{r} = \frac{-x}{1} = -x, \\ \tan \theta &= \frac{-y}{-x} = \frac{y}{x}. \end{aligned} \right\} \text{Only Tangent is Positive; Sine and Cosine are Negative.}$$

QUADRANT IV: For all angles between 270° and 360° :

$$\left. \begin{aligned} \sin \theta &= \frac{-y}{r} = \frac{-y}{1} = -y, \\ \cos \theta &= \frac{x}{r} = \frac{x}{1} = x, \\ \tan \theta &= \frac{-y}{x}. \end{aligned} \right\} \text{Only Cosine is Positive; Sine and Tangent are Negative.}$$

The above facts also corresponds the reciprocal functions associated with such primary ones. That is, the quadrants in which Sine, Cosine and Tangent are positive, Cosecant, Secant and Cotangent is also positive respectively.

SUMMERY

Here, we will add labels for the quadrant for easy follow up and recalling when solving equations that demands these quadrant signs of trig functions.

- I. In **QUADRANT I**, all the trig funcs are **positive**. we will, therefore, label this with **A**.
- II. In **QUADRANT II**, only **Sine** and its reciprocal(Cosecant) are positive. We will label this with **S**
- III. In **QUADRANT III**, only tangent**tangent** and its reciprocal (Cotangent) are positive. We will label this with **T**
- IV In **QUADRANT IV**, only **Cosine** and its reciprocal (Secant) are positive. We will label this **C**

NOTE: If you arrange the labels in the order, IV $\rightarrow$ I $\rightarrow$ II $\rightarrow$ III, we spell out **C.A.S.T** as shown in **Figure 1.21**.

QUADRANT	RANGE	sin	cos	tan	csc	sec	cot
I	0° to 90°	+	+	+	+	+	+
II	90° to 180°	+	-	-	+	-	-
III	180° to 270°	-	-	+	-	-	+
IV	270° to 360°	-	+	-	-	+	-

Table 1.1: The Table Showing Signs of Trigonometric Functions Across Quadrants

The Range of Trigonometric Functions

Definition 1.4.1 Recall;

- **Domain:** The complete set of all possible input values (usually represented by x) for which the function is defined.
- **Range:** The complete set of all possible output values (usually represented by y) that the function can produce.

From College Algebra, we know that for sine and cosine, every known angle is part of the domain. Consider **Figure 1.22**

Using Pythagoras theorem, $x^2 + y^2 = r^2$. From this, we know that

$$x^2 \leq r^2 \text{ and } y^2 \leq r^2$$

If we take roots everywhere;

$$-r \leq x \leq r \text{ and } -r \leq y \leq r$$

Which when divided by r

$$-1 \leq \frac{x}{r} \leq 1 \text{ and } -1 \leq \frac{y}{r} \leq 1$$

From definitions of trigonometric functions, $\sin \alpha = \frac{y}{r}$ and $\cos \alpha = \frac{x}{r}$. Thus

$$-1 \leq \cos \alpha \leq 1 \text{ and } -1 \leq \sin \alpha \leq 1$$

From these, you can easily derive the ranges of $\csc$ and $\sec$ by applying reciprocals and basic algebraic manipulations.

For **tangent**, we know that it can be expressed as a ratio of **sine** and **cosine**. Hence, its value won't exist for any angle whose cosine is zero. This is because for those angles, we will be dividing by zero which is impossible and tangent will increase or decrease without bounds towards infinity (∞).

Therefore, for tangent, the domain consists of all angles α , such that $\cos \alpha \neq 0$ while on the converse, the range is $-\infty < \tan \alpha < +\infty$.

Cofunctions

Definition 1.4.2 Cofunctions are pairs of trigonometric functions that have equal values when their arguments are complementary angles. Cofunction identities are trigonometric identities that show a relationship between trigonometric functions and complementary angles (angles that add up to 90°).

Consider **Figure 1.23**; Since the angle sum of a triangle is 180°, $\theta + \alpha =$

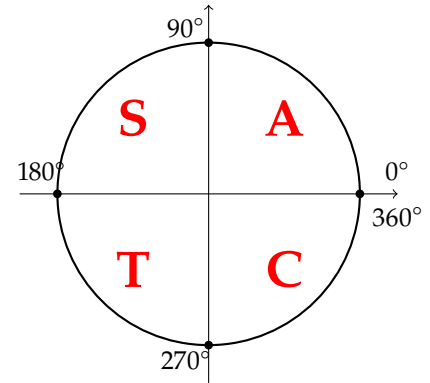


Figure 1.21: Signs of Trigonometric Functions Across the Quadrants

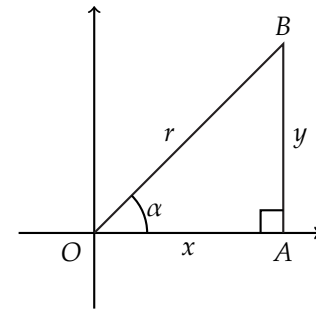


Figure 1.22: The Range of Trigonometric Functions

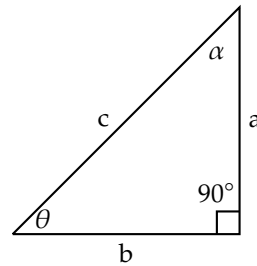


Figure 1.23: Confunctions and Complementary Angles

90° . That is, $\theta = 90^\circ - \alpha$ and $\alpha = 90^\circ - \theta$. From Figure 1.23;

$$\sin \theta = \frac{a}{c} \text{ and } \cos \alpha = \frac{a}{b}$$

. From here we will notice that $\sin \theta = \cos \alpha$. Thus

$$\text{Substituting } \alpha = 90^\circ - \theta \text{ and } \theta = 90^\circ - \alpha \begin{cases} \sin \theta = \cos(90^\circ - \theta) \\ \cos \alpha = \sin(90^\circ - \alpha) \end{cases}$$

This is the first set of confunctions. and from this we can create 6 sets of confunctions in total.

(a) Sine and Cosine

$$\begin{aligned} \sin \theta &= \cos(90^\circ - \theta) \\ \cos \theta &= \sin(90^\circ - \theta) \end{aligned} \tag{1.9}$$

(b) Secant and Cosecant

$$\begin{aligned} \sec \theta &= \csc(90^\circ - \theta) \\ \csc \theta &= \sec(90^\circ - \theta) \end{aligned} \tag{1.10}$$

(b) Tangent and Cotangent

$$\begin{aligned} \tan \theta &= \cot(90^\circ - \theta) \\ \cot \theta &= \tan(90^\circ - \theta) \end{aligned} \tag{1.11}$$

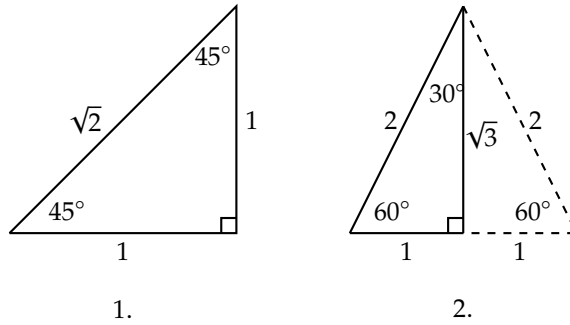
NOTE: The **CO** in the trig functions means that those various functions are confunctions. If you are handling angles in radian measure, you need to just replace 90° with $\frac{\pi}{2}$.

Trigonometric Functions and Quadrantal Angles

In the early sections of this document, we looked at what quadrantal angles are (**Figure 1.4**). You may use a unit circle to determine these values. The outcome is presented in a table below. If you consider the unit circle in standard position, you will notice that the values of the quadrantal angles will only contain the radius since the values of x or y depending on the angle you are dealing with at that particular time.

Table 1.2: Values of Trigonometric Functions for Quadrantal Angles

θ	$\sin \theta$	$\cos \theta$	$\tan \theta$	$\sec \theta$	$\csc \theta$	$\cot \theta$
0°	0	1	0	1	Undefined	Undefined
90°	1	0	undefined	Undefined	1	1
180°	0	-1	0	-1	Undefined	Undefined
270°	-1	0	Undefined	Undefined	-1	1
360°	0	1	0	1	Undefined	Undefined

**Figure 1.24:** Special Angles

Values of Trig Functions for Special Angles

Special Angles consists of 30° , 45° and 60° . These are widely and commonly used in Mathematics. Their ratios are neat and in their worst form, include at most one root. The process of finding the values of these special angles is done from two different types of triangles.

1. **Isosceles Right-Angles Triangle** of adjacent sides 1 unit each and the other 2 angles are 45° each. Here, you will also use the Pythagoras theorem to find the length of the hypotenuse.
2. **Equilateral Triangle** of sides 2 units long. Keeping in mind that the angle sum of a triangle is 180° , each angle of this triangle type measures 60° . Then, a bisector of one angle bisects the line opposite to it. Using the Pythagoras theorem, you will find the length of the new line and keep it rolling.

Consider **Figure 1.24**:

- (a) Values of 45° or $\frac{\pi}{4}$ rad

Figure 1.24 (a) is a right-angled isosceles triangle. In this case we will notice the following.

$$\begin{aligned}\sin 45^\circ &= \frac{1}{\sqrt{2}} \\ \cos 45^\circ &= \frac{1}{\sqrt{2}} \\ \tan 45^\circ &= 1\end{aligned}\tag{1.12}$$

- (b) Values of 30° and 60° or $\frac{\pi}{6}$ and $\frac{\pi}{3}$

Figure 1.24(b) is an equilateral triangle that has been bisected. From this after knowing that 30° and 60° are complementary angles and

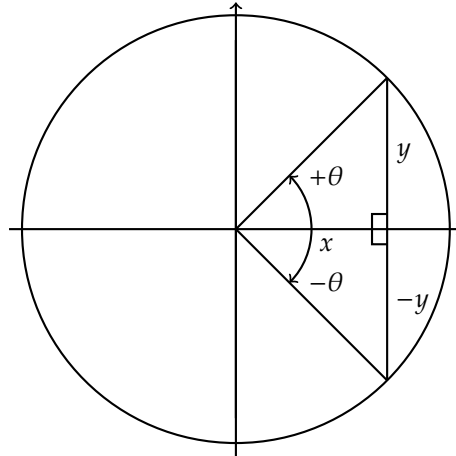


Figure 1.25: Negative angles

Sine and Cosine being confunctors, we can see that:

$$\begin{aligned}\sin 30^\circ &= \cos 60^\circ = \frac{1}{2} \\ \sin 60^\circ &= \cos 30^\circ = \frac{\sqrt{3}}{2} \\ \tan 30^\circ &= \frac{1}{\sqrt{3}}, \tan 60^\circ = \sqrt{3}\end{aligned}\tag{1.13}$$

From these, the reciprocal of each ratio, gives the values of secant, cosecant and cotangent for the special angles.

Triks and General Knowledge

Deriving these values of special angles from scratch is fun and smart but in an exam, you might not have all the time to do that, therefore, you might be forced to memorise them which is also not reliable. So, why don't you use a calculator instead?

1. Three-line-display calculators like **CASIO** (not every version of Casio) has an advanced programming that gives you all the values of special angles in fraction and root forms.
2. The other basic calculators provide these values in long decimals which are not easy to convert to fraction and root form.
 - a) Start by finding the ratio in your calculator. For example, $\sin 60^\circ = 0.86602540378443864676372317075294$
 - b) Without cancelling click **ANS** button then **X²** to square the result. This removes the conceptual roots entirely and gives you a fraction. In this case, $\frac{3}{4}$
 - c) Assign your own roots for the new fraction for both the numerator and denominator. In this example, $\frac{\sqrt{3}}{\sqrt{4}} = \frac{\sqrt{3}}{2}$

Negative Angles

Negative Angles are angles that are measured in a clockwise direction. Consider **Figure 1.25** of a unit circle.

From the figure, we see that:

$$\begin{aligned}\cos(-\theta) &= x = \cos \theta, \\ \sin(-\theta) &= -y = -\sin \theta \quad \text{since } \sin \theta = y, \\ \tan(-\theta) &= \frac{-y}{x} = -\tan \theta \quad \text{since } \tan \theta = \frac{y}{x}.\end{aligned}$$

In general,

$$\begin{aligned}\cos(-\theta) &= \cos \theta, \\ \sin(-\theta) &= -\sin \theta, \\ \tan(-\theta) &= -\tan \theta.\end{aligned}\tag{1.14}$$

From the results in 1.14, we can conclude that a cosine function is an **even** function, whereas sine and tangent functions are **odd** functions.

Solving Trigonometric Equations

In secondary school mathematics, we learnt how to find an angle that corresponds to a certain ratio using the inverse of a given trigonometric function. For example, given that $\cos \theta = \frac{1}{\sqrt{2}}$, we found that $\theta = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = 45^\circ$. This process normally involves a calculator and the angles are only found from the first quadrant.

But, trigonometric functions are periodic functions, that is, their values repeat themselves and for sine and cosine, the values resonate between -1 and 1 as the angles increase from 0° to 360° ¹⁰. Because of this fact, there exists angles in other quadrants that have the same ratios with the angles we find in the first quadrant¹¹. For example, $\sin 30^\circ = \sin 150^\circ = \frac{1}{2}$ despite the fact that 30° and 150° are from two different quadrants. Finding these angles from other quadrants that relates to angles in the first quadrant is the new challenge to deal with and we will be using the **Theorem 1.4.1**

10: More about this on graphs of trigonometric functions and periodicity

11: Some documents provide this concept separately under **Reference Angles**

Theorem 1.4.1 Given an angle $\theta^{(I)}$ from quadrant I, below is how you find all the other angles in each and every quadrant.

$$\begin{aligned}\theta^{(II)} &= 180^\circ - \theta^{(I)} \quad \text{or} \quad \theta^{(II)} = \pi - \theta^{(I)} \\ \theta^{(III)} &= 180^\circ + \theta^{(I)} \quad \text{or} \quad \theta^{(III)} = \pi + \theta^{(I)} \\ \theta^{(IV)} &= 360^\circ - \theta^{(I)} \quad \text{or} \quad \theta^{(IV)} = 2\pi - \theta^{(I)}\end{aligned}$$

We apply everything in the above segment of this section to solve the trig equations and we follow the following simple points.

Solving Trig Equations

This process is for solving single trig equations like $\sin \theta = k$, $\cos \theta = k$ and $\tan \theta = k$ where k is a constant.

Step 1. find the trig inverse of k . This will give you an angle in the first quadrant. If k is negative, remove the negative first and find

the angle in first quadrant (this angle is not part of the final solution).

Step 2. check the quadrants that correspond to the trig function given and the sign of k .

Step 3. using the quadrantal equations on **Theorem 1.4.1**, Find the other angles and present your solutions with all angles in a given range.

NOTE: If the equations are compound and not single, simplify them using any suitable algebraic manipulations into their simplest single equations.

1.4.2 Worked Examples

Example 1.4.1 Solve $\sin \theta = \frac{1}{3}$ for $0^\circ \leq \theta \leq 360^\circ$

Solution:

Firstly, we will take the inverse sine of $\frac{1}{3}$:

$$\theta_1 = \sin^{-1}\left(\frac{1}{3}\right) = 19.47^\circ$$

Since sine is positive in the first and second quadrants and using $\theta^{(II)} = 180^\circ - \theta^{(I)}$ from **Theorem 1.4.1**:

$$\theta_2 = 180^\circ - 19.47^\circ = 160.53^\circ$$

Therefore, the solutions are:

$$\theta = 19.47^\circ, 160.53^\circ$$

Example 1.4.2 Solve $\cos \theta = -0.35$ for $0^\circ \leq \theta \leq 360^\circ$

Solution:

Remove the negative and take the inverse cosine for 0.35:

$$\theta_1 = \cos^{-1}(0.35) \approx 69.51^\circ$$

It's very important to note that this is not part of the solution but a reference to help us find the solutions. Since the ratio -0.35 is negative and cosine is negative in the second and third quadrants, we will use **Theorem 1.4.1** to find angles that corresponds to 69° in second and third quadrants:

$$\theta_2 = 180^\circ - \theta_1 = 180^\circ - 69.51^\circ = 110.49^\circ$$

$$\theta_3 = 180^\circ + \theta_1 = 180^\circ + 69.51^\circ = 249.51^\circ$$

Therefore, the solutions are:

$$\theta = 110.49^\circ, 249.51^\circ$$

12

12: You can check your solution by taking the cosine of these angles, they should all be -0.35

Example 1.4.3 Solve $\tan \theta = 1.15$ for $0^\circ \leq \theta \leq 360^\circ$

Solution:

Taking the inverse tangent:

$$\theta_1 = \tan^{-1}(1.15) \approx 48.99^\circ$$

Since tangent is positive in the first and third quadrants using the formula for the third quadrant:

$$\theta_2 = 180^\circ + 48.99^\circ = 228.99^\circ$$

Therefore, the solutions are:

$$\theta = 48.99^\circ, 228.99^\circ$$

Example 1.4.4 Solve $5 \sin^2 \theta = 2 \sin \theta$ for $0^\circ \leq \theta \leq 360^\circ$

Solution:

This is not a single equation. As such, we will simplify it. Firstly, Rewrite the equation:

$$5 \sin^2 \theta - 2 \sin \theta = 0$$

¹³ This is a quadratic equation since the highest power is 2. Therefore, Factorize

$$\sin \theta (5 \sin \theta - 2) = 0$$

Thus:

$$\sin \theta = 0 \quad \text{or} \quad \sin \theta = \frac{2}{5} = 0.4$$

At this point, we have two sine equations. This means that the final solution will include angles that satisfy both of them.

If $\sin \theta = 0$, we can use the values of quadrantal angles from Table 1.2:

$$\theta = 0^\circ, 180^\circ$$

On the other hand, if $\sin \theta = 0.4$:

$$\theta_1 = \sin^{-1}(0.4) \approx 23.58^\circ$$

13: Please never think of dividing by $\sin \theta$.

Since sine is positive in the first and second quadrants:

$$\theta_2 = 180^\circ - 23.58^\circ = 156.42^\circ$$

Therefore, the solutions are:

$$\theta = 0^\circ, 23.58^\circ, 156.42^\circ, 180^\circ$$

Example 1.4.5 Solve $\sec \theta = 2.07$ for $0^\circ \leq \theta \leq 360^\circ$

Solution:

Recall **Identity 1.2** $\sec \theta = \frac{1}{\cos \theta}$. Thus:

$$\cos \theta = \frac{1}{2.07} \approx 0.4831$$

Taking the inverse cosine of 0.4831:

$$\theta_1 = \cos^{-1}(0.4831) \approx 61.11^\circ$$

Since cosine is positive in the first and fourth quadrants:

$$\theta_2 = 360^\circ - 61.16^\circ = 298.84^\circ$$

Therefore, the solutions are:

$$\theta = 61.16^\circ, 298.84^\circ$$

Example 1.4.6 Solve $6 \sin^2 \theta + 7 \sin \theta + 2 = 0$ for $0^\circ \leq \theta \leq 360^\circ$

Solution:

Rewrite the quadratic equation:

$$6 \sin^2 \theta + 7 \sin \theta + 2 = 0$$

This equation can be factorised since factors of 12 that add up to 7 are 4 and 3 but lets try something different. Use the quadratic formula:

$$\sin \theta = \frac{-7 \pm \sqrt{7^2 - 4(6)(2)}}{2(6)}$$

Simplify:

$$\begin{aligned} \sin \theta &= \frac{-7 \pm \sqrt{49 - 48}}{12} = \frac{-7 \pm 1}{12} \\ \sin \theta &= -\frac{6}{12} = -\frac{1}{2}, \quad \sin \theta = -\frac{8}{12} = -\frac{2}{3} \end{aligned}$$

For $\sin \theta = -\frac{1}{2}$, firstly remove the negative sign:

$$\theta = \sin^{-1} \frac{1}{2} = 30^\circ$$

Since sine is negative in third and fourth quadrants,

$$\theta_3 = 180^\circ + 30^\circ = 210^\circ$$

$$\theta_4 = 360^\circ - 30^\circ = 330^\circ$$

For $\sin \theta = -\frac{2}{3}$:

$$\theta_1 = \sin^{-1} \left(\frac{2}{3} \right) \approx 41.81^\circ$$

Finding negative equivalents:

$$\theta_2 = 180^\circ + 41.81^\circ = 221.81^\circ$$

$$\theta_4 = 360^\circ - 41.81^\circ = 318.19^\circ$$

Therefore,

$$\theta = 210^\circ, 330^\circ, 221.81^\circ, 318.19^\circ$$

Example 1.4.7 Solve $2 \sec^2 x = 3 - 5 \sec x$ for $0 \leq x \leq 2\pi$.

Solution:

First, rewrite the equation as:

$$2 \sec^2 x + 5 \sec x - 3 = 0.$$

Let $y = \sec x$. The equation becomes a quadratic:

$$2y^2 + 5y - 3 = 0.$$

Factorize $2y^2 + 5y - 3$. Find two numbers that multiply to -6 and add up to 5 : these are 6 and -1 . Rewrite $5y$:

$$2y^2 + 6y - y - 3 = 0.$$

Group and factorize:

$$(2y^2 + 6y) - (y + 3) = 0 \implies 2y(y + 3) - 1(y + 3) = 0.$$

Factor out $(y + 3)$:

$$(2y - 1)(y + 3) = 0.$$

Solve for y :

$$2y - 1 = 0 \implies y = \frac{1}{2}, \quad y + 3 = 0 \implies y = -3.$$

Return to $\sec x$:

$$\sec x = \frac{1}{2} \quad (\text{invalid, as } \sec x \geq 1 \text{ or } \sec x \leq -1).$$

$$\sec x = -3 \implies \cos x = -\frac{1}{3}.$$

Find x using $\cos x = -\frac{1}{3}$. The cosine function is negative in the second and third quadrants. First, find the reference angle:

$$\theta_{\text{ref}} = \cos^{-1}\left(\frac{1}{3}\right).$$

Using a calculator:

$$\theta_{\text{ref}} \approx 1.2310 \text{ radians.}$$

Now find the solutions:

$$x = \pi - \theta_{\text{ref}} \quad (\text{second quadrant}) \implies x \approx \pi - 1.2310 \approx 1.9106.$$

$$x = \pi + \theta_{\text{ref}} \quad (\text{third quadrant}) \implies x \approx \pi + 1.2310 \approx 4.3726.$$

Final Answer:

$x \approx 1.9106 \text{ radians, } x \approx 4.3726 \text{ radians.}$

Example 1.4.8 Solve the following equations for θ in the range $0 \leq \theta \leq 2\pi$:

- (a) $2 \sin \theta = \tan \theta$
- (b) $2 \sin \theta \cos \theta + \sin \theta = 0$
- (c) $5 \cos \theta - 4 \sin^2 \theta = 2$
- (d) $4 \csc^2 \theta - 7 = 4 \cot \theta$
- (e) $5 \cos \theta - \sec \theta = 4$
- (f) $4 \sec^2 \theta - 3 \tan \theta = 5$
- (g) $3 \cot 2\theta + 2 \sin 2\theta = 0$
- (h) $\cos\left(2\theta - \frac{\pi}{4}\right) = 0$
- (i) $\tan(\theta - \pi/3) = 0$

Solutions

- (a) $2 \sin \theta = \tan \theta$

We know that $\tan \theta = \frac{\sin \theta}{\cos \theta}$. Substituting this gives:

$$2 \sin \theta = \frac{\sin \theta}{\cos \theta}.$$

Multiplying through by $\cos \theta$ (for $\cos \theta \neq 0$):

$$2 \sin \theta \cos \theta = \sin \theta.$$

Rearrange and factorise

$$\sin \theta (2 \cos \theta - 1) = 0.$$

Thus:

$$\sin \theta = 0 \quad \text{or} \quad 2 \cos \theta - 1 = 0.$$

For $\sin \theta = 0$:

$$\theta = 0, \pi, 2\pi.$$

For $\cos \theta = \frac{1}{2}$:

$$\theta = \frac{\pi}{3}, \frac{5\pi}{3}.$$

Final solution:

$$\theta = 0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}, 2\pi.$$

(b) $2 \sin \theta \cos \theta + \sin \theta = 0$

Factorize:

$$\sin \theta (2 \cos \theta + 1) = 0.$$

Thus:

$$\sin \theta = 0 \quad \text{or} \quad 2 \cos \theta + 1 = 0.$$

For $\sin \theta = 0$:

$$\theta = 0, \pi, 2\pi.$$

For $\cos \theta = -\frac{1}{2}$:

$$\theta = \pi - \frac{\pi}{3}, \pi + \frac{\pi}{3} = \frac{2\pi}{3}, \frac{4\pi}{3}.$$

Final solution:

$$\theta = 0, \pi, 2\pi, \frac{2\pi}{3}, \frac{4\pi}{3}.$$

(c) $5 \cos \theta - 4 \sin^2 \theta = 2$

Use $\sin^2 \theta = 1 - \cos^2 \theta$:

$$5 \cos \theta - 4(1 - \cos^2 \theta) = 2.$$

Simplify:

$$5 \cos \theta - 4 + 4 \cos^2 \theta = 2.$$

Rearrange:

$$4 \cos^2 \theta + 5 \cos \theta - 6 = 0.$$

Factorize:

$$(4 \cos \theta - 3)(\cos \theta + 2) = 0.$$

Thus:

$$\cos \theta = \frac{3}{4}, \quad \cos \theta = -2 \quad (\text{not valid}).$$

For $\cos \theta = \frac{3}{4}$:

$$\theta = \cos^{-1} \left(\frac{3}{4} \right), \quad 2\pi - \cos^{-1} \left(\frac{3}{4} \right).$$

Final solution:

$$\theta \approx 0.722, 5.561.$$

(d) $4 \csc^2 \theta - 7 = 4 \cot \theta$

Substitute $\csc^2 \theta = 1 + \cot^2 \theta$:

$$4(1 + \cot^2 \theta) - 7 = 4 \cot \theta.$$

Simplify:

$$4 \cot^2 \theta - 4 \cot \theta - 3 = 0.$$

Factorize:

$$(2 \cot \theta + 1)(2 \cot \theta - 3) = 0.$$

Thus:

$$\cot \theta = -\frac{1}{2}, \quad \cot \theta = \frac{3}{2}.$$

Solve for θ using the unit circle:

$$\theta = \tan^{-1}(-2), \quad \theta = \tan^{-1}\left(\frac{2}{3}\right).$$

(e) $5 \cos \theta - \sec \theta = 4$

Substitute $\sec \theta = \frac{1}{\cos \theta}$:

$$5 \cos \theta - \frac{1}{\cos \theta} = 4.$$

Multiply through by $\cos \theta$ (for $\cos \theta \neq 0$):

$$5 \cos^2 \theta - 1 = 4 \cos \theta.$$

Rearrange:

$$5 \cos^2 \theta - 4 \cos \theta - 1 = 0.$$

Factorize:

$$(5 \cos \theta + 1)(\cos \theta - 1) = 0.$$

Thus:

$$\cos \theta = -\frac{1}{5}, \quad \cos \theta = 1.$$

For $\cos \theta = -\frac{1}{5}$:

$$\theta = \cos^{-1}\left(-\frac{1}{5}\right), \quad 2\pi - \cos^{-1}\left(-\frac{1}{5}\right).$$

For $\cos \theta = 1$:

$$\theta = 0.$$

Final solution:

$$\theta = 0, \cos^{-1}\left(-\frac{1}{5}\right), 2\pi - \cos^{-1}\left(-\frac{1}{5}\right).$$

(f) $4 \sec^2 \theta - 3 \tan \theta = 5$

Substitute $\sec^2 \theta = 1 + \tan^2 \theta$:

$$4(1 + \tan^2 \theta) - 3 \tan \theta = 5.$$

Simplify:

$$4 \tan^2 \theta - 3 \tan \theta - 1 = 0.$$

Factorize:

$$(4 \tan \theta + 1)(\tan \theta - 1) = 0.$$

Thus:

$$\tan \theta = -\frac{1}{4}, \quad \tan \theta = 1.$$

For $\tan \theta = -\frac{1}{4}$:

$$\theta = \tan^{-1}\left(-\frac{1}{4}\right), \quad \pi + \tan^{-1}\left(-\frac{1}{4}\right).$$

For $\tan \theta = 1$:

$$\theta = \frac{\pi}{4}, \frac{5\pi}{4}.$$

Final solution:

$$\theta = \tan^{-1}\left(-\frac{1}{4}\right), \pi + \tan^{-1}\left(-\frac{1}{4}\right), \frac{\pi}{4}, \frac{5\pi}{4}.$$

(g) $3 \cot 2\theta + 2 \sin 2\theta = 0$

Factorize:

$$\cot 2\theta(3 + 2 \sin 2\theta) = 0.$$

Thus:

$$\cot 2\theta = 0 \quad \text{or} \quad 3 + 2 \sin 2\theta = 0.$$

For $\cot 2\theta = 0$:

$$2\theta = \frac{\pi}{2}, \frac{3\pi}{2}.$$

$$\theta = \frac{\pi}{4}, \frac{3\pi}{4}.$$

For $2 \sin 2\theta = -3$:

$$\sin 2\theta = -\frac{3}{2} \quad (\text{not possible since } |\sin x| \leq 1).$$

Final solution:

$$\theta = \frac{\pi}{4}, \frac{3\pi}{4}.$$

(h) $\cos\left(2\theta - \frac{\pi}{4}\right) = 0$

The cosine function is zero when:

$$2\theta - \frac{\pi}{4} = \frac{\pi}{2}, \frac{3\pi}{2}.$$

Solve for θ :

$$2\theta = \frac{\pi}{2} + \frac{\pi}{4}, \frac{3\pi}{2} + \frac{\pi}{4}.$$

$$2\theta = \frac{3\pi}{4}, \frac{7\pi}{4}.$$

$$\theta = \frac{3\pi}{8}, \frac{7\pi}{8}.$$

Final solution:

$$\theta = \frac{3\pi}{8}, \frac{7\pi}{8}.$$

(i) $\tan(\theta - \pi/3) = 0$

The tangent function is zero when:

$$\theta - \frac{\pi}{3} = n\pi, \quad n \in \mathbb{Z}.$$

Solve for θ :

$$\theta = \frac{\pi}{3} + n\pi.$$

In the range $0 \leq \theta \leq 2\pi$, the solutions are:

$$\theta = \frac{\pi}{3}, \frac{4\pi}{3}.$$

Final solution:

$$\theta = \frac{\pi}{3}, \frac{4\pi}{3}.$$

1.5 Compound Angle Formulae

1.5.1 Core Insights

Sum and Difference Formulae

In the study of trigonometry, it is very essential to note that $\sin(\alpha + \beta) \neq \sin \alpha + \sin \beta$. A special set of equations called **Additional Angle Formulae** is defined to ensure that these computations are handled correctly. On the same concept, $\sin(\alpha - \beta) \neq \sin \alpha - \sin \beta$

Consider Figure 1.26

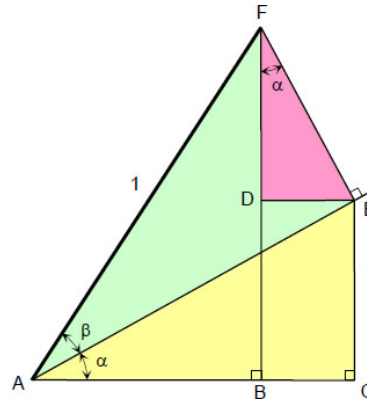


Figure 1.26: Deriving Sum and Difference Formulae of two Angles

Firstly, let's do some coordinate geometry to show why $\angle EAC = \angle DFE$ as shown on the figure. From Figure 1.26, line AC is parallel to line DE . As such;

$$\angle EAC = \angle AED \quad (\text{Alternate angles also known as Z-angles.})$$

Since $\angle AEF = 90^\circ$, then, $\angle DEF = 90^\circ - \alpha$. From here, we will see that $\angle EDF = 90^\circ$. Logically,

$$\angle DFE = \angle EAC = \alpha$$

Now that we have seen where this is coming from, we will proceed to the derivation process of the additional and subtraction angle formulae.

Let's say $\theta = (\alpha + \beta)$, using the definitions of trig functions on Section 1.3.1,

$$\begin{aligned}\sin(\alpha + \beta) &= \frac{BF}{AF} \quad (\text{but } BF = FD + DB) \\ \sin(\alpha + \beta) &= \frac{FD + DB}{AF} \quad \text{And } DB = EC \text{ (opposite sides of a rectangle.)} \\ \sin(\alpha + \beta) &= \frac{FD + EC}{AF} = \frac{FD}{AF} + \frac{EC}{AF}\end{aligned}$$

At this point we have to link the auxiliary sides on both fractions (numerators) to their hypotenuses. FD is an adjacent to angle α and the hypotenuse is FE . On the other hand, EC is the opposite side of angle α and its hypotenuse is AE . Then we will introduce and distribute these hypotenuses as 1 such that the original statement remains unchanged.

$$\begin{aligned}\sin(\alpha + \beta) &= \frac{FD}{AF} \cdot \frac{FE}{FE} + \frac{EC}{AF} \cdot \frac{AE}{AE} \\ \sin(\alpha + \beta) &= \frac{FD}{FE} \cdot \frac{FE}{AF} + \frac{EC}{AE} \cdot \frac{AE}{AF}\end{aligned}$$

From here, we can see that $\frac{FD}{FE} = \cos \alpha$, $\frac{FE}{AF} = \sin \beta$, $\frac{EC}{AE} = \sin \alpha$ and $\frac{AE}{AF} = \cos \beta$. Therefore the additional angle formula for Sine is,

$$\boxed{\sin(\alpha + \beta) = \cos \alpha \sin \beta + \sin \alpha \cos \beta}$$

For Cosine,

$$\begin{aligned}\cos(\alpha + \beta) &= \frac{AB}{AF} \quad \text{but } AB = AC - BC \text{ and } BC = DE \\ \cos(\alpha + \beta) &= \frac{AC - BC}{AF} = \frac{AC - DE}{AF} = \frac{AC}{AF} - \frac{DE}{AF}\end{aligned}$$

Introduce and distribute hypotenuses.

$$\begin{aligned}\cos(\alpha + \beta) &= \frac{AC}{AF} \cdot \frac{AE}{AE} - \frac{DE}{AF} \cdot \frac{FE}{FE} \\ \cos(\alpha + \beta) &= \frac{AC}{AE} \cdot \frac{AE}{AF} - \frac{DE}{FE} \cdot \frac{FE}{AF}\end{aligned}$$

And from here, we can see that $\frac{AC}{AE} = \cos \alpha$, $\frac{AE}{AF} = \cos \beta$, $\frac{DE}{FE} = \sin \alpha$ and $\frac{FE}{AF} = \sin \beta$. Therefore the additional angle formula for Cosine is,

$$\boxed{\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta}$$

For tangent, we will use these two formulae to derive the additional angle formula for tangent. From **identity 1.1**, $\tan x = \frac{\sin x}{\cos x}$. That means,

$$\tan(\alpha + \beta) = \frac{\sin(\alpha + \beta)}{\cos(\alpha + \beta)} = \frac{\cos \alpha \sin \beta + \sin \alpha \cos \beta}{\cos \alpha \cos \beta - \sin \alpha \sin \beta}$$

Divide by $\cos \alpha \cos \beta$ everywhere:

$$\begin{aligned}\tan(\alpha + \beta) &= \frac{\frac{\cancel{\cos \alpha} \sin \beta}{\cancel{\cos \alpha} \cos \beta} + \frac{\sin \alpha \cancel{\cos \beta}}{\cos \alpha \cancel{\cos \beta}}}{\frac{\cancel{\cos \alpha} \cancel{\cos \beta}}{\cancel{\cos \alpha} \cancel{\cos \beta}} - \frac{\sin \alpha \sin \beta}{\cos \alpha \cos \beta}} \\ \tan(\alpha + \beta) &= \frac{\frac{\sin \alpha}{\cos \alpha} + \frac{\sin \beta}{\cos \beta}}{1 - \frac{\sin \alpha}{\cos \alpha} \frac{\sin \beta}{\cos \beta}}\end{aligned}$$

Therefore, the additional angle formula for tangent is

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

Lastly; the additional angle formulae are:

$$\begin{aligned}\sin(\alpha + \beta) &= \sin \alpha \cos \beta + \cos \alpha \sin \beta \\ \cos(\alpha + \beta) &= \cos \alpha \cos \beta - \sin \alpha \sin \beta \\ \tan(\alpha + \beta) &= \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}\end{aligned} \tag{1.15}$$

Assuming angle β is negative, we get the difference angle formulae as follows. **For sine:**

$$\sin(\alpha + (-\beta)) = \sin \alpha \cos(-\beta) + \cos \alpha \sin(-\beta)$$

From Section 1.4 Subsection 1.4.1 (Negative Angles), we noticed that sine function is odd and cosine is even. Thus, $\sin(-\beta) = -\sin \beta$ and $\cos(-\beta) = \cos \beta$. If we substitute, we get the following:

$$\begin{aligned}\sin(\alpha - \beta) &= \sin \alpha \cos \beta + \cos \alpha (-\sin \beta) \\ \sin(\alpha - \beta) &= \sin \alpha \cos \beta - \cos \alpha \sin \beta\end{aligned}$$

If you follow the same process for cosine and tangent, we get the following difference formulae

$$\begin{aligned}\sin(\alpha - \beta) &= \sin \alpha \cos \beta - \cos \alpha \sin \beta \\ \cos(\alpha - \beta) &= \cos \alpha \cos \beta + \sin \alpha \sin \beta \\ \tan(\alpha - \beta) &= \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}\end{aligned} \tag{1.16}$$

Double and Half Angle Formulae

From the additional angle formulae, if the two angles are equal, the formulae are transformed to double angle formulae.

Consider $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$. If $\alpha = \beta$,

$$\sin(\alpha + \alpha) = \sin \alpha \cos \alpha + \cos \alpha \sin \alpha$$

$$\boxed{\sin 2\alpha = 2 \sin \alpha \cos \alpha}$$

For Cosine, $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$. Thus,

$$\cos(\alpha + \alpha) = \cos \alpha \cos \alpha - \sin \alpha \sin \alpha$$

$$\boxed{\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha}$$

From **identity 1.6**, $\cos^2 \alpha + \sin^2 \alpha = 1$. If we substitute $\cos^2 \alpha = 1 - \sin^2 \alpha$,

$$\cos 2\alpha = 1 - \sin^2 \alpha - \sin^2 \alpha$$

Thus,

$$\boxed{\cos 2\alpha = 1 - 2 \sin^2 \alpha}$$

And if you substitute $\sin^2 \alpha = 1 - \cos^2 \alpha$, into $\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$,

$$\cos 2\alpha = \cos^2 \alpha - (1 - \cos^2 \alpha)$$

$$\cos 2\alpha = \cos^2 \alpha - 1 + \cos^2 \alpha$$

And, therefore, another double angle formula for Cosine is

$$\boxed{\cos 2\alpha = 2 \cos^2 \alpha - 1}$$

And lastly, for Tangent, $\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$, that means

$$\tan(\alpha + \alpha) = \frac{\tan \alpha + \tan \alpha}{1 - \tan \alpha \tan \alpha}$$

$$\boxed{\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}}$$

That means the Double angle formulae are

$$\begin{aligned} \sin 2\alpha &= 2 \sin \alpha \cos \alpha \\ \cos 2\alpha &= \cos^2 \alpha - \sin^2 \alpha \\ \cos 2\alpha &= 1 - 2 \sin^2 \alpha \\ \cos 2\alpha &= 2 \cos^2 \alpha - 1 \\ \tan 2\alpha &= \frac{2 \tan \alpha}{1 - \tan^2 \alpha} \end{aligned} \quad (1.17)$$

14:

14

From the Cosine double angle formulae $\cos 2\alpha = 2 \cos^2 \alpha - 1$, we can make $\cos^2 \alpha$ subject of the formula.

$$\cos^2 \alpha = \frac{\cos 2\alpha + 1}{2}$$

And if we divide the angle by 2, only the angle will be affected hence the

Double Angle For Cosine

Cosine has three different double angle formulae which can be used to ease the work of solving a question depending on the nature of questions asked. But at the same time, each and every one of them produce the same answer if used simultaneously.

identity will remain true since the domain isn't changing. Hence,

$$\cos^2 \frac{\alpha}{2} = \frac{\cos \frac{\alpha}{2} - 1}{2} = \frac{\cos \alpha - 1}{2}$$

And if we take square roots both sides, the half angle formula for cosine will be

$$\cos \frac{\alpha}{2} = \sqrt{\frac{\cos \alpha - 1}{2}}$$

And from $\cos 2\alpha = 1 - 2\sin^2 \alpha$, if we make $\sin^2 \alpha$ subject, we get,

$$\sin^2 \alpha = \frac{1 - \cos 2\alpha}{2}$$

And following the other steps,

$$\sin^2 \frac{\alpha}{2} = \frac{1 - \cos \frac{\alpha}{2}}{2} = \frac{1 - \cos \alpha}{2}$$

$$\sin \frac{\alpha}{2} = \sqrt{\frac{1 - \cos \alpha}{2}}$$

For tangent, we will use the fact that $\tan^2 \frac{\alpha}{2} = \frac{\sin^2 \frac{\alpha}{2}}{\cos^2 \frac{\alpha}{2}}$. If we substitute,

$$\tan^2 \frac{\alpha}{2} = \frac{\frac{1 - \cos \alpha}{2}}{\frac{\cos \alpha + 1}{2}}$$

After simplifying this algebraically,

$$\tan^2 \frac{\alpha}{2} = \frac{\cos \alpha + 1}{1 - \cos \alpha}$$

Therefore,

$$\tan \frac{\alpha}{2} = \sqrt{\frac{1 + \cos \alpha}{1 - \cos \alpha}}$$

Lastly, the half angle formulae are

$$\begin{aligned} \cos \frac{\alpha}{2} &= \sqrt{\frac{\cos \alpha - 1}{2}} \\ \sin \frac{\alpha}{2} &= \sqrt{\frac{1 - \cos \alpha}{2}} \\ \tan \frac{\alpha}{2} &= \sqrt{\frac{1 + \cos \alpha}{1 - \cos \alpha}} \end{aligned} \quad (1.18)$$

Product and Factor Formulae

So far we have looked at the sum and difference equations: $\sin(\alpha \pm \beta)$ and $\cos(\alpha \pm \beta)$ where we looked at the fact that $\sin(\alpha + \beta) \neq \sin \alpha + \sin \beta$ and was true for all the sum and difference formulae. So, what if we have $\sin \alpha \pm \sin \beta$ or $\cos \alpha \pm \cos \beta$? In this section we will go ahead and derive the formulae for expressions of this nature.

Recall from Equation 1.15, additional angle formulae, and Equation 1.16, Subtraction angle formulae;

$$\begin{aligned}\sin(\alpha + \beta) &= \sin \alpha \cos \beta + \cos \alpha \sin \beta \quad \text{and} \\ \sin(\alpha - \beta) &= \sin \alpha \cos \beta - \cos \alpha \sin \beta\end{aligned}$$

If we add these two equations, we get

$$\begin{aligned}\sin(\alpha + \beta) + \sin(\alpha - \beta) &= \sin \alpha \cos \beta + \sin \alpha \cos \beta + \cancel{\cos \alpha \sin \beta} - \cancel{\cos \alpha \sin \beta} \\ \sin(\alpha + \beta) + \sin(\alpha - \beta) &= 2 \sin \alpha \cos \beta\end{aligned}$$

On the converse, subtracting them yields the following.

$$\sin(\alpha + \beta) - \sin(\alpha - \beta) = 2 \cos \alpha \sin \beta$$

For Cosine, we know that $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$ and $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$. If we add these two equations together,

$$\cos(\alpha + \beta) + \cos(\alpha - \beta) = 2 \cos \alpha \cos \beta$$

And if we subtract them,

$$\cos(\alpha + \beta) - \cos(\alpha - \beta) = -2 \sin \alpha \sin \alpha \quad \text{or} \quad \cos(\alpha - \beta) - \cos(\alpha + \beta) = 2 \sin \alpha \sin \beta$$

Therefore, the product formulae for Sine and Cosine are

$\begin{aligned}\sin \alpha \cos \beta &= \frac{\sin(\alpha + \beta) + \sin(\alpha - \beta)}{2} \\ \cos \alpha \sin \alpha &= \frac{\sin(\alpha + \beta) - \sin(\alpha - \beta)}{2} \\ \cos \alpha \cos \beta &= \frac{\cos(\alpha + \beta) + \cos(\alpha - \beta)}{2} \\ \sin \alpha \sin \beta &= \frac{\cos(\alpha - \beta) - \sin(\alpha + \beta)}{2}\end{aligned}$	(1.19)
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From Equation 1.19, let $\alpha + \beta$ be u and $\alpha - \beta$ be v . That means we will have simultaneous equations as follows.

$$\alpha + \beta = u \quad (i) \quad \text{and} \quad \alpha - \beta = v \quad (ii)$$

Solving these two equations using any suitable method yields

$$\alpha = \frac{u + v}{2} \quad \text{and} \quad \beta = \frac{u - v}{2}$$

. From here we will substitute these into the product formulae and do a little manipulation to obtain the factor formulae.

Thus, the factor formulae for sin and cosine are

$$\begin{aligned}
 \sin u + \sin v &= 2 \sin \left(\frac{u+v}{2} \right) \cos \left(\frac{u-v}{2} \right) \\
 \sin u - \sin v &= 2 \cos \left(\frac{u+v}{2} \right) \sin \left(\frac{u-v}{2} \right) \\
 \cos u + \cos v &= 2 \cos \left(\frac{u+v}{2} \right) \cos \left(\frac{u-v}{2} \right) \\
 \cos u - \cos v &= -2 \sin \left(\frac{u+v}{2} \right) \sin \left(\frac{u-v}{2} \right)
 \end{aligned} \tag{1.20}$$

Points to Note

1. It is advisable to not memorise any of the identities or compound angle formulae. Rather, you should be able to walk through the derivation process alone with ease.
2. These compound angle formulae are used for, proving identities involving them, evaluating values of trigonometric functions of compound angles and solving equations.

1.5.2 Worked Examples

Example 1.5.1 Evaluate the following leaving your answer in surd form

- (a) $\csc 105^\circ$
- (b) $\sin \left(\frac{2\pi}{3} \right)$
- (c) $\sin \left(\frac{\pi}{12} \right)$

Solution

- (a) $\csc 105^\circ$

First of all, we will write the given angle as a sum or difference of any two special angles. That is, $105^\circ = 60^\circ + 45^\circ$. Since $\csc \alpha = \frac{1}{\sin \alpha}$, we first find the value of $\sin 105^\circ$. Thus:

$$\sin 105^\circ = \sin (60^\circ + 45^\circ)$$

. Using the additional angle formula for Sine, $\sin(A + B) = \sin A \cos B + \cos A \sin B$ We will expand it as follows:

$$\sin 105^\circ = \sin 60^\circ \cos 45^\circ + \cos 60^\circ \sin 45^\circ$$

Substitute the trigonometric values of special angles from Section 1.4.1:

$$\sin 60^\circ = \frac{\sqrt{3}}{2}, \quad \cos 60^\circ = \frac{1}{2}, \quad \sin 45^\circ = \frac{1}{\sqrt{2}}, \quad \cos 45^\circ = \frac{1}{\sqrt{2}}$$

$$\sin 105^\circ = \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} + \frac{1}{2} \cdot \frac{1}{\sqrt{2}}$$

Simplify:

$$\sin 105^\circ = \frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} = \frac{\sqrt{3}+1}{2\sqrt{2}}$$

Lastly, we will re-apply the definition of cosecant, $\csc 105^\circ = \frac{1}{\sin 105^\circ}$.
Therefore,

$$\csc 105^\circ = \frac{1}{\frac{\sqrt{3}+1}{2\sqrt{2}}} = \frac{2\sqrt{2}}{\sqrt{3}+1}$$

(b) $\sin\left(\frac{2\pi}{3}\right)$
 $\frac{2\pi}{3} = \frac{\pi}{3} + \frac{\pi}{3}$, and hence:

$$\sin\left(\frac{2\pi}{3}\right) = \sin\left(\frac{\pi}{3} + \frac{\pi}{3}\right)$$

Expand using $\sin(A+B) = \sin A \cos B + \cos A \sin B$ Therefore:

$$\sin\left(\frac{2\pi}{3}\right) = \sin\left(\frac{\pi}{3}\right) \cos\left(\frac{\pi}{3}\right) + \cos\left(\frac{\pi}{3}\right) \sin\left(\frac{\pi}{3}\right)$$

Substitute the values:

$$\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}, \quad \cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$$

$$\sin\left(\frac{2\pi}{3}\right) = \frac{\sqrt{3}}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{\sqrt{3}}{2}$$

Simplify:

$$\sin\left(\frac{2\pi}{3}\right) = \frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} = \frac{\sqrt{3}}{2}$$

(c) $\sin\left(\frac{\pi}{12}\right)$
 $\frac{\pi}{12} = \frac{\pi}{3} + \frac{\pi}{6}$. Therefore:

$$\sin\left(\frac{\pi}{12}\right) = \sin\left(\frac{\pi}{3} + \frac{\pi}{6}\right)$$

Expand using $\sin(A+B) = \sin A \cos B + \cos A \sin B$;

$$\sin\left(\frac{\pi}{12}\right) = \sin\left(\frac{\pi}{3}\right) \cos\left(\frac{\pi}{6}\right) + \cos\left(\frac{\pi}{3}\right) \sin\left(\frac{\pi}{6}\right)$$

Substitute the values:

$$\sin\left(\frac{\pi}{12}\right) = \frac{\sqrt{3}}{2}, \quad \cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}, \quad \cos\left(\frac{\pi}{3}\right) = \frac{1}{2}, \quad \sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$$

$$\sin\left(\frac{\pi}{12}\right) = \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{1}{2} \cdot \frac{1}{2}$$

After simplifying we get

$$\sin\left(\frac{\pi}{12}\right) = \frac{3}{4} + \frac{1}{4} = 1$$

Example 1.5.2 Evaluate the exact values of the following compound angles

- (a) $\cos 75^\circ$
 (b) $\cos \left(\frac{\pi}{12} \right)$

Solution

- (a) $\cos 75^\circ$

First of all, we have to break 75° into either a sum or difference of two special angles. Hence,

$$75^\circ = 30^\circ + 45^\circ$$

. This will be expanded by the additional angle formulae for cosine, $\cos(A + B) = \cos A \cos B - \sin A \sin B$ (Equation 1.15), as follows:

$$\cos 75^\circ = \cos 30^\circ \cos 45^\circ - \sin 30^\circ \sin 45^\circ$$

Using the values of special angles (Section 1.4.1), we will get the following

$$\cos 75^\circ = \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} - \frac{1}{2} \cdot \frac{1}{\sqrt{2}}$$

After simplifying;

$$\cos 75^\circ = \frac{\sqrt{3}}{2\sqrt{2}} - \frac{1}{2\sqrt{2}} = \frac{\sqrt{3} - 1}{2\sqrt{2}}$$

- (b) $\cos \left(\frac{\pi}{12} \right) = \cos \left(\frac{\pi}{3} - \frac{\pi}{4} \right)$

In this case,

$$\cos \left(\frac{\pi}{12} \right) = \cos \left(\frac{\pi}{4} \right) \cos \left(\frac{\pi}{3} \right) + \sin \left(\frac{\pi}{4} \right) \sin \left(\frac{\pi}{3} \right)$$

Substituting the values

$$\cos \left(\frac{\pi}{12} \right) = \frac{1}{2} \cdot \frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}}$$

$$\cos \left(\frac{\pi}{12} \right) = \frac{1}{2\sqrt{2}} + \frac{\sqrt{3}}{2\sqrt{2}} = \frac{1 + \sqrt{3}}{2\sqrt{2}}$$

15:

Value for $\sin 45^\circ$ and $\cos 45^\circ$

In Section 1.4.1, we have seen that $\sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}}$. But in some books, you will find that this is not the value used. Instead, $\frac{\sqrt{2}}{2}$ is used. These values are mathematically the same since the second value is found by rationalising the denominator of the first one. You can use them interchangeably and there is no significant problem.

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Example 1.5.3 Find the exact value of $\tan \left(\frac{7\pi}{12} \right)$.

Solution

$$\frac{7\pi}{12} = \frac{\pi}{3} + \frac{\pi}{4}.$$

Therefore,

$$\tan\left(\frac{7\pi}{12}\right) = \tan\left(\frac{\pi}{3} + \frac{\pi}{4}\right),$$

using the formula for tangent of a sum,

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B},$$

we get:

$$\tan\left(\frac{7\pi}{12}\right) = \frac{\tan\left(\frac{\pi}{3}\right) + \tan\left(\frac{\pi}{4}\right)}{1 - \tan\left(\frac{\pi}{3}\right)\tan\left(\frac{\pi}{4}\right)}.$$

Substituting the values:

$$\tan\left(\frac{\pi}{3}\right) = \sqrt{3}, \quad \tan\left(\frac{\pi}{4}\right) = 1,$$

we have:

$$\tan\left(\frac{7\pi}{12}\right) = \frac{\sqrt{3} + 1}{1 - \sqrt{3}}.$$

Example 1.5.4 Use the appropriate addition formulae to evaluate the following. Leave your answer in surd form.

- (a) $\sec 105^\circ$
- (b) $\tan\left(\frac{11\pi}{6}\right)$
- (c) $\cot\left(\frac{\pi}{12}\right)$
- (d) $\cos^2(75^\circ)$
- (e) $\sin\left(\frac{4\pi}{3}\right)$
- (f) $\csc\left(\frac{7\pi}{4}\right)$

Solution

$$(a) \sec 105^\circ = \frac{1}{\cos 105^\circ}$$

$$105^\circ = 60^\circ + 45^\circ.$$

Using the addition formula for cosine:

$$\cos(105^\circ) = \cos(60^\circ)\cos(45^\circ) - \sin(60^\circ)\sin(45^\circ).$$

Substituting values:

$$\cos(105^\circ) = \frac{1}{2} \cdot \frac{\sqrt{2}}{2} - \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{2} - \sqrt{6}}{4}.$$

Thus,

$$\sec(105^\circ) = \frac{1}{\cos(105^\circ)} = \frac{1}{\frac{\sqrt{2} - \sqrt{6}}{4}} = \frac{4}{\sqrt{2} - \sqrt{6}}.$$

You can stop here if you like. But, what if we go a little further and

rationalise the denominator here

$$= \frac{4}{\sqrt{2} - \sqrt{6}} \cdot \frac{\sqrt{2} + \sqrt{6}}{\sqrt{2} + \sqrt{6}} = \frac{4(\sqrt{2} + \sqrt{6})}{(\sqrt{2})^2 - (\sqrt{6})^2} = \frac{4(\sqrt{2} + \sqrt{6})}{2 - 6}$$

Therefore,

$$\sec 105^\circ = \frac{4(\sqrt{2} + \sqrt{6})}{-4} = -(\sqrt{2} + \sqrt{6})$$

(b) $\tan\left(\frac{11\pi}{6}\right)$

$$\frac{11\pi}{6} = \pi + \frac{5\pi}{6}$$

Using the additional angle formula for tangent

$$\tan\left(\frac{11\pi}{6}\right) = \frac{\tan \pi + \tan\left(\frac{5\pi}{6}\right)}{1 - \tan \pi \cdot \tan\left(\frac{5\pi}{6}\right)}$$

Since π is a quadrantal angle, we will use the values from **Table 1.2** where $\tan \pi = 0$. On the other hand, $\frac{5\pi}{6}$ is in the second quadrant where tangent is negative and its reference angle is $\frac{\pi}{6}$. Thus, $\tan\left(\frac{5\pi}{6}\right) = -\frac{\sqrt{3}}{3}$. Substituting these values gives

$$\tan\left(\frac{11\pi}{6}\right) = \frac{0 + \left(-\frac{\sqrt{3}}{3}\right)}{1 - (0)\left(-\frac{\sqrt{3}}{3}\right)} = \frac{-\frac{\sqrt{3}}{3}}{1} = -\frac{\sqrt{3}}{3}$$

(c) $\cot\left(\frac{\pi}{12}\right)$ We know (from Example 1.5.2) that $\frac{\pi}{12} = \frac{\pi}{4} - \frac{\pi}{6}$. Using the formula for $\tan(A - B)$:

$$\tan\left(\frac{\pi}{12}\right) = \frac{\tan\left(\frac{\pi}{4}\right) - \tan\left(\frac{\pi}{6}\right)}{1 + \tan\left(\frac{\pi}{4}\right)\tan\left(\frac{\pi}{6}\right)}$$

Substituting values:

$$\tan\left(\frac{\pi}{12}\right) = \frac{1 - \frac{1}{\sqrt{3}}}{1 + 1 \cdot \frac{1}{\sqrt{3}}} = \frac{\frac{\sqrt{3}-1}{\sqrt{3}}}{\frac{\sqrt{3}+1}{\sqrt{3}}} = \frac{\sqrt{3}-1}{\sqrt{3}+1}$$

Therefore:

$$\cot\left(\frac{\pi}{12}\right) = \frac{1}{\tan\left(\frac{\pi}{12}\right)} = \frac{\sqrt{3}+1}{\sqrt{3}-1}$$

(d) $\cos^2(75^\circ)$

$$75^\circ = 45^\circ + 30^\circ$$

Using the addition formula:

$$\cos(75^\circ) = \cos(45^\circ)\cos(30^\circ) - \sin(45^\circ)\sin(30^\circ).$$

Substituting values:

$$\cos(75^\circ) = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{6} - \sqrt{2}}{4}.$$

Hence:

$$\cos^2(75^\circ) = \left(\frac{\sqrt{6} - \sqrt{2}}{4} \right)^2 = \frac{(\sqrt{6} - \sqrt{2})^2}{16}.$$

(e) $\sin\left(\frac{4\pi}{3}\right)$

$$\sin\left(\frac{4\pi}{3}\right) = -\sin\left(\frac{\pi}{3}\right).$$

Using $\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$, we get:

$$\sin\left(\frac{4\pi}{3}\right) = -\frac{\sqrt{3}}{2}.$$

(f) $\csc\left(\frac{7\pi}{4}\right)$

$$\sin\left(\frac{7\pi}{4}\right) = -\sin\left(\frac{\pi}{4}\right) = -\frac{\sqrt{2}}{2}.$$

Hence:

$$\csc\left(\frac{7\pi}{4}\right) = \frac{1}{\sin\left(\frac{7\pi}{4}\right)} = -\sqrt{2}.$$

Example 1.5.5 Prove the identity $2 \cos^2(x) = 1 + \cos(2x)$.

Solution

Take RHS:

$$1 + \cos(2x)$$

Substitute the double angle formula for cosine. Since there are three different formulae, you choose the one with similar properties to the other side of your identity. In this case, we will substitute $\cos 2x = 2 \cos^2 x - 1$ since the other side has cosine only. Thus;

$$1 + (2 \cos^2 x - 1) = 2 \cos^2 x + 1 - 1 = 2 \cos^2 x \cong \text{LHS}$$

Example 1.5.6 Show that $\sin(A+B+C) = \sin A \cos B \cos C + \cos A \sin B \cos C + \cos A \cos B \sin C - \sin A \sin B \sin C$.

Solution

LHS:

Expand $\sin(A + B + C)$ as $\sin((A + B) + C)$:

$$\sin((A + B) + C) = \sin(A + B) \cos C + \cos(A + B) \sin C.$$

Using the addition formulas:

$$\sin(A + B) = \sin A \cos B + \cos A \sin B, \quad \cos(A + B) = \cos A \cos B - \sin A \sin B$$

substitute these into the equation:

$$\sin(A+B+C) = (\sin A \cos B + \cos A \sin B) \cos C + (\cos A \cos B - \sin A \sin B) \sin C.$$

Distribute terms:

$$\sin(A + B + C) = \sin A \cos B \cos C + \cos A \sin B \cos C + \cos A \cos B \sin C - \sin A \sin B \sin C$$

Example 1.5.7 Prove that $\sin(\pi - x) = \sin(x)$.

Solution

We will take LHS to prove RHS:

Using the subtraction angle formula for sine:

$$\sin(\pi - x) = \sin(\pi) \cos x - \cos \pi \sin x.$$

Recall, Table 1.2, $\sin \pi = 0$ and $\cos \pi = -1$. Substituting the values,

$$\sin(\pi - x) = 0 \cdot \cos x - (-1) \sin x = -(-1) \sin x = \sin x$$

Example 1.5.8 Given that $\sin A = \frac{3}{5}$ and $\cos B = \frac{12}{13}$, find the exact values for the following:

- (a) $\sin(A + B)$
- (b) $\cos \frac{B}{2}$
- (c) $\tan(2B)$
- (d) $\cos(A - B)$

Solution

Step 1: Define the triangles for A and B

For A: $\sin A = \frac{3}{5}$, which implies the opposite side is 3 and the hypotenuse is 5. Using the Pythagorean theorem:

$$\text{Adjacent side} = \sqrt{5^2 - 3^2} = \sqrt{16} = 4.$$

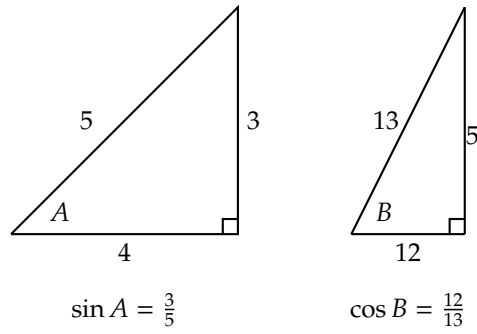


Figure 1.27: Illustration for Example 1.5.8.

Thus:

$$\cos A = \frac{4}{5}, \quad \tan A = \frac{3}{4}.$$

For B : $\cos B = \frac{12}{13}$, which implies the adjacent side is 12 and the hypotenuse is 13. Using the Pythagorean theorem:

$$\text{Opposite side} = \sqrt{13^2 - 12^2} = \sqrt{25} = 5.$$

Thus:

$$\sin B = \frac{5}{13}, \quad \tan B = \frac{5}{12}.$$

Step 2: Solve each part

(a) $\sin(A + B)$

The formula is:

$$\sin(A + B) = \sin A \cos B + \cos A \sin B.$$

Substituting the values:

$$\sin(A + B) = \left(\frac{3}{5}\right)\left(\frac{12}{13}\right) + \left(\frac{4}{5}\right)\left(\frac{5}{13}\right).$$

Simplify:

$$\sin(A + B) = \frac{36}{65} + \frac{20}{65} = \frac{56}{65}.$$

(b) $\cos \frac{B}{2}$

The formula is:

$$\cos \frac{B}{2} = \sqrt{\frac{1 + \cos B}{2}}.$$

Substituting $\cos B = \frac{12}{13}$:

$$\cos \frac{B}{2} = \sqrt{\frac{1 + \frac{12}{13}}{2}} = \sqrt{\frac{\frac{25}{13}}{2}} = \sqrt{\frac{25}{26}} = \frac{5}{\sqrt{26}} = \frac{5\sqrt{26}}{26}.$$

(c) $\tan(2B)$

The formula is:

$$\tan(2B) = \frac{2 \tan B}{1 - \tan^2 B}.$$

Substituting $\tan B = \frac{5}{12}$:

$$\tan^2 B = \left(\frac{5}{12}\right)^2 = \frac{25}{144}.$$

Then:

$$\tan(2B) = \frac{2 \cdot \frac{5}{12}}{1 - \frac{25}{144}} = \frac{\frac{10}{12}}{\frac{119}{144}}.$$

Simplify:

$$\tan(2B) = \frac{10}{12} \cdot \frac{144}{119} = \frac{120}{119}.$$

(d) $\cos(A - B)$

The formula is:

$$\cos(A - B) = \cos A \cos B + \sin A \sin B.$$

Substituting:

$$\cos(A - B) = \left(\frac{4}{5}\right)\left(\frac{12}{13}\right) + \left(\frac{3}{5}\right)\left(\frac{5}{13}\right).$$

Simplify:

$$\cos(A - B) = \frac{48}{65} + \frac{15}{65} = \frac{63}{65}.$$

Example 1.5.9 Prove that $2 \sin^2 x = \tan x \sin 2x$.

Solution

We simplify the RHS and show that it is equal to the LHS.

RHS = $\tan x \sin 2x$ expand $\sin 2x$ using the double-angle identity.

$$= \tan x (2 \sin x \cos x)$$

$$= 2 \frac{\sin x}{\cos x} \sin x \cos x$$

$$= 2 \sin^2 x = \text{LHS}.$$

Example 1.5.10 Show that $\tan \theta = \frac{\sin 2\theta}{\cos 2\theta + 1}$.

Solution

We simplify the RHS and show that it is equal to the LHS. Thus,

$$\text{RHS} = \frac{\sin 2\theta}{\cos 2\theta + 1} \quad \text{expand } \sin 2\theta \text{ and } \cos 2\theta.$$

$$\begin{aligned}
 &= \frac{2 \sin \theta \cos \theta}{2 \cos^2 \theta - 1 + 1} \\
 &= \frac{2 \sin \theta \cos \theta}{2 \cos^2 \theta} \\
 &= \frac{\sin \theta}{\cos \theta} \\
 &= \tan \theta = \text{LHS.}
 \end{aligned}$$

Example 1.5.11 Prove that $\sin 4A = 4 \sin A \cos A \cos 2A$

Solution

Using the identity $\sin 4A = \sin(2A + 2A)$, using the additional angle formulae for Sine

$$\begin{aligned}
 \sin 4A &= \sin 2A \cos 2A + \cos 2A \sin 2A \\
 &= 2 \sin A \cos A \cdot \cos 2A + 2 \sin A \cos A \cdot \cos 2A \\
 &= 4 \sin A \cos A \cos 2A
 \end{aligned}$$

Example 1.5.12 Prove that $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$

Solution

Using the identity $\cos 3\theta = \cos(2\theta + \theta)$,

$$\cos 3\theta = \cos 2\theta \cos \theta - \sin 2\theta \sin \theta$$

Expanding $\cos 2\theta$ and $\sin 2\theta$,

$$\begin{aligned}
 &= (2 \cos^2 \theta - 1) \cos \theta - (2 \sin \theta \cos \theta) \sin \theta \\
 &= 2 \cos^3 \theta - \cos \theta - 2 \sin^2 \theta \cos \theta
 \end{aligned}$$

Substituting $\sin^2 \theta = 1 - \cos^2 \theta$,

$$\begin{aligned}
 &= 2 \cos^3 \theta - \cos \theta - 2(1 - \cos^2 \theta) \cos \theta \\
 &= 2 \cos^3 \theta - \cos \theta - 2 \cos \theta + 2 \cos^3 \theta \\
 &= 4 \cos^3 \theta - 3 \cos \theta
 \end{aligned}$$

Example 1.5.13 Prove that $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$

Solution

Using the identity $\sin 3\theta = \sin(2\theta + \theta)$,

$$\sin 3\theta = \sin 2\theta \cos \theta + \cos 2\theta \sin \theta$$

Expanding $\sin 2\theta$ and $\cos 2\theta$,

$$\begin{aligned} &= (2 \sin \theta \cos \theta) \cos \theta + (2 \cos^2 \theta - 1) \sin \theta \\ &= 2 \sin \theta \cos^2 \theta + 2 \cos^2 \theta \sin \theta - \sin \theta \end{aligned}$$

Substituting $\cos^2 \theta = 1 - \sin^2 \theta$,

$$\begin{aligned} &= 2 \sin \theta (1 - \sin^2 \theta) + (1 - \sin^2 \theta) \sin \theta \\ &= 2 \sin \theta - 2 \sin^3 \theta + \sin \theta - \sin^3 \theta \\ &= 3 \sin \theta - 4 \sin^3 \theta \end{aligned}$$

Example 1.5.14 Let $\sin \theta = \frac{3}{5}$ and $\cos \theta = \frac{4}{5}$. Find:

- (a) $\sin 2\theta$
- (b) $\cos 2\theta$
- (c) $\tan 2\theta$
- (d) $\cos^2 \frac{\theta}{2}$
- (e) $\tan^2 \frac{\theta}{2}$
- (e) $\csc 2\theta$

Solution

- (a) $\sin 2\theta$

Using $\sin 2\theta = 2 \sin \theta \cos \theta$,

$$\sin 2\theta = 2 \cdot \frac{3}{5} \cdot \frac{4}{5} = \frac{24}{25}$$

- (b) $\cos 2\theta$

Using $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$,

$$\cos 2\theta = \left(\frac{4}{5}\right)^2 - \left(\frac{3}{5}\right)^2 = \frac{16}{25} - \frac{9}{25} = \frac{7}{25}$$

- (c) $\tan 2\theta$

Using $\tan 2\theta = \frac{\sin 2\theta}{\cos 2\theta}$,

$$\tan 2\theta = \frac{\frac{24}{25}}{\frac{7}{25}} = \frac{24}{7}$$

- (d) $\cos^2 \frac{\theta}{2}$

Using $\cos^2 \frac{\theta}{2} = \frac{1 + \cos \theta}{2}$,

$$\cos^2 \frac{\theta}{2} = \frac{1 + \frac{4}{5}}{2} = \frac{\frac{9}{5}}{2} = \frac{9}{10}$$

(e) $\tan^2 \frac{\theta}{2}$

Using $\tan^2 \frac{\theta}{2} = \frac{1 - \cos \theta}{1 + \cos \theta}$,

$$\tan^2 \frac{\theta}{2} = \frac{1 - \frac{4}{5}}{1 + \frac{4}{5}} = \frac{\frac{1}{5}}{\frac{9}{5}} = \frac{1}{9}$$

(e) $\csc 2\theta$

Using $\csc 2\theta = \frac{1}{\sin 2\theta}$,

$$\csc 2\theta = \frac{1}{\frac{24}{25}} = \frac{25}{24}$$

Example 1.5.15 Find the exact value of $\sin \frac{\theta}{2}$ and $\cos \frac{\theta}{2}$ given $\cos \theta = -\frac{1}{8}$.

Solution

First, calculate $\sin^2 \theta$:

$$\sin^2 \theta = 1 - \cos^2 \theta = 1 - \left(-\frac{1}{8}\right)^2 = 1 - \frac{1}{64} = \frac{63}{64}$$

$$\text{Thus, } \sin \theta = \pm \sqrt{\frac{63}{64}} = \pm \frac{\sqrt{63}}{8}.$$

Using the half-angle formulas:

$$\sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}}, \quad \cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}}$$

Substitute $\cos \theta = -\frac{1}{8}$:

$$\sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - (-\frac{1}{8})}{2}} = \pm \sqrt{\frac{1 + \frac{1}{8}}{2}} = \pm \sqrt{\frac{\frac{9}{8}}{2}} = \pm \sqrt{\frac{9}{16}} = \pm \frac{3}{4}$$

$$\cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + (-\frac{1}{8})}{2}} = \pm \sqrt{\frac{1 - \frac{1}{8}}{2}} = \pm \sqrt{\frac{\frac{7}{8}}{2}} = \pm \sqrt{\frac{7}{16}} = \pm \frac{\sqrt{7}}{4}$$

Example 1.5.16 Without using a calculator, evaluate the following leaving your answer in surd form where possible:

(a) $\sin 75^\circ + \sin 15^\circ$

(b) $\cos 105^\circ - \cos 15^\circ$

Solution

(a) $\sin 75^\circ + \sin 15^\circ$

$$\begin{aligned}\sin 75^\circ + \sin 15^\circ &= 2 \sin \frac{75^\circ + 15^\circ}{2} \cos \frac{75^\circ - 15^\circ}{2} \\ &= 2 \sin 90^\circ \cos 30^\circ = 2 \times 1 \times \frac{\sqrt{3}}{2} = \sqrt{3}.\end{aligned}$$

(b) $\cos 105^\circ - \cos 15^\circ$

$$\begin{aligned}\cos 105^\circ - \cos 15^\circ &= -2 \sin \frac{105^\circ + 15^\circ}{2} \sin \frac{105^\circ - 15^\circ}{2} \\ &= -2 \sin 60^\circ \sin 45^\circ = -2 \times \frac{\sqrt{3}}{2} \times \frac{\sqrt{2}}{2} = -\frac{\sqrt{6}}{2}.\end{aligned}$$

Example 1.5.17 Find the exact value of $\sin 52.5^\circ \cos 7.5^\circ$.**Solution** $\sin 52.5^\circ \cos 7.5^\circ$ = Exact value is calculated as shown.

Applying the formula, we obtain:

$$\begin{aligned}\sin(52.5^\circ \cos 7.5^\circ) &= \frac{\sin(52.5^\circ + 7.5^\circ) + \sin(52.5^\circ - 7.5^\circ)}{2} \\ &= \frac{\sin 60^\circ + \sin 45^\circ}{2} = \frac{\frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2}}{2} \\ &= \frac{\sqrt{3} + \sqrt{2}}{4}.\end{aligned}$$

Example 1.5.18 Prove that $\sin^2 x - \sin^2 y = \sin(x + y) \sin(x - y)$.**Solution**

We simplify the LHS and show that it is equal to the RHS.

$$\begin{aligned}\text{LHS} &= \sin^2 x - \sin^2 y \quad (\text{apply difference of two squares}) \\ &= (\sin x + \sin y)(\sin x - \sin y)\end{aligned}$$

Applying the respective factor formulae

$$\begin{aligned}&= \left[2 \sin \left(\frac{x+y}{2} \right) \cos \left(\frac{x-y}{2} \right) \right] \left[2 \sin \left(\frac{x-y}{2} \right) \cos \left(\frac{x+y}{2} \right) \right] \\ &= 4 \sin \left(\frac{x+y}{2} \right) \cos \left(\frac{x+y}{2} \right) \sin \left(\frac{x-y}{2} \right) \cos \left(\frac{x-y}{2} \right).\end{aligned}$$

Simplifying using the double angle formula:

$$= \sin(x + y) \sin(x - y).$$

Thus,

$$\text{LHS} = \text{RHS}.$$

If we prove this identity from RHS to LHS,

$$\text{RHS} = \sin(x + y) \sin(x - y)$$

. But $\sin(x + y) = \sin x \cos y + \cos x \sin y$ and $\sin(x - y) = \sin x \cos y - \cos x \sin y$. If we substitute

$$\sin(x + y) \sin(x - y) = [\sin x \cos y + \cos x \sin y] [\sin x \cos y - \cos x \sin y]$$

What we have here is a difference of squares which will then be simplified to

$$(\sin x \cos y)^2 - (\cos x \sin y)^2$$

$$\sin^2 x \cos^2 y - \cos^2 x \sin^2 y$$

At this point we will eliminate cosine using **Identity 1.6**. We know that $\cos^2 A = 1 - \sin^2 A$. Thus

$$= \sin^2 x (1 - \sin^2 y) - (1 - \sin^2 x) \sin^2 y$$

$$= \sin^2 x - \cancel{\sin^2 x \sin^2 y} + \cancel{\sin^2 x \sin^2 y} - \sin^2 y = \sin^2 x - \sin^2 y = \text{LHS}$$

Example 1.5.19 Write the expression $\cos 3x \cos 5x$ as a sum of two trigonometric functions.

Solution

$$\begin{aligned} \cos 3x \cos 5x &= \frac{\cos(3x + 5x) + \cos(3x - 5x)}{2} \\ &= \frac{\cos 8x + \cos(-2x)}{2}. \end{aligned}$$

Since $\cos(-2x) = \cos 2x$, we have:

$$\cos 3x \cos 5x = \frac{\cos 8x + \cos 2x}{2}.$$

Example 1.5.20 Express the following as a product of sine and cosine:

(a) $\sin 5x + \sin 3x$

(b) $\cos 2t - \cos 3t$

Solution

(a) $\sin 5x + \sin 3x$

$$\begin{aligned}\sin 5x + \sin 3x &= 2 \sin \frac{5x+3x}{2} \cos \frac{5x-3x}{2} \\ &= 2 \sin 4x \cos x.\end{aligned}$$

(b) $\cos 2t - \cos 3t$

$$\begin{aligned}\cos 2t - \cos 3t &= -2 \sin \frac{2t+3t}{2} \sin \frac{2t-3t}{2} \\ &= -2 \sin \frac{5t}{2} \sin \frac{-t}{2}.\end{aligned}$$

Simplifying further:

$$= 2 \sin \frac{5t}{2} \sin \frac{t}{2}.$$

Example 1.5.21 Evaluate the following, leaving your answer in surd form:

(a) $\sin 75^\circ - \sin 15^\circ$

(b) $\cos 75^\circ \cos 15^\circ$

Solution

(a) $\sin 75^\circ - \sin 15^\circ$

$$\begin{aligned}\sin 75^\circ - \sin 15^\circ &= 2 \cos \frac{75^\circ + 15^\circ}{2} \sin \frac{75^\circ - 15^\circ}{2} \\ &= 2 \cos 45^\circ \sin 30^\circ = 2 \cdot \frac{\sqrt{2}}{2} \cdot \frac{1}{2} \\ &= \frac{\sqrt{2}}{2}.\end{aligned}$$

(b) $\cos 75^\circ \cos 15^\circ$

$$\begin{aligned}\cos 75^\circ \cos 15^\circ &= \frac{\cos(75^\circ + 15^\circ) + \cos(75^\circ - 15^\circ)}{2} \\ &= \frac{\cos 90^\circ + \cos 60^\circ}{2} = \frac{0 + \frac{1}{2}}{2} \\ &= \frac{1}{4}.\end{aligned}$$

Example 1.5.22 Prove that $\frac{\sin 4x + \sin 6x}{\cos 4x - \cos 6x} = \cot x$ **Solution**

$$\text{LHS} = \frac{2 \sin \frac{4x+6x}{2} \cos \frac{4x-6x}{2}}{-2 \sin \frac{4x+6x}{2} \sin \frac{4x-6x}{2}}$$

$$= -\cot \frac{4x - 6x}{2} = -\cot(-x) = \cot x.$$

Hence, LHS = RHS.

Example 1.5.23 Prove that $\frac{\sin x + 2 \sin 2x + \sin 3x}{\cos x + 2 \cos 2x + \cos 3x} = \tan 2x$

Solution

Using the sum-to-product identities:

$$\text{Numerator: } \sin x + \sin 3x + 2 \sin 2x = 2 \sin 2x \cos x + 2 \sin 2x = 2 \sin 2x (\cos x + 1).$$

$$\text{Denominator: } \cos x + \cos 3x + 2 \cos 2x = 2 \cos 2x \cos x + 2 \cos 2x = 2 \cos 2x (\cos x + 1).$$

Simplifying:

$$\text{LHS} = \frac{2 \sin 2x (\cos x + 1)}{2 \cos 2x (\cos x + 1)} = \tan 2x.$$

Hence, LHS = RHS.

1.6 Graphs of Trigonometric Functions

1.6.1 Core Insights

Definition 1.6.1 (Periodicity of Trigonometric Functions) *A function is said to be periodic if*

$$f(x) = f(x + c)$$

), where c is a constant that we call the period of the function. Graphically, this means that if we shift the function by a constant "distance" along the horizontal axis, we see the same picture again.

The periods of the **primary trigonometric functions** are as follows:

- **Sine and Cosine:** The period is 2π .

$$\sin(x + 2\pi) = \sin x, \quad \cos(x + 2\pi) = \cos x$$

- **Tangent:** The period is π .

$$\tan(x + \pi) = \tan x$$

This means that the pattern of the sine and cosine waves repeat after every interval of length 2π . And for tangent, the pattern repeats after every interval of length π .

Graphs of $y = \sin x$, $y = \cos x$ and $y = \tan x$

Here, we will look at the basic forms of trigonometric functions. These forms determine the general traits of a trig graphs before any graph shifting agents (constants) are introduced. The angle used for the plotting

Table 1.3: Values and intervals for a Co-sine graph

Sub-Interval	Values		Change in $\sin x$
0 to $\frac{\pi}{2}$	$\sin 0 = 0$	$\sin \frac{\pi}{2} = 1$	Increases from 0 to 1
$\frac{\pi}{2}$ to π	$\sin \frac{\pi}{2} = 1$	$\sin \pi = 0$	Decreases from 1 to 0
π to $\frac{3\pi}{2}$	$\sin \pi = 0$	$\sin \frac{3\pi}{2} = -1$	Decreases from 0 to -1
$\frac{3\pi}{2}$ to 2π	$\sin \frac{3\pi}{2} = -1$	$\sin 2\pi = 0$	Increases from -1 to 0

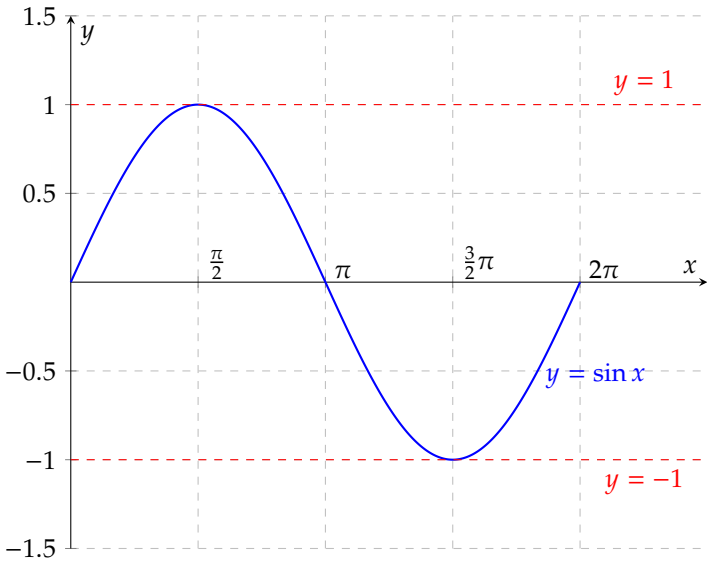


Figure 1.28: Graph of $y = \sin x$

of these graphs are always measured in **radians**. In addition, we mostly consider the values of quadrantal angles (Section 1.4.1) when determining the primary coordinates of the respective graphs.

1. **Graph of $y = \sin x$**

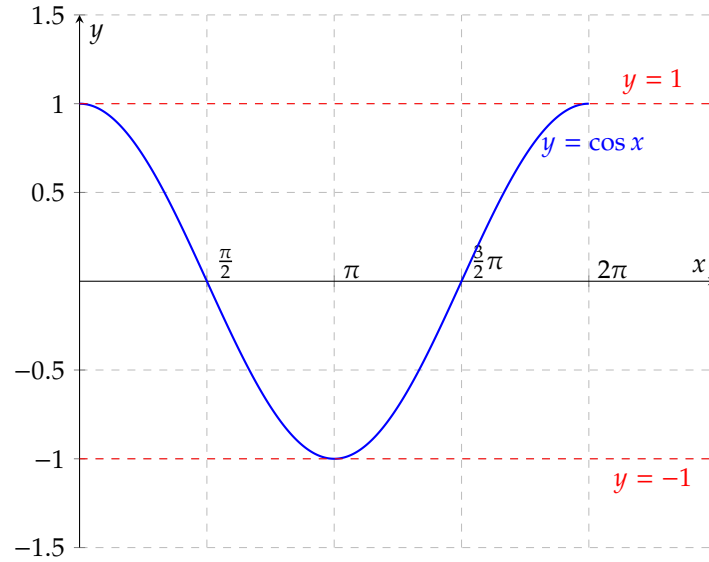
Recall Section 1.4.1, we saw that the range of $\sin x$ is $-1 \leq \sin x \leq 1$. As such, the graph of sine oscillates between -1 and 1 and it is enclosed between lines $y = -1$ and $y = 1$.

When plotting the graph of sine, it is enough to only plot the graph for $0 \leq x \leq 2\pi$ because, according to periodicity, after this interval the graph will begin to repeat the same pattern. To extract the pattern, we will divide the interval into four sub-intervals according to the quadrants and use the values of quadrantal angles to do it. These sub-intervals are,

- (i) 0 to $\frac{\pi}{2}$
- (ii) $\frac{\pi}{2}$ to π
- (iii) π to $\frac{3\pi}{2}$
- (iv) $\frac{3\pi}{2}$ to 2π

According to Table??, we can plot a smooth **Sine wave** as shown below.

Sub-Interval	Values		Change in $\sin x$
0 to $\frac{\pi}{2}$	$\cos 0 = 1$	$\cos \frac{\pi}{2} = 0$	Decreases from 1 to 0
$\frac{\pi}{2}$ to π	$\cos \frac{\pi}{2} = 0$	$\cos \pi = -1$	Decreases from 0 to -1
π to $\frac{3\pi}{2}$	$\cos \pi = -1$	$\cos \frac{3\pi}{2} = 0$	Increases from -1 to 0
$\frac{3\pi}{2}$ to 2π	$\cos \frac{3\pi}{2} = -1$	$\cos 2\pi = 0$	Increases from 0 to 1

Table 1.4: Values and intervals for a Cosine graph**Figure 1.29:** Graph of $y = \cos x$

2. Graph of $y = \cos x$

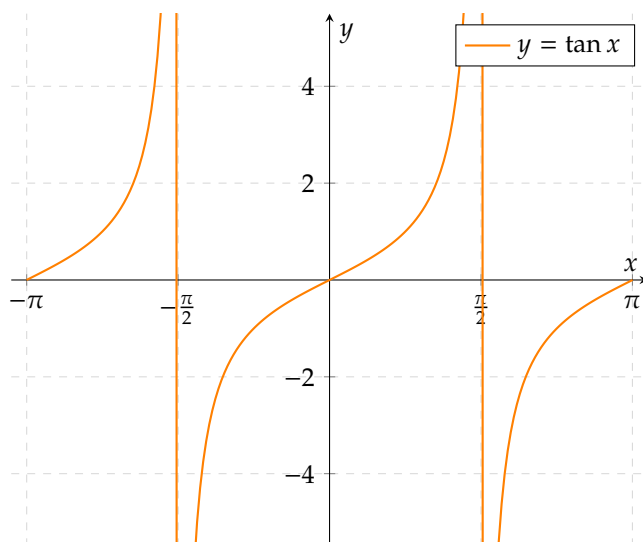
Everything from range to period is the same except the values of quadrantal angles. As such we will go straight to determining the pattern of the graph. And its graphs is as follows

3. Graph of $y = \tan x$

The tangent function $y = \tan x$ has the following properties:

The function $\tan x$ increases without bound (approaches $+\infty$) as $x \rightarrow \frac{\pi}{2}^-$ and $x \rightarrow \frac{3\pi}{2}^-$. Similarly, $\tan x$ decreases without bound (approaches $-\infty$) as $x \rightarrow \frac{\pi}{2}^+$ and $x \rightarrow \frac{3\pi}{2}^+$. These vertical asymptotes occur wherever $\cos x = 0$, because $\tan x = \frac{\sin x}{\cos x}$. This is due to the fact that $\tan x$ is undefined at $x = \frac{\pi}{2} + k\pi$ for all integers k , due to division by zero. The domain of $y = \tan x$ is therefore:

$$\text{Domain} = \{x \in \mathbb{R} \mid x \neq \frac{\pi}{2} + k\pi, k \in \mathbb{Z}\}.$$

Figure 1.30: Graph of $y = \tan x$

Since the tangent function is periodic with a period of π . This means:

$$\tan(x + \pi) = \tan(x) \quad \text{for all } x \in \mathbb{R}.$$

The function crosses the x-axis whenever $\tan x = 0$, which occurs when $\sin x = 0$. These x-intercepts are located at:

$$x = k\pi, \quad k \in \mathbb{Z}.$$

The vertical asymptotes of tangent occur at all points where the function is undefined: As such, asymptotes will occur at:

$$x = \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z}.$$

These asymptotes divide the graph into repeating segments over intervals of π and between each pair of consecutive asymptotes, $y = \tan x$ is strictly increasing.

The range of $y = \tan x$ is all real numbers:

$$\text{Range} = (-\infty, \infty).$$

The graph of $y = \tan x$ exhibits vertical asymptotes at $x = \frac{\pi}{2} + k\pi$, x-intercepts at $x = k\pi$, and repeats every π . The function increases monotonically within each interval, transitioning from $-\infty$ to $+\infty$ as x approaches the asymptotes.

Graphs of $y = \sec x$, $y = \csc x$ and $y = \cot x$

1. Graph of $y = \sec x$

The secant function is the reciprocal of the cosine function:

$$\sec(x) = \frac{1}{\cos(x)}$$

- **Domain:** All real numbers except where $\cos(x) = 0$, which occurs at $x = \frac{\pi}{2} + n\pi$, where n is an integer.
- **Range:** $(-\infty, -1] \cup [1, \infty)$

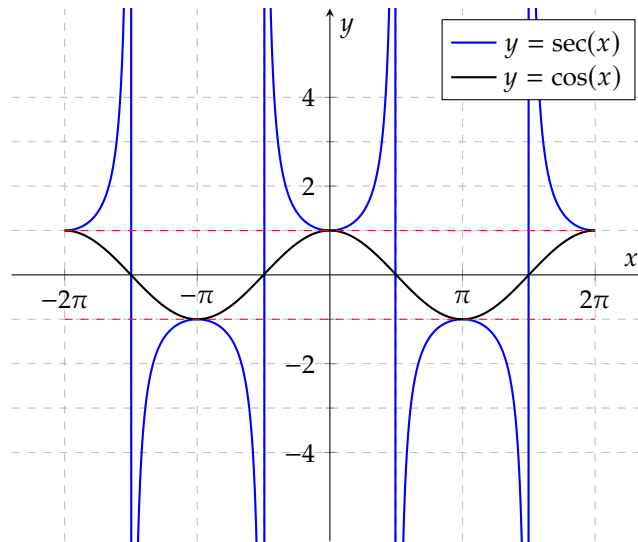


Figure 1.31: Graph of $y = \sec x$

- **Periodicity:** The function has a period of 2π .
- **Vertical Asymptotes:** At $x = \frac{\pi}{2} + n\pi$, due to $\cos(x) = 0$ at these points, making $\sec(x)$ undefined.

The graph of $\sec(x)$ consists of a series of U-shaped curves opening upwards or downwards, depending on the sign of $\cos(x)$. Between each pair of vertical asymptotes, the secant graph mirrors the behaviour of the cosine graph, but with values inverted and stretched away from the x-axis.

2. Cosecant Function $y = \csc x$

The cosecant function is the reciprocal of the sine function:

$$\csc(x) = \frac{1}{\sin(x)}$$

- **Domain:** All real numbers except where $\sin(x) = 0$, which occurs at $x = n\pi$, where n is an integer.
- **Range:** $(-\infty, -1] \cup [1, \infty)$
- **Periodicity:** The function has a period of 2π .
- **Vertical Asymptotes:** At $x = n\pi$, due to $\sin(x) = 0$ at these points, making $\csc(x)$ undefined.

The graph of $\csc(x)$ consists of a series of U-shaped curves similar to those of the secant function. These curves open upwards or downwards, depending on the sign of $\sin(x)$. Between each pair of vertical asymptotes, the cosecant graph mirrors the behaviour of the sine graph, but with values inverted and stretched away from the x-axis.

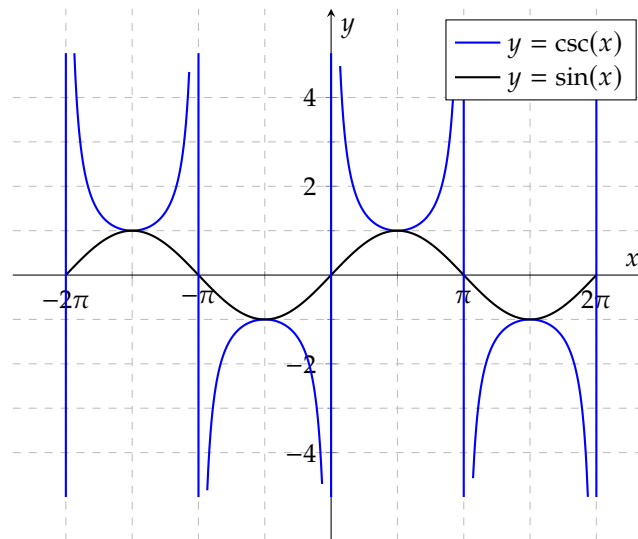


Figure 1.32: Graph of $y = \csc x$

3. Cotangent Function (cot)

The cotangent function is the reciprocal of the tangent function:

$$\cot(x) = \frac{1}{\tan(x)} = \frac{\cos(x)}{\sin(x)}$$

- **Domain:** All real numbers except where $\sin(x) = 0$, which occurs at $x = n\pi$, where n is an integer.
- **Range:** All real numbers $(-\infty, \infty)$
- **Periodicity:** The function has a period of π .
- **Vertical Asymptotes:** At $x = n\pi$, due to $\sin(x) = 0$ at these points, making $\cot(x)$ undefined.

The graph of $\cot(x)$ consists of a series of decreasing curves that approach vertical asymptotes at $x = n\pi$. Unlike the tangent function, which increases from negative to positive infinity between asymptotes, the cotangent function decreases from positive to negative infinity.

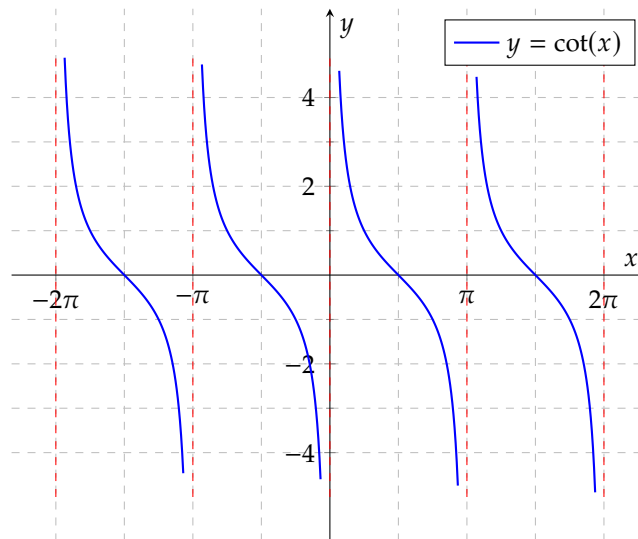


Figure 1.33: Graph of $y = \cot x$

Graph Shifting

From College Algebra, we looked at how basic graphs of different functions change when constants are introduced into the functions. These changes of form were referred to as **graph shifting** and they include:

1. Vertical shifts

This is a change in which a graph is pushed upwards or downwards of the original position depending on the sign of a constant in play. In this shift, the basic graph is shifted vertically upwards if a constant is added to the output of the basic function and it is shifted vertically downwards if a constant is subtracted. Therefore, the shift is defined as

$$\begin{aligned}\text{Upward Shift: } & y = f(x) + c \\ \text{Downward Shift: } & y = f(x) - c\end{aligned}$$

In order to visualise how this works in terms of trigonometric functions, let's look at the graphs of $y = \cos(x) + 2$ and $y = \cos(x) - 2$ on Figure 1.34

2. Horizontal Shifts

In this change, a graph is pushed left or right of the original position. This occurs when a constant is added or subtracted from the input of the original function. The graph is therefore pushed to the **right when a constant is subtracted and to the left when the constant is added**.

$$\begin{aligned}\text{Left Shift: } & y = f(x + c) \\ \text{Right Shift: } & y = f(x - c)\end{aligned}$$

Let's visualise by looking at the graphs of $y = \sin(x + \frac{\pi}{5})$ and $y = \sin(x - \frac{\pi}{3})$. Clearly Figure 1.35 might not be easily understood. But the only thing you need to look at is the idea. Using some graphing calculators like [DESMOS](https://www.desmos.com/calculator) which can be found on <https://www.desmos.com/calculator> or [GEOGEBRA](https://www.geogebra.org/classic) which can be found on <https://www.geogebra.org/classic>, you can try

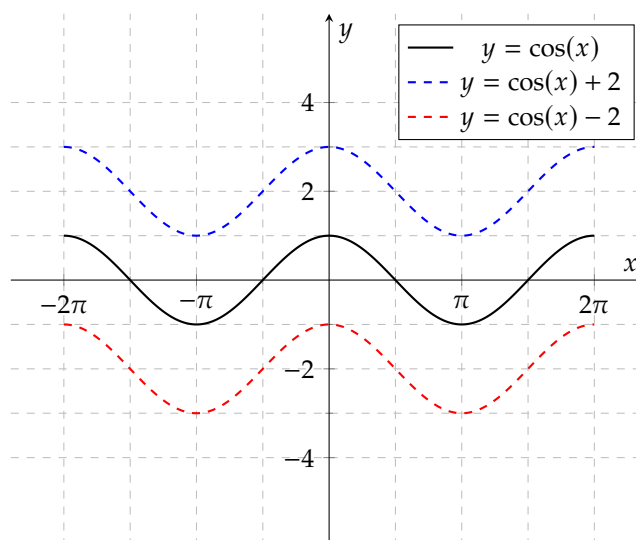


Figure 1.34: Graph of $y = \cos(x) + 2$ and $y = \cos(x) - 2$

out these plots to visualise how they switch positions when different constants are introduced. For better understanding, review the work on Functions from college algebra.

In addition, it can be noted that the graph of Sine is the graph of cosine shifted to the right by $\frac{\pi}{2}$ units and the graph of Cosine is the graph of sine that has been shifted $\frac{\pi}{2}$ units to the left. For example if you consider the graph of $y = \cos\left(x - \frac{\pi}{2}\right)$ on Figure 1.36, it is exactly the same as the graph of $y = \sin x$ on Figure 1.28.

3. Stretching and Shrinking

In this graph shift, the period is either increase or reduced. This occurs when the input angle changes, thereby, it is effected by multiplying the input angles (which involves the domain, x) by a constant.

- If it is greater than 1, the graph shrinks. This occurs because the period reduces as input angles increase by greater multiples and they reach the maximum point of the period (say 2π) as a multiple of a lower angle.
- If the constant is a fraction ($0 < c < 1$), the graph stretches. In this case, the input angles are divided continuously and the period is increased.

Consider Figure 1.37 where the graphs of $y = \sin\left(\frac{1}{2}x\right)$ and $y = \sin(2x)$ have been plotted in relation to the graph of $y = \sin x$. Here, you will note that the graph of $y = \sin(2x)$ used a domain $[0, \pi]$ to complete one sine wave. That's because the period shrinks to π and if you multiply the new period, π , by the constant 2 you get the original period 2π . On the other hand, the graph of $y = \sin\left(\frac{1}{2}x\right)$ needed a whole domain $[0, 4\pi]$ just to complete one sine wave. And here, the period stretched to 4π and if you do the multiplication with its constant you still get 2π .

4. Period and Inversion

Period is affected by the range of trigonometric functions just because the highest value a graph can reach is just the maximum value the function can produce. That means, if we want to alter

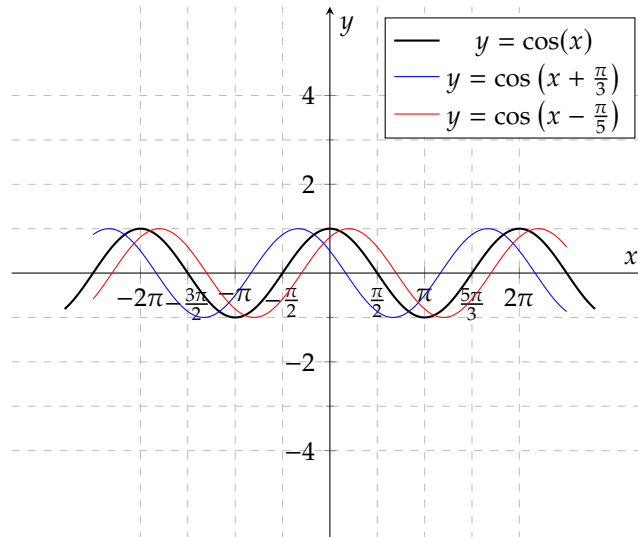


Figure 1.35: Graph of $y = \cos(x + \frac{\pi}{3})$ and $y = \cos(x - \frac{\pi}{5})$

the period, we should multiply a constant to the output of the functions. Depending on the sign of the constant, the behaviour of the resulting graphs differ.

- **Positive Constants** simply increase or decrease (whole number and fraction respectively) the amplitude of the graphs and the don't affect the original shape of the functions.
- However, **Negative Constants** do the same task as the positive ones only that they go ahead and invert the graphs.

Consider Figure 1.38

Graphs of $y = a \sin(bx + c) + d$ and $y = a \cos(bx + c) + d$

Here, we will try to combine all the concepts we have seen on graph shifting (Section 1.6 Subsection 1.6.1). Here we will consider the graphs in the form

$$y = a \sin(bx + c) + d \text{ and } y = a \cos(bx + c) + d$$

$$\forall a, b, c, d \in \mathbb{R}$$

For now we already know that a affects the period no matter the sign. Hence, period will always be $|a|$. We also know that d will show whether we are pushing the graph upwards, downwards or we are leaving it in its original position. That means, the real drama occurs inside the brackets on $bx + c$. Since the sine wave for the graph of $y = \sin(bx + c)$ is obtained if $bx + c$ varies from 0 to 2π , that is, if x ranges from

$$-\frac{c}{b} \quad \text{to} \quad \frac{2\pi - c}{b} = \frac{2\pi}{b} - \frac{c}{b}.$$

Going ahead with this approach and derive the phase shifts is complicated and mostly not clear (check your course modules and lecture notes). Since we know what every constant affects, let's keep it simple and choose the easy approach. **Keep things simple and chill guys!**

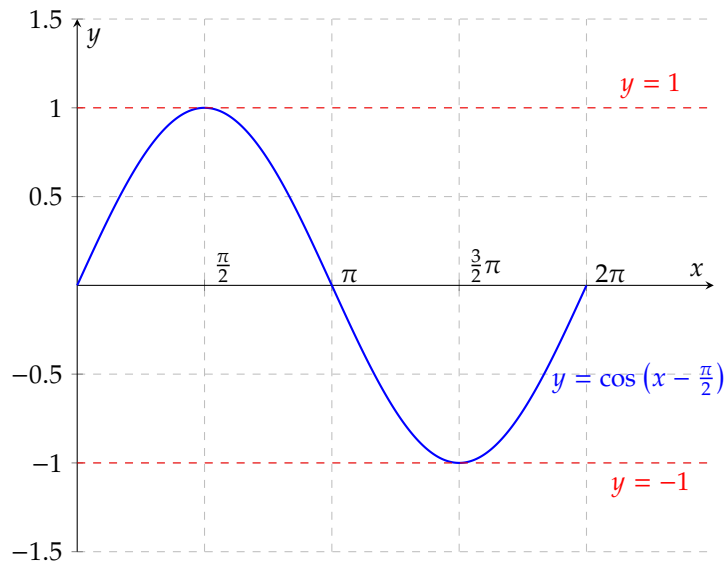


Figure 1.36: Graph of $y = \cos\left(x - \frac{\pi}{2}\right)$

1. **Amplitude:**

This the easiest. As shown already, amplitude is $|a|$.

2. **Period:**

The period of sine and cosine is 2π . And we said that the product of the constant on x and the new period gives us the original period, 2π . Hence

$$\text{Period} \times b = 2\pi$$

$$\text{Period} = \frac{2\pi}{b}$$

3. **Step:**

Step is a distance from one point of plot to another. Since we use quadrantal angles, we technically divide the period into 4 parts. Therefore,

$$\text{Step} = \frac{\text{Period}}{4}$$

4. **Domain:**

Since we have seen that $bx + c$ varies from 0 to 2π , our domain will run from where $bx + c = 0$ to where $bx + c = 2\pi$. Solving these two equations for x will give us the domain, $[\text{Min}, \text{Max}]$. Then, we will be adding the step to min-value recursively until max-value is obtained and we create the set form of the domain. The angles in the domain set correspond to the values of quadrantal angles from Table ?? and Table?? for the trends sine and cosine respectively.

Phase Shifts High Jacked!

Knowing where the graph will be shifted to is useless with this approach since you know where your plot is starting from and you know where it ends. Therefore, we don't need to apply any crazy concept here and we can happily go ahead and make our plots.

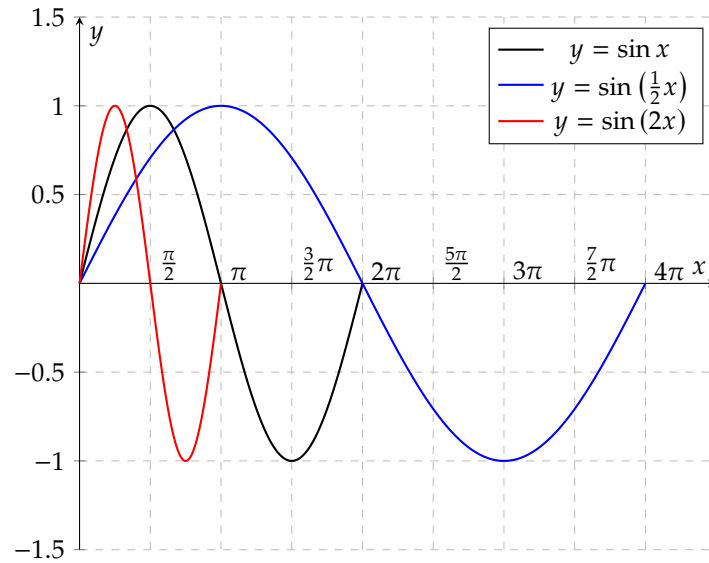


Figure 1.37: Graphs of $y = \sin\left(\frac{1}{2}x\right)$ and $y = \sin(2x)$ in relation to the graph of $y = \sin x$

1.6.2 Worked Examples

Example 1.6.1 Sketch the graph of $y = 4 \cos\left(\frac{x}{2}\right)$

Solution

- The amplitude is $|4| = 4$.
- The period is given by:

$$\frac{2\pi}{b} = \frac{2\pi}{\frac{1}{2}} = 4\pi$$

- Step:

$$\text{Step} = \frac{\text{Period}}{4} = \pi$$

- Domain: $[0, 4\pi]$. And after adding step to the minimum value in x continuously, we get the domain set of the function. Thus, $0 + \pi$ gives π , $\pi + \pi$ gives 2π , $2\pi + \pi$ gives 3π and finally $3\pi + \pi$ gives 4π . This gives the domain set as follows: $x = \{0, \pi, 2\pi, 3\pi, 4\pi\}$.
- Values:

Here, we will look at the combination of these value to the values of their corresponding quadrantal angles according to the amplitude.

$$\begin{aligned} x = 0, \quad y &= 4 \quad (\text{maximum}) \\ x = \pi, \quad y &= 0 \quad (\text{crosses axis}) \\ x = 2\pi, \quad y &= -4 \quad (\text{minimum}) \\ x = 3\pi, \quad y &= 0 \quad (\text{crosses axis}) \\ x = 4\pi, \quad y &= 4 \quad (\text{maximum}) \end{aligned}$$

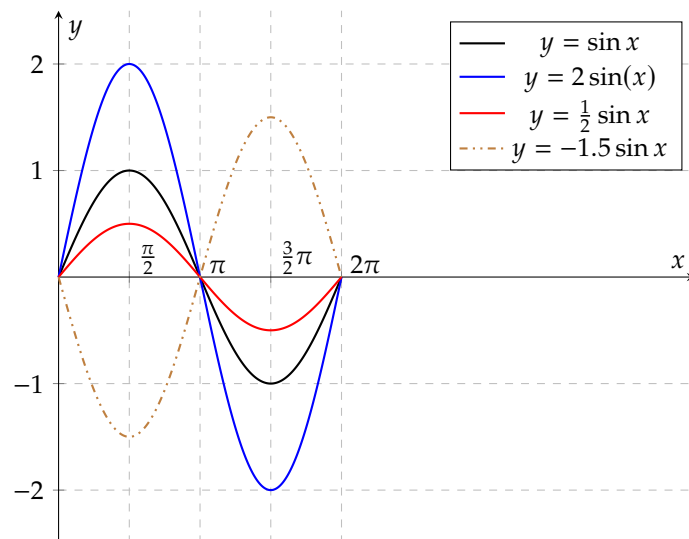
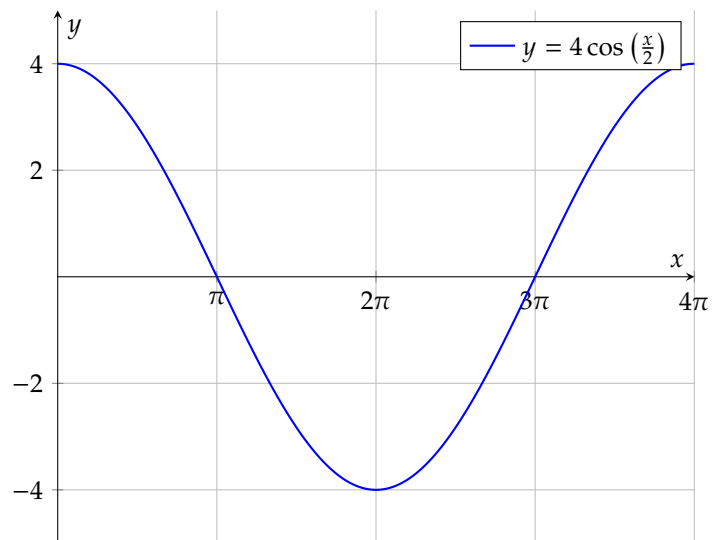


Figure 1.38: Graphs of $y = \frac{1}{2} \sin(x)$, $y = 2 \sin(x)$ and $y = -\frac{3}{2} \sin x$



Example 1.6.2 Sketch the graph of $y = -\sin(4x)$

Solution

- The amplitude is $|-1| = 1$. Since it's negative, the graph is inverted.
- The period is given by:

$$\frac{2\pi}{b} = \frac{2\pi}{4} = \frac{\pi}{2}$$

- Step:

$$\text{Step} = \frac{\text{Period}}{4} = \frac{\pi}{8}$$

- Domain: $[0, \frac{\pi}{2}]$. Key points: $x = 0, \frac{\pi}{8}, \frac{\pi}{4}, \frac{3\pi}{8}, \frac{\pi}{2}$.

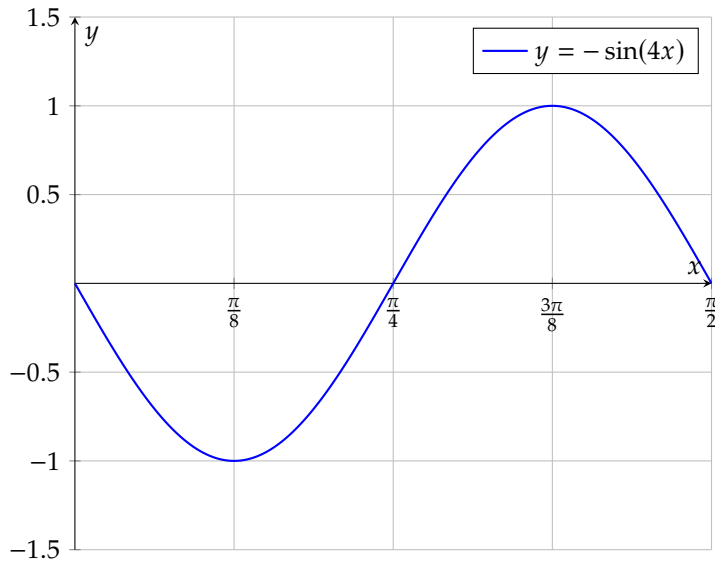


Figure 1.39: Graph of $y = -\sin(4x)$ - Example 1.6.2

► Values:

$$\begin{aligned}
 x = 0, \quad y &= 0 \\
 x = \frac{\pi}{8}, \quad y &= -1 \quad (\text{minimum}) \\
 x = \frac{\pi}{4}, \quad y &= 0 \quad (\text{crosses axis}) \\
 x = \frac{3\pi}{8}, \quad y &= 1 \quad (\text{maximum}) \\
 x = \frac{\pi}{2}, \quad y &= 0
 \end{aligned}$$

Example 1.6.3 Sketch the graph of $y = 2 + \tan(x)$

Solution

The graph of tangent has different characteristics as the others. So, always be careful!

- Amplitude: Not applicable.
- Period:

$$\pi$$

- Step: This is also not technically applicable but we always just need to know where the x – intercept will be. But since we have a vertical shift, instead of having a normal intercept, the graph will cut the line $y = 2$ at the point of a normal intercept (without shift) $x = 0$
- Domain: Exclude vertical asymptotes ($x = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \dots$).

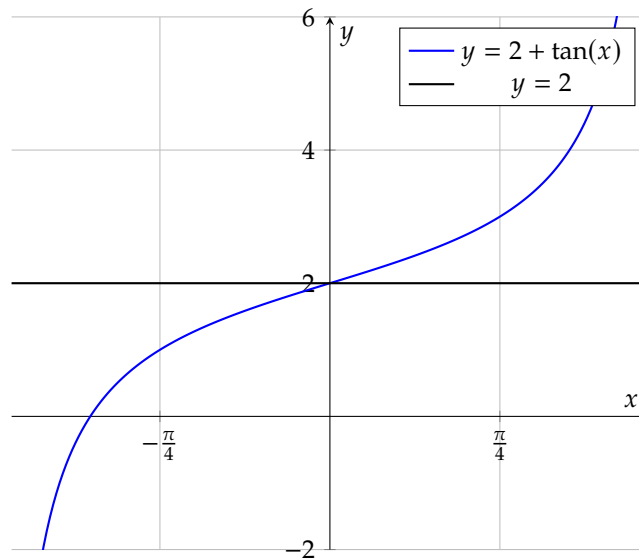


Figure 1.40: Graph of $y = 2 + \tan(x)$ - Example 1.6.3

► Values (with vertical shift +2):

$$x = -\frac{\pi}{4}, \quad y = 1$$

$$x = 0, \quad y = 2$$

$$x = \frac{\pi}{4}, \quad y = 3$$

Example 1.6.4 Sketch the graph of $y = -6 \cos(2x)$

Solution

- The amplitude is $|-6| = 6$.
- The period is given by:

$$\frac{2\pi}{b} = \frac{2\pi}{2} = \pi$$

► Step:

$$\text{Step} = \frac{\text{Period}}{4} = \frac{\pi}{4}$$

- Domain: $[0, \pi]$. Key points: $x = 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi$.
- Values:

$$x = 0, \quad y = -6 \quad (\text{minimum})$$

$$x = \frac{\pi}{4}, \quad y = 0 \quad (\text{crosses axis})$$

$$x = \frac{\pi}{2}, \quad y = 6 \quad (\text{maximum})$$

$$x = \frac{3\pi}{4}, \quad y = 0 \quad (\text{crosses axis})$$

$$x = \pi, \quad y = -6 \quad (\text{minimum})$$

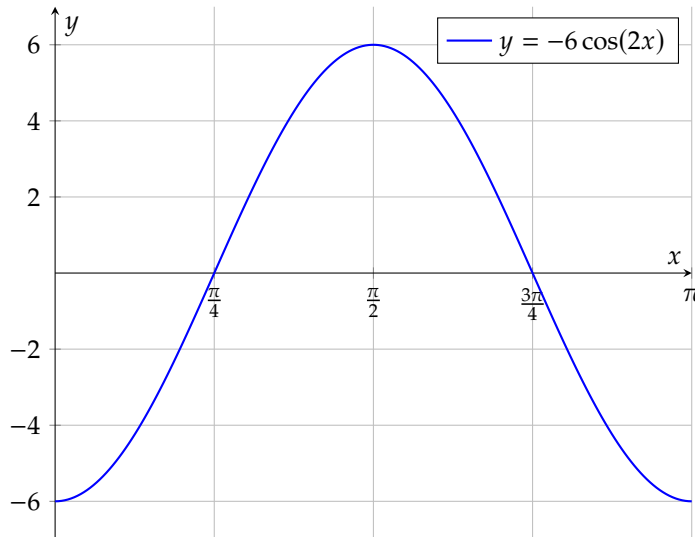


Figure 1.41: Graph of $y = -6 \cos(2x)$ - Example 1.6.4

Example 1.6.5 Sketch the graphs of the following functions.

- (a) $f(x) = 2 \sin\left(2x - \frac{\pi}{2}\right)$
 (b) $y = \cos\left(x + \frac{\pi}{3}\right)$

Solution

(a) Consider $f(x) = 2 \sin\left(2x - \frac{\pi}{2}\right)$:

- The amplitude is $|2| = 2$.
- The period is given by $\frac{2\pi}{b} = \frac{2\pi}{2} = \pi$.
- Step is $\frac{\text{Period}}{4} = \frac{\pi}{4}$
- Domain

Minimum value: $2x - \frac{\pi}{2} = 0$, if we solve, $x = \frac{\pi}{4}$

Maximum value: $2x - \frac{\pi}{2} = 2\pi$, after solving, $x = \frac{5\pi}{4}$

Therefore, Domain = $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$.

Here, we will find the domain in set form as follows: $\frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}$,
 $\frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4}$, and so on. Hence the domain is:

$$x = \left\{ \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi, \frac{5\pi}{4} \right\}$$

From this, we have to always understand that these angles in the domain correspond to the values of quadrantal angles. Thus, 0 corresponds to $\frac{\pi}{4}$ and the value is 0, $\frac{\pi}{2}$ corresponds to $\frac{\pi}{2}$ and the value is 1, π corresponds to $\frac{3\pi}{4}$ and the value is -1 and so on. Then you multiply each of these values by the amplitude to determine the range of y .

After all this, we will plot the graph (see **Figure 1.42**).

(b) Consider $y = \cos\left(x + \frac{\pi}{3}\right)$:

- The amplitude is 1 (default for cosine).
- The period is $\frac{2\pi}{1} = 2\pi$.

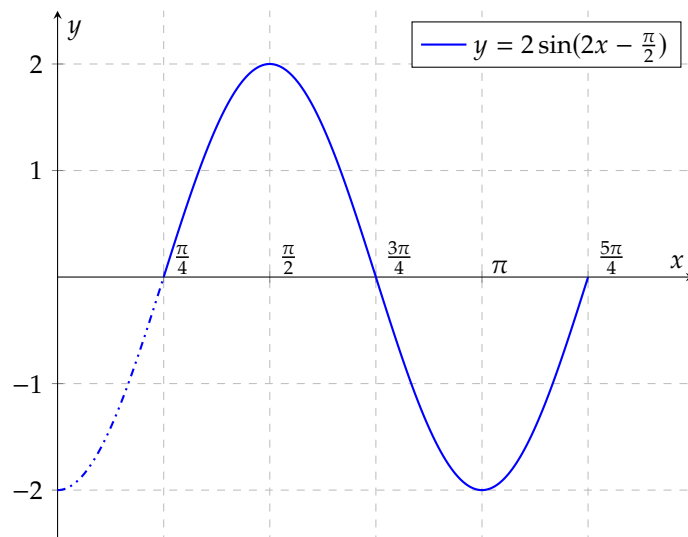


Figure 1.42: Graphs of $y = 2 \sin(2x - \frac{\pi}{2})$ - Example 1.6.5 (a)

17:

Phase Shift if Asked

Simply find the minimum value of the domain. If it is positive, the has been pushed to the right with that units. And if it's negative, it has been shifted to the left.

► Step is $\frac{\text{Period}}{4} = \frac{\pi}{4}$

► Domain:

Minimum Value: $x + \frac{\pi}{3} = 0$, which gives $x = -\frac{\pi}{3}$.¹⁷

Maximum Value: $x + \frac{\pi}{3} = 2\pi$, which gives $x = \frac{5\pi}{3}$

Domain Set = $-\frac{\pi}{3} + \frac{\pi}{2}, -\frac{\pi}{3} + \pi$, etc. Therefore Domain

$$x = \left\{ -\frac{\pi}{3}, \frac{\pi}{6}, \frac{2\pi}{3}, \frac{7\pi}{6}, \frac{5\pi}{3} \right\}$$

and the coordinates to plot are $(-\frac{\pi}{3}, 1)$, $(\frac{\pi}{6}, 0)$, $(\frac{2\pi}{6}, -1)$, $(\frac{7\pi}{6}, 0)$, and $(\frac{5\pi}{3}, 1)$.

Now, let's plot the graph. (see Figure 1.43)

Example 1.6.6 Sketch the graph of $y = 3 \sin(2x + \frac{\pi}{2}) - 1$

Solution

- The amplitude is $|3| = 3$.
- The period is given by:

$$\frac{2\pi}{b} = \frac{2\pi}{2} = \pi$$

- Vertical shift: Down by 1 unit ($d = -1$)¹⁸.
- Step:

$$\text{Step} = \frac{\text{Period}}{4} = \frac{\pi}{4}$$

- Domain:

$$\text{Min Value, } x: 2x + \frac{\pi}{2} = 0; x = -\frac{\pi}{4}$$

$$\text{Max Value, } x: 2x + \frac{\pi}{2} = 2\pi; x = \frac{3\pi}{4}$$

18: You will note that the line of focus is no longer the x -axis ($y = 0$) but the the line of shift, in this case $y = -1$ and we can say that the new axis is $y = -1$ and plot normally.

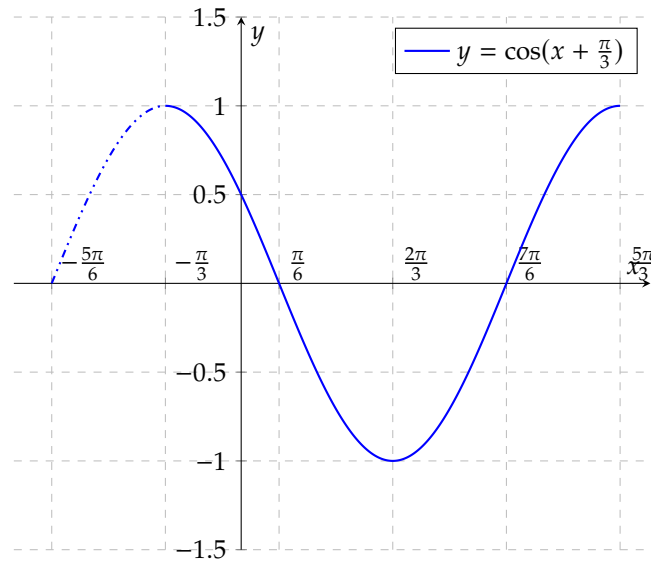


Figure 1.43: Graphs of $y = \cos(x + \frac{\pi}{3})$ - Example 1.6.5 (b)

Therefore, the domain is $[-\frac{\pi}{4}, \frac{3\pi}{4}]$. Key points:

$$x = \left\{ -\frac{\pi}{4}, 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4} \right\}$$

► Values:

$$x = -\frac{\pi}{4}, \quad y = -1 \quad (\text{crosses axis})$$

$$x = 0, \quad y = 2 \quad (\text{maximum})$$

$$x = \frac{\pi}{4}, \quad y = -1 \quad (\text{crosses axis})$$

$$x = \frac{\pi}{2}, \quad y = -4 \quad (\text{minimum})$$

$$x = \frac{3\pi}{4}, \quad y = -1 \quad (\text{crosses axis})$$

Example 1.6.7 Sketch the graph of $y = -2 \cos\left(\frac{x}{3} - \frac{\pi}{6}\right) + 4$

Solution

- The amplitude is $|-2| = 2$.
- The period is given by:

$$\frac{2\pi}{b} = \frac{2\pi}{\frac{1}{3}} = 6\pi$$

- Vertical shift: Up by 4 units ($d = 4$).
- Step:

$$\text{Step} = \frac{\text{Period}}{4} = \frac{6\pi}{4} = \frac{3\pi}{2}$$

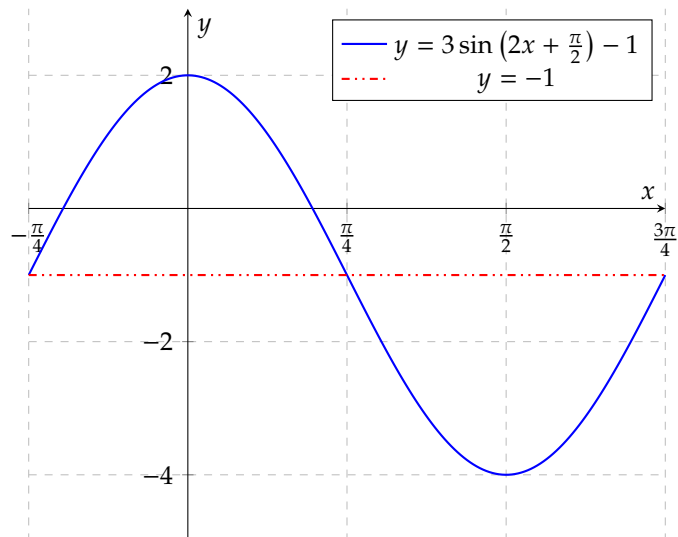


Figure 1.44: Graph of $y = 3 \sin(2x + \frac{\pi}{2}) - 1$ - Example 1.6.6

► Domain:

$$\text{Min Value, } x: \frac{x}{3} - \frac{\pi}{6} = 0; x = \frac{\pi}{2}$$

$$\text{Max Value, } x: \frac{x}{3} - \frac{\pi}{6} = 2\pi; x = \frac{13\pi}{6}$$

Therefore, the Domain is $[\frac{\pi}{2}, \frac{13\pi}{2}]$. Key points:

$$x = \left\{ \frac{\pi}{2}, 2\pi, \frac{7\pi}{2}, 5\pi, \frac{13\pi}{2} \right\}$$

► Values:

$$x = \frac{\pi}{2}, y = 2 \quad (\text{minimum})$$

$$x = 2\pi, y = 4 \quad (\text{crosses axis})$$

$$x = \frac{7\pi}{2}, y = 6 \quad (\text{maximum})$$

$$x = 5\pi, y = 4 \quad (\text{crosses axis})$$

$$x = \frac{13\pi}{2}, y = 2 \quad (\text{minimum})$$

Example 1.6.8 Sketch the graph of $y = 4 \sin(3x - \pi) + 2$

Solution

► The amplitude is $|4| = 4$.

► The period is given by:

$$\frac{2\pi}{b} = \frac{2\pi}{3}$$

► Vertical shift: Up by 2 units ($d = 2$), axis is $y = 2$.

► Step:

$$\text{Step} = \frac{\text{Period}}{4} = \frac{2\pi}{12} = \frac{\pi}{6}$$

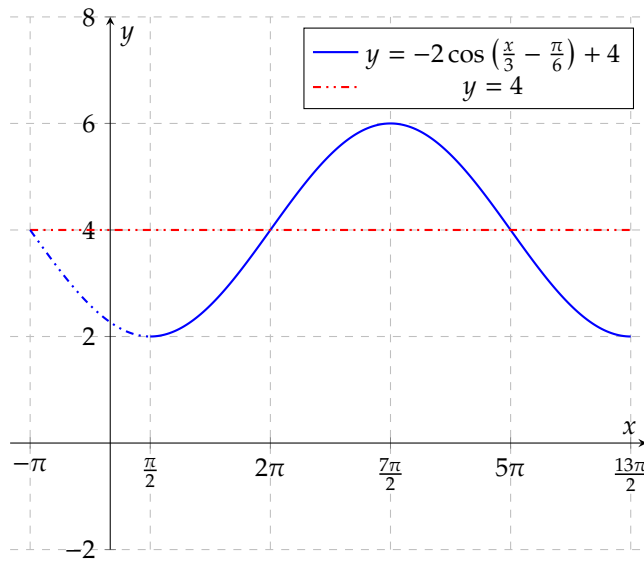


Figure 1.45: Graph of $y = -2 \cos\left(\frac{x}{3} - \frac{\pi}{6}\right) + 4$ - Example 1.6.7

► Domain:

$$\text{Min Value, } x: 3x - \pi = 0; x = \frac{\pi}{3}$$

$$\text{Max Value, } x: 3x - \pi = 2\pi; x = \pi$$

Hence, the domain is $\left[\frac{\pi}{3}, \pi\right]$. Key points:

$$x = \left\{ \frac{\pi}{3}, \frac{\pi}{2}, \frac{2\pi}{3}, \frac{5\pi}{6}, \pi \right\}$$

► Values:

$$x = \frac{\pi}{3}, \quad y = 2 \quad (\text{crosses axis})$$

$$x = \frac{\pi}{2}, \quad y = 6 \quad (\text{maximum})$$

$$x = \frac{2\pi}{3}, \quad y = 2 \quad (\text{crosses axis})$$

$$x = \frac{5\pi}{6}, \quad y = -2 \quad (\text{minimum})$$

$$x = \pi, \quad y = 2 \quad (\text{crosses axis})$$

Example 1.6.9 Sketch the graph of $y = 5 \sin\left(\frac{\pi}{2}x + \frac{\pi}{4}\right)$

Solution

► The amplitude is $|5| = 5$.

► The period is given by:

$$\frac{2\pi}{b} = \frac{2\pi}{\frac{\pi}{2}} = 4$$

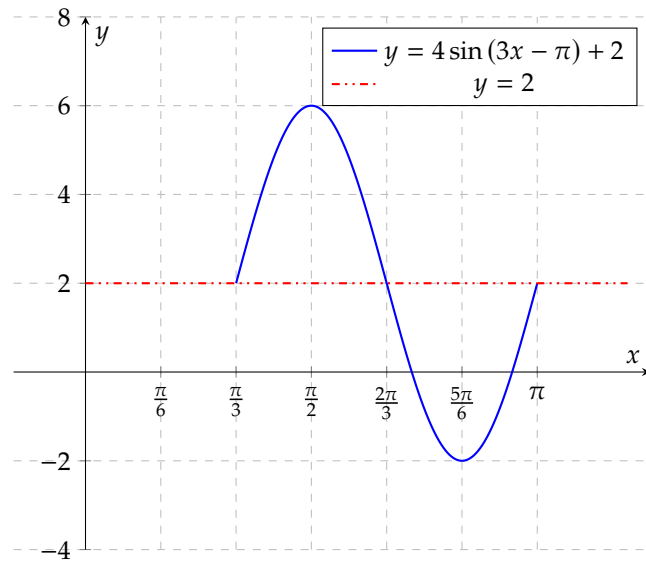


Figure 1.46: Graph of $y = 4 \sin(3x - \pi) + 2$ - Example 1.6.8

► Step:

$$\text{Step} = \frac{\text{Period}}{4} = \frac{4}{4} = 1$$

► Domain:

$$\text{Min Value, } x: \frac{\pi}{2}x + \frac{\pi}{4} = 0; x = -\frac{1}{2}$$

$$\text{Max Value, } x: \frac{\pi}{2}x + \frac{\pi}{4} = 2\pi; x = \frac{7}{2}$$

Thus, the domain is $[-\frac{1}{2}, \frac{7}{2}]$. Key points:

$$x = \left\{ -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \frac{7}{2} \right\}$$

► Values:

$$x = -\frac{1}{2}, \quad y = 0 \quad (\text{crosses axis})$$

$$x = \frac{1}{2}, \quad y = 5 \quad (\text{maximum})$$

$$x = \frac{3}{2}, \quad y = 0 \quad (\text{crosses axis})$$

$$x = \frac{5}{2}, \quad y = -5 \quad (\text{minimum})$$

$$x = \frac{7}{2}, \quad y = 0 \quad (\text{crosses axis})$$

Example 1.6.10 Sketch the graph of $y = -3 \cos(4x - \frac{\pi}{3})$

Solution

► The amplitude is $|-3| = 3$ (inverted).

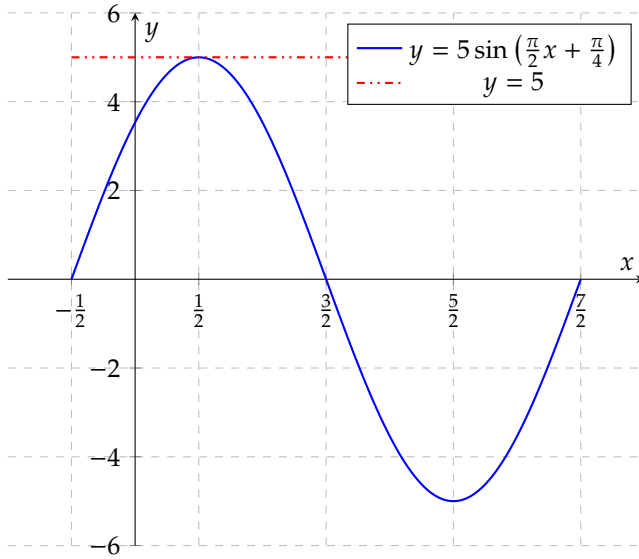


Figure 1.47: Graph of $y = 5 \sin\left(\frac{\pi}{2}x + \frac{\pi}{4}\right)$
- Example 1.6.9

- The period is given by:

$$\frac{2\pi}{b} = \frac{2\pi}{4} = \frac{\pi}{2}$$

- Step:

$$\text{Step} = \frac{\text{Period}}{4} = \frac{\pi}{8}$$

- Domain:

$$\text{Min Value, } x: 4x - \frac{\pi}{3} = 0; x = \frac{\pi}{12}$$

$$\text{Max Value, } x: 4x - \frac{\pi}{3} = 2\pi; x = \frac{7\pi}{12}$$

Hence, the domain is $\left[\frac{\pi}{12}, \frac{7\pi}{12}\right]$. Key points:

$$x = \left\{ \frac{\pi}{12}, \frac{5\pi}{24}, \frac{\pi}{3}, \frac{11\pi}{24}, \frac{7\pi}{12} \right\}$$

- Values:

$$x = \frac{\pi}{12}, \quad y = -3 \quad (\text{minimum})$$

$$x = \frac{5\pi}{24}, \quad y = 0 \quad (\text{crosses axis})$$

$$x = \frac{\pi}{3}, \quad y = 3 \quad (\text{maximum})$$

$$x = \frac{11\pi}{24}, \quad y = 0 \quad (\text{crosses axis})$$

$$x = \frac{7\pi}{12}, \quad y = -3 \quad (\text{minimum})$$

Example 1.6.11 Sketch the graph of $y = 2 \sin\left(6x - \frac{\pi}{2}\right)$

Solution

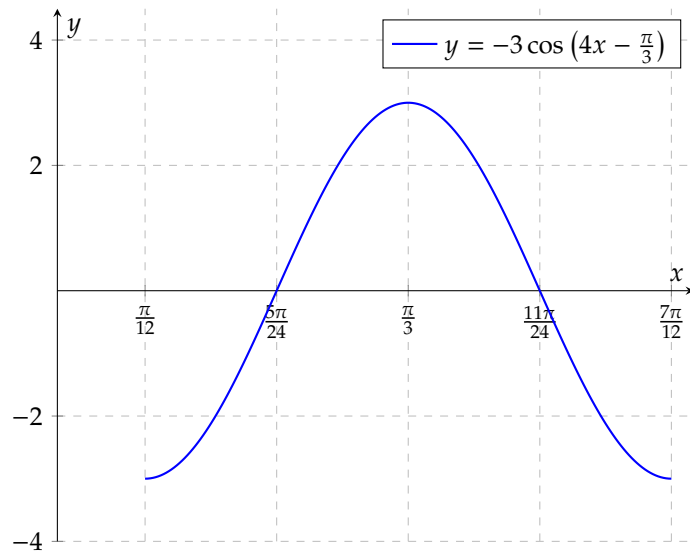


Figure 1.48: Graph of $y = -3 \cos(4x - \frac{\pi}{3})$ - Example 1.6.10

- The amplitude is $|2| = 2$.
- The period is given by:

$$\frac{2\pi}{b} = \frac{2\pi}{6} = \frac{\pi}{3}$$

- Step:

$$\text{Step} = \frac{\text{Period}}{4} = \frac{\pi}{12}$$

- Domain:

$$\text{Min Value, } x: 6x - \frac{\pi}{2} = 0; x = \frac{\pi}{12}$$

$$\text{Max Value, } x: 6x - \frac{\pi}{2} = 2\pi; x = \frac{5\pi}{12}$$

Hence, the domain is $[\frac{\pi}{12}, \frac{5\pi}{12}]$. Key points:

$$x = \left\{ \frac{\pi}{18}, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{5\pi}{12} \right\}$$

- Values:

$$x = \frac{\pi}{12}, \quad y = 0 \quad (\text{crosses axis})$$

$$x = \frac{\pi}{6}, \quad y = 2 \quad (\text{maximum})$$

$$x = \frac{\pi}{4}, \quad y = 0 \quad (\text{crosses axis})$$

$$x = \frac{\pi}{3}, \quad y = -2 \quad (\text{minimum})$$

$$x = \frac{5\pi}{12}, \quad y = 0 \quad (\text{crosses axis})$$

Example 1.6.12 Sketch the graph of $y = \tan(2x - \frac{\pi}{6}) - 2$.

Solution

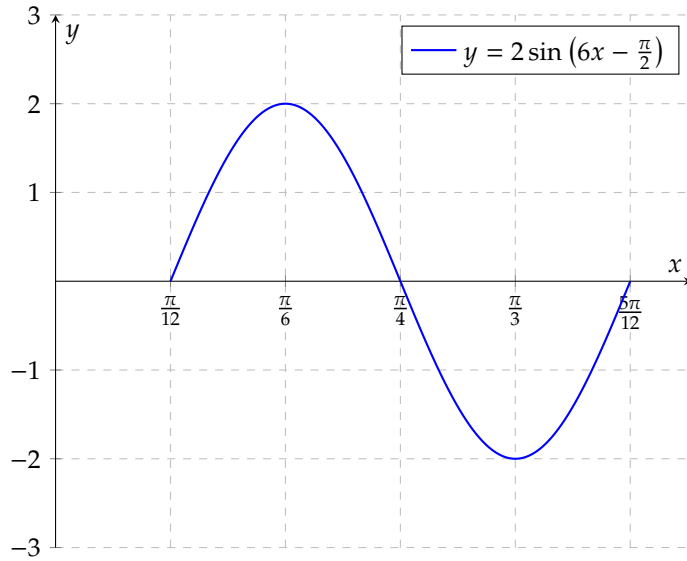


Figure 1.49: Graph of $y = 2 \sin(6x - \frac{\pi}{2})$
- Example 1.6.11

We know that the graph of tangent is a little different from the graphs of sine and cosine. Therefore, the approach will be different but not complicated. We will explore very simple ideas to manage a plot as follows:

- Amplitude: Not applicable
- Period: As we have looked so far, the coefficient of x affects the period. And since the period of tangent is π , the period for this graph will be

$$\frac{\pi}{b} = \frac{\pi}{2}$$

- Domain: The graph of tangent is sandwiched between two vertical asymptotes at points where tangent is undefined. For a most basic cycle, these points are $-\frac{\pi}{2}$ and $\frac{\pi}{2}$. Therefore, to determine a domain, we will find points at which $2x - \frac{\pi}{6}$ is equal to $-\frac{\pi}{2}$ and $\frac{\pi}{2}$ respectively.

$$x - \text{Min: } 2x - \frac{\pi}{6} = -\frac{\pi}{2}; x = -\frac{\pi}{6}$$

$$x - \text{Max: } 2x - \frac{\pi}{6} = \frac{\pi}{2}; x = \frac{\pi}{3}$$

Therefore, the domain is $(-\frac{\pi}{6}, \frac{\pi}{3})$.

- x - Intercept: Here, we don't need step because the key points are just three; the left asymptote, right asymptote and the x - intercept. The asymptotes are just the limit points of the domain. and the intercept lies at the middle point of the domain. Thus,

$$x - \text{Intercept} = (x - \text{Min}) + \frac{\text{Period}}{2} = -\frac{\pi}{6} + \frac{\pi}{4} = \frac{\pi}{12}$$

- Vertical Shift: -2 . This means that the horizontal resting point of the graph is no longer the x - axis, ($y = 0$) the new shifting point $y = -2$. Therefore, at the x - Intercept, the graph will cross $y = -2$ and not $y = 0$.

Consider **Figure 1.50**

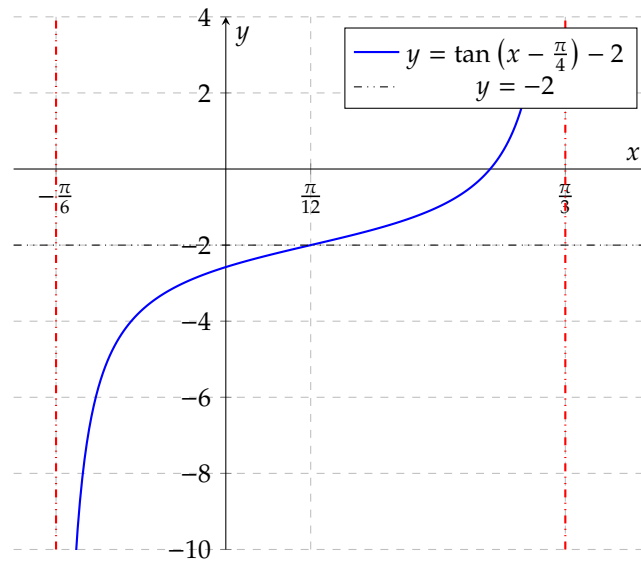


Figure 1.50: The graph of $y = \tan\left(2x - \frac{\pi}{6}\right) - 2$ - Example 1.6.12

Summary

This chapter has covered a number of concepts that are very useful, not only to trigonometry, but also other incoming challenges like Calculus, for example. We have looked at like the art of measuring angles in degrees and radians, the six trigonometric functions, identities and graphs. It is very crucial to know that no formula or theorem derived in this book is to be memorised but understood as that helps when the question changes form but stays in the same context since you can easily spot the relationships and apply the concepts better. And for proving identities, note that there is no fixed solution to the questions. The choices of the identities used in the process of proving determines the nature of the solution.

Chapter Review

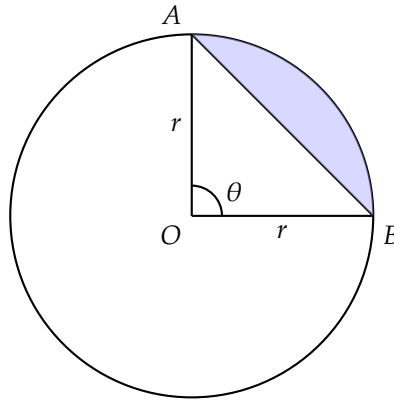


Figure 1.51: Diagram for exercise 24 and 25

In Exercises 1 to 8, convert the angle from degree measure into radian measure, giving the exact value in terms of π .

- | | | | |
|----------------|-----------------|-----------------|-----------------|
| 1. 0° | 3. 135° | 5. -315° | 7. 45° |
| 2. 240° | 4. -270° | 6. 150° | 8. -225° |

In Exercises 9 to 16, convert the angle from radian measure into degree measure.

- | | | | |
|-----------------------|-----------------------|----------------------|----------------------|
| 9. π | 11. $\frac{7\pi}{6}$ | 13. $\frac{\pi}{3}$ | 15. $-\frac{\pi}{6}$ |
| 10. $-\frac{2\pi}{3}$ | 12. $\frac{11\pi}{6}$ | 14. $\frac{5\pi}{3}$ | 16. $\frac{\pi}{2}$ |

In Exercises 17 to 22, calculate the lengths of arcs and the areas of sectors with the following measurements

- | | |
|--|-------------------------------------|
| 17. $\theta = \frac{\pi}{6}$ $r = 12$ cm | 20. $\theta = \pi$ $r = 1$ cm |
| 18. $\theta = \frac{5\pi}{4}$ $r = 100$ cm | 21. $\theta = 240^\circ$ $r = 5$ cm |
| 19. $\theta = 330^\circ$ $r = 9.3$ cm | 22. $\theta = 1^\circ$ $r = 117$ cm |

23. A sector has a perimeter of 2 cm, radius of length r and angle θ radians.

- Show that $\theta = 2 \left(\frac{1-r}{r} \right)$
- Also show that area A cm² of the sector is given by $A = r(1 - r)$

24. Study Figure 1.51 of a sector AOB inside a circle with centre O . If AO and OB are radii of length 5 cm and arc AB is 7 cm long, Find

- The angle θ
- The area of sector AOB
- The difference in length between arc AB and chord AB
- The area of shaded part
(Hint: Area of triangle $OAB = \frac{1}{2}AB \sin \theta$)

25. Figure 1.51 shows a circle of centre O , radius r and a chord AB that subtends an angle θ at the centre.

- Area of the shaded part in terms of r and θ
- Given that area of the shaded part is one-third of the area of triangle AOB , show that $3\theta - 4 \sin \theta = 0$

26. In Figure 1.51 the sector OAB has centre O and radius 6 cm, and $\angle AOB = \frac{\pi}{3}$ radians. OC is the bisector of $\angle AOB$, and P is the midpoint of OC . An arc DE of a circle is drawn with centre P to meet OA and OB at D and E , respectively.

- Find the size of $\angle OPD$.
- Calculate the area of the shaded region.

In Exercise 27 to 73, prove the identities

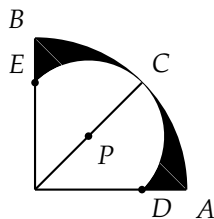


Figure 1.52: Intersected sectors

27. $\cos(\theta) \sec(\theta) = 1$
28. $\tan(\theta) \cos(\theta) = \sin(\theta)$
29. $\sin(\theta) \csc(\theta) = 1$
30. $\tan(\theta) \cot(\theta) = 1$
31. $\csc(\theta) \cos(\theta) = \cot(\theta)$
32. $\frac{\sin(\theta)}{\cos^2(\theta)} = \sec(\theta) \tan(\theta)$
33. $\frac{\cos(\theta)}{\sin^2(\theta)} = \csc(\theta) \cot(\theta)$
34. $\frac{1+\sin(\theta)}{\cos(\theta)} = \sec(\theta) + \tan(\theta)$
35. $\frac{1-\cos(\theta)}{\sin(\theta)} = \csc(\theta) - \cot(\theta)$
36. $\frac{\cos(\theta)}{1-\sin^2(\theta)} = \sec(\theta)$
37. $\frac{\sin(\theta)}{1-\cos^2(\theta)} = \csc(\theta)$
38. $\frac{\sec(\theta)}{1+\tan^2(\theta)} = \cos(\theta)$
39. $\frac{\csc(\theta)}{1+\cot^2(\theta)} = \sin(\theta)$
40. $\frac{\tan(\theta)}{\sec^2(\theta)-1} = \cot(\theta)$
41. $\frac{\cot(\theta)}{\csc^2(\theta)-1} = \tan(\theta)$
42. $4 \cos^2(\theta) + 4 \sin^2(\theta) = 4$
43. $9 - \cos^2(\theta) - \sin^2(\theta) = 8$
44. $\tan^3(\theta) = \tan(\theta) \sec^2(\theta) - \tan(\theta)$
45. $\sin^5(\theta) = (1 - \cos^2(\theta))^2 \sin(\theta)$
46. $\sec^{10}(\theta) = (1 + \tan^2(\theta))^4 \sec^2(\theta)$
47. $\cos^2(\theta) \tan^3(\theta) = \tan^2(\theta) + \tan^4(\theta)$
48. $\sec^4(\theta) - \sec^2(\theta) = \tan^2(\theta) + \tan^4(\theta)$
49. $\frac{\cos(\theta)+1}{\cos(\theta)-1} = \frac{1+\sec(\theta)}{1-\sec(\theta)}$
50. $\frac{\sin(\theta)+1}{\sin(\theta)-1} = \frac{1+\csc(\theta)}{1-\csc(\theta)}$
51. $\frac{1-\cot(\theta)}{1+\cot(\theta)} = \frac{\tan(\theta)-1}{\tan(\theta)+1}$
52. $\frac{1-\tan(\theta)}{1+\tan(\theta)} = \frac{\cos(\theta)-\sin(\theta)}{\cos(\theta)+\sin(\theta)}$
53. $\tan(\theta) + \cot(\theta) = \sec(\theta) \csc(\theta)$
54. $\csc(\theta) - \sin(\theta) = \cot(\theta) \cos(\theta)$
55. $\cos(\theta) - \sec(\theta) = -\tan(\theta) \sin(\theta)$
56. $\cos(\theta)(\tan(\theta) + \cot(\theta)) = \csc(\theta)$
57. $\sin(\theta)(\tan(\theta) + \cot(\theta)) = \sec(\theta)$
58. $\frac{1}{1-\cos(\theta)} + \frac{1}{1+\cos(\theta)} = 2 \csc^2(\theta)$
59. $\frac{1}{\sec(\theta)+1} + \frac{1}{\sec(\theta)-1} = 2 \csc(\theta) \cot(\theta)$
60. $\frac{1}{\csc(\theta)+1} + \frac{1}{\csc(\theta)-1} = 2 \sec(\theta) \tan(\theta)$
61. $\frac{1}{\csc(\theta)-\cot(\theta)} - \frac{1}{\csc(\theta)+\cot(\theta)} = 2 \cot(\theta)$
62. $\frac{\cos(\theta)}{1-\tan(\theta)} + \frac{\sin(\theta)}{1-\cot(\theta)} = \sin(\theta) + \cos(\theta)$
63. $\frac{1}{\sec(\theta)+\tan(\theta)} = \sec(\theta) - \tan(\theta)$
64. $\frac{1}{\sec(\theta)-\tan(\theta)} = \sec(\theta) + \tan(\theta)$
65. $\frac{1}{\csc(\theta)-\cot(\theta)} = \csc(\theta) + \cot(\theta)$
66. $\frac{1}{\csc(\theta)+\cot(\theta)} = \csc(\theta) - \cot(\theta)$
67. $\frac{1}{1-\sin(\theta)} = \sec^2(\theta) + \sec(\theta) \tan(\theta)$
68. $\frac{1}{1+\sin(\theta)} = \sec^2(\theta) - \sec(\theta) \tan(\theta)$
69. $\frac{1}{1-\cos(\theta)} = \csc^2(\theta) + \csc(\theta) \cot(\theta)$
70. $\frac{1}{1+\cos(\theta)} = \csc^2(\theta) - \csc(\theta) \cot(\theta)$
71. $\frac{\cos(\theta)}{1+\sin(\theta)} = \frac{1-\sin(\theta)}{\cos(\theta)}$
72. $1 - \sin(\theta) = (\sec(\theta) - \tan(\theta))^2$
73. $\csc(\theta) - \cot(\theta) = \frac{\sin(\theta)}{1+\cos(\theta)}$

In Exercises 74 to 79, use the Even / Odd Identities (Negative Angles) to verify the identity. Assume all quantities are defined.

74. $\sin(3\pi - 2\theta) = -\sin(2\theta - 3\pi)$
75. $\cos\left(-\frac{\pi}{4} + 5t\right) = \cos\left(5t + \frac{\pi}{4}\right)$
76. $\tan(-t^2 + 1) = -\tan(t^2 - 1)$
77. $\csc(-\theta - 5) = -\csc(\theta + 5)$
78. $\sec(-6t) = \sec(6t)$
79. $\cot(9 - 7\theta) = -\cot(7\theta - 9)$

In Exercises 80 to 94, use the Sum and Difference Identities to find the exact value. You may have need of all the other identities as well.

80. $\cos(75^\circ)$
81. $\sec(165^\circ)$
82. $\sin(105^\circ)$
83. $\csc(195^\circ)$
84. $\cot(255^\circ)$
85. $\tan(375^\circ)$
86. $\cos\left(\frac{13\pi}{12}\right)$
87. $\sin\left(\frac{11\pi}{12}\right)$
88. $\tan\left(\frac{7\pi}{12}\right)$
89. $\cos\left(\frac{17\pi}{12}\right)$
90. $\sec\left(\frac{5\pi}{12}\right)$
91. $\sin\left(\frac{11\pi}{12}\right)$
92. $\cot\left(\frac{11\pi}{12}\right)$
93. $\csc\left(\frac{5\pi}{12}\right)$
94. $\sec\left(-\frac{\pi}{12}\right)$

95. If α is a Quadrant IV angle with $\cos(\alpha) = \frac{\sqrt{5}}{5}$, and $\sin(\beta) = \frac{\sqrt{10}}{10}$ where $\frac{\pi}{2} < \beta < \pi$, find

- | | | |
|----------------------------|----------------------------|----------------------------|
| (a) $\cos(\alpha + \beta)$ | (c) $\tan(\alpha + \beta)$ | (e) $\sin(\alpha - \beta)$ |
| (b) $\sin(\alpha + \beta)$ | (d) $\cos(\alpha - \beta)$ | (f) $\tan(\alpha - \beta)$ |

96. If $\csc(\alpha) = 3$, where $0 < \alpha < \frac{\pi}{2}$, and β is a Quadrant II angle with $\tan(\beta) = -7$, find

- | | | |
|----------------------------|----------------------------|----------------------------|
| (a) $\cos(\alpha + \beta)$ | (c) $\tan(\alpha + \beta)$ | (e) $\sin(\alpha - \beta)$ |
| (b) $\sin(\alpha + \beta)$ | (d) $\cos(\alpha - \beta)$ | (f) $\tan(\alpha - \beta)$ |

97. If $\sin(\alpha) = \frac{3}{5}$, where $0 < \alpha < \frac{\pi}{2}$, and $\cos(\beta) = \frac{12}{13}$ where $\frac{3\pi}{2} < \beta < 2\pi$, find

- | | | |
|----------------------------|----------------------------|----------------------------|
| (a) $\sin(\alpha + \beta)$ | (c) $\tan(\alpha + \beta)$ | (e) $\cos(\alpha - \beta)$ |
| (b) $\cos(\alpha + \beta)$ | (d) $\sin(\alpha - \beta)$ | |

98. If $\sec(\alpha) = \frac{5}{3}$, where $\frac{\pi}{2} < \alpha < \pi$, and $\tan(\beta) = \frac{24}{7}$, where $\pi < \beta < \frac{3\pi}{2}$, find

- | | | |
|----------------------------|----------------------------|----------------------------|
| (a) $\csc(\alpha - \beta)$ | (b) $\sec(\alpha + \beta)$ | (b) $\cot(\alpha + \beta)$ |
|----------------------------|----------------------------|----------------------------|

In Exercises 99 to 111, verify the identity.

- | | |
|--|--|
| 99. $\cos(\theta - \pi) = -\cos(\theta)$ | 107. $\frac{\cos(\alpha + \beta)}{\cos(\alpha - \beta)} = \frac{1 + \tan(\alpha) \tan(\beta)}{1 - \tan(\alpha) \tan(\beta)}$ |
| 100. $\sin(\pi - \theta) = \sin(\theta)$ | 108. $\frac{\sin(t+h)}{h} = \frac{\sin(t)\cos(h) + \sin(h)\cos(t)}{h}$ |
| 101. $\tan(\theta + \frac{\pi}{2}) = -\cot(\theta)$ | 109. $\frac{\sin(t+h)}{h} = \left(\frac{\sin(h)}{h}\right)\cos(t) + \left(\frac{\cos(h)-1}{h}\right)\sin(t)$ |
| 102. $\sin(\alpha + \beta) + \sin(\alpha - \beta) = 2\sin(\alpha)\cos(\beta)$ | 110. $\frac{\cos(t+h) - \cos(t)}{h} = \cos(t)\left(\frac{\cos(h)-1}{h}\right) - \sin(t)\left(\frac{\sin(h)}{h}\right)$ |
| 103. $\sin(\alpha + \beta) - \sin(\alpha - \beta) = 2\cos(\alpha)\sin(\beta)$ | 111. $\frac{\tan(t+h) - \tan(t)}{h} = \frac{\sec^2(t)\left(\frac{\tan(h)}{h}\right)}{1 - \tan(t)\tan(h)}$ |
| 104. $\cos(\alpha + \beta) + \cos(\alpha - \beta) = 2\cos(\alpha)\cos(\beta)$ | |
| 105. $\cos(\alpha + \beta) - \cos(\alpha - \beta) = -2\sin(\alpha)\sin(\beta)$ | |
| 106. $\tan(\alpha + \beta) = \frac{\tan(\alpha) + \tan(\beta)}{1 - \tan(\alpha)\tan(\beta)}$ | |

In Exercises 112 to 120, use the Half Angle Formulas to find the exact value. You may have need of all the other Identities as well.

- | | | |
|--------------------------|---|--|
| 111. $\cos(75^\circ)$ | 115. $\tan(112.5^\circ)$ | 119. $\sin\left(\frac{5\pi}{8}\right)$ |
| 112. $\sin(105^\circ)$ | 116. $\cos\left(\frac{7\pi}{12}\right)$ | 120. $\tan\left(\frac{7\pi}{8}\right)$ |
| 113. $\cos(67.5^\circ)$ | 117. $\sin\left(\frac{\pi}{12}\right)$ | |
| 114. $\sin(157.5^\circ)$ | 118. $\cos\left(\frac{5\pi}{8}\right)$ | |

In Exercises 121 to 134, verify the identity. Assume all quantities are defined.

120. $\cos(\theta) + \sin(\theta))^2 = 1 + \sin(2\theta)$
 121. $\cos(\theta) - \sin(\theta))^2 = 1 - \sin(2\theta)$
 122. $\tan(2\theta) = \frac{1}{\tan(\theta)} = \frac{1 + \tan(\theta)}{1 - \tan(\theta)}$
 123. $\cot(2\theta) = \frac{\cot(\theta) + \tan(\theta)}{2}$
 124. $\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta)$
 125. $\tan(2\theta) = \frac{2\tan(\theta)}{1 - \tan^2(\theta)}$
 126. $\cot(2\theta) = \frac{1 - \tan^2(\theta)}{2\tan(\theta)}$
 127. $\sin(3\theta) = 3\sin(\theta) - 4\sin^3(\theta)$
 128. $\cos(3\theta) = 4\cos^3(\theta) - 3\cos(\theta)$
 129. $\sin(4\theta) = 4\sin(\theta)\cos(\theta) - 4\sin^3(\theta)\cos(\theta)$
 130. $\cos(4\theta) = 8\cos^4(\theta) - 8\cos^2(\theta) + 1$
 131. $\sin(5\theta) = 5\sin(\theta) - 20\sin^3(\theta) + 16\sin^5(\theta)$
 132. $\cos(5\theta) = 16\cos^5(\theta) - 20\cos^3(\theta) + 5\cos(\theta)$
 133. $\sin(6\theta) = 6\sin(\theta) - 32\sin^3(\theta) + 48\sin^5(\theta) - 32\sin^7(\theta)$
 134. $\cos(6\theta) = 32\cos^6(\theta) - 48\cos^4(\theta) + 18\cos^2(\theta) - 1$

In Exercises 135 to 146, graph one cycle of the given function. State the period, amplitude, domain(set of keypoints) and vertical shift of the function.

$$\begin{aligned} 135. y &= 3 \sin(x) \\ 136. y &= \sin(3x) \\ 137. y &= -2 \cos(x) \\ 138. y &= \cos\left(x - \frac{\pi}{2}\right) \end{aligned}$$

$$\begin{aligned} 139. y &= \sin\left(x + \frac{\pi}{3}\right) \\ 140. y &= \sin(2x - \pi) \\ 141. y &= -\frac{1}{3} \cos\left(\frac{1}{2}x + \frac{\pi}{3}\right) \\ 142. y &= \cos(3x - 2\pi) + 4 \end{aligned}$$

$$\begin{aligned} 143. y &= \sin\left(-x - \frac{\pi}{4}\right) - 2 \\ 144. y &= \frac{2}{3} \cos\left(\frac{\pi}{2} - 4x\right) + 1 \\ 145. y &= -\frac{3}{2} \cos\left(2x + \frac{\pi}{3}\right) - \frac{1}{2} \\ 146. y &= 4 \sin(2\pi x + \pi) \end{aligned}$$

In Exercises **147 to 149**, graph one cycle of the given functions. State period, domain and x – intercept.

$$147. y = \tan\left(x - \frac{\pi}{4}\right)$$

$$148. y = 2 \tan\left(\frac{1}{4}x\right) - 3$$

$$149. y = \frac{1}{3} \tan(-2x - \pi) + 1$$

Answers on Page 107 (Proving identities questions do not need answers)

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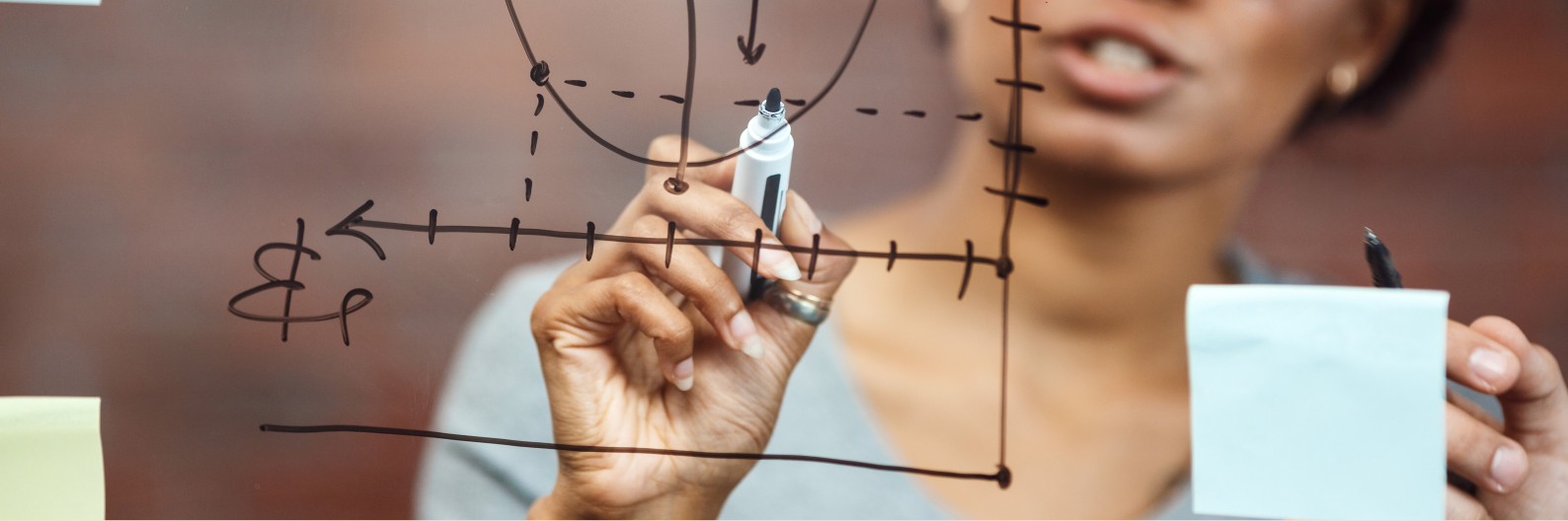
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ELEMENTARY CALCULUS



2 Limits and Continuity

Introduction

The concept of **limits** is a cornerstone of calculus, forming the foundation for understanding differentiation and integration. Historically, the study of limits can be traced back to ancient Greek mathematicians like Archimedes, who used early notions of limits to measure curved surfaces and volumes. However, it was not until the 19th century that mathematicians such as Augustin-Louis Cauchy and Karl Weierstrass formalized the definition of limits, replacing earlier, less rigorous approaches. This advancement provided a robust framework for calculus, addressing geometric problems like finding tangent lines to curves and determining areas under curves.

Limits enable us to precisely describe how functions behave and vary. Some functions change continuously, meaning small changes in x lead to small changes in $f(x)$, while others exhibit discontinuities, erratic behaviour, or unbounded growth. These distinctions can be analysed rigorously using the concept of limits.

Additionally, limits serve as the basis for defining the **derivative**, which represents rates of change, and the **integral**, which quantifies accumulation. These tools are indispensable for solving real-world problems across fields such as physics, engineering, biology, and economics. Mastering the concept of limits is therefore essential for understanding the advanced topics of differentiation and integration, as well as the broader study of continuous change and accumulation.

Success Criteria

By the end of this chapter you should be able to

- ▶ define a limit of a function
- ▶ define a continuous function
- ▶ evaluate limits of various algebraic functions
- ▶ evaluate limits of trigonometric functions
- ▶ determine the continuity of functions at a given point
- ▶ prove limit function using the formal definition of a limit

2.1	Limit of a Function	98
2.1.1	Core Insights	98
2.2	Limits of Trigonometric Functions	98
2.3	Limits Involving Infinity	98
2.4	Formal Definition of a Limit	98

2.1 Limit of a Function

2.1.1 Core Insights

2.2 Limits of Trigonometric Functions

2.3 Limits Involving Infinity

2.4 Formal Definition of a Limit

A hand holding a white marker points to the derivative term in the first equation on a whiteboard. The whiteboard contains two equations: $a = \frac{dv}{dt} = 2At + BC \cos(\omega t) = v'(t)$ and $x(t) = x(0) + \int_0^t v(t) dt$.
$$a = \frac{dv}{dt} = 2At + BC \cos(\omega t) = v'(t)$$

$$x(t) = x(0) + \int_0^t v(t) dt$$

3 Differentiation

$$f(t) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cdot \cos\left(\frac{n\pi t}{L}\right) + b_n \cdot \sin\left(\frac{n\pi t}{L}\right) \right]$$

$$= a_0 + a_1 \cdot \cos\left(\frac{\pi t}{L}\right) + b_1 \cdot \sin\left(\frac{\pi t}{L}\right) + a_2 \cdot \cos\left(\frac{2\pi t}{L}\right) + b_2 \cdot \sin\left(\frac{2\pi t}{L}\right) + \dots$$



$$L=2 \Rightarrow L=1 \quad a_n \approx \frac{1}{n^2}$$

4 Integration

$$a_0 = \frac{1}{2L} \int_{-L}^L f(t) dt = \frac{1}{2} \int_{-1}^1 f(t) dt$$

$$= \frac{1}{2} \int_{-1}^0 f(t) dt + \frac{1}{2} \int_0^1 f(t) dt$$

$$= \frac{1}{2} \int_{-1}^0 1 dt + \frac{1}{2} \int_0^1 1 dt$$

$$= \frac{1}{2} \left[t \right]_{-1}^0 + \frac{1}{2} \left[t \right]_0^1$$

$$= -\frac{1}{2} + \frac{1}{2} = 0$$

$$a_n \rightarrow a(\omega)$$

$$b_n \rightarrow b(\omega)$$



5 Applications

APPENDIX

Answers to Chapter Review Exercises A

Chapter 1 Review Exercises

- | | | |
|---|--|--|
| 1. 0
2. $\frac{4\pi}{3}$
3. $\frac{3\pi}{4}$
4. $-\frac{3\pi}{2}$
5. $-\frac{7\pi}{4}$
6. $\frac{5\pi}{6}$ | 7. $\frac{\pi}{4}$
8. $-\frac{5\pi}{4}$
9. 180°
10. -120°
11. 210°
12. 330° | 13. 60°
14. 300°
15. -30°
16. 90°
17. 12π units ²
18. 6250π units ² |
|---|--|--|

- | |
|---|
| 19. 79.2825π units ²
20. $\frac{\pi}{2}$ units ²
21. $\frac{50\pi}{3}$ units ²
22. 38.025π units ² |
|---|

80. $\cos(75^\circ) = \frac{\sqrt{6}-\sqrt{2}}{4}$
 81. $\sec(165^\circ) = \frac{-4}{\sqrt{2}+\sqrt{6}} = \sqrt{2} - \sqrt{6}$
 82. $\sin(105^\circ) = \frac{\sqrt{6}+\sqrt{2}}{4}$
 83. $\csc(195^\circ) = \frac{4}{\sqrt{2}-\sqrt{6}} = -(\sqrt{2} + \sqrt{6})$
 84. $\cot(255^\circ) = \frac{\sqrt{3}-1}{\sqrt{3}+1} = 2 - \sqrt{3}$
 85. $\tan(375^\circ) = \frac{3-\sqrt{3}}{3+\sqrt{3}} = 2 - \sqrt{3}$
 86. $\cos\left(\frac{13\pi}{12}\right) = \frac{-\sqrt{6}+\sqrt{2}}{4}$

87. $\sin\left(\frac{11\pi}{12}\right) = \frac{\sqrt{6}-\sqrt{2}}{4}$
 88. $\tan\left(\frac{13\pi}{12}\right) = \frac{3-\sqrt{3}}{3+\sqrt{3}} = 2 - \sqrt{3}$
 89. $\cos\left(\frac{7\pi}{12}\right) = \frac{\sqrt{2}-\sqrt{6}}{4}$
 90. $\tan\left(\frac{17\pi}{12}\right) = 2 + \sqrt{3}$
 91. $\sin\left(\frac{\pi}{12}\right) = \frac{\sqrt{6}-\sqrt{2}}{4}$
 92. $\cot\left(\frac{11\pi}{12}\right) = -(2 + \sqrt{3})$
 93. $\csc\left(\frac{5\pi}{12}\right) = \sqrt{6} - \sqrt{2}$
 94. $\sec\left(-\frac{\pi}{12}\right) = \sqrt{6} - \sqrt{2}$

- | | | |
|--|---|--|
| 95. (a) $\cos(\alpha + \beta) = -\frac{\sqrt{2}}{10}$
(b) $\sin(\alpha + \beta) = \frac{7\sqrt{2}}{10}$ | (c) $\tan(\alpha + \beta) = -7$
(d) $\cos(\alpha - \beta) = -\frac{\sqrt{2}}{2}$ | (e) $\sin(\alpha - \beta) = \frac{\sqrt{2}}{2}$
(f) $\tan(\alpha - \beta) = -1$ |
|--|---|--|

- | | | |
|--|---|--|
| 96. (a) $\cos(\alpha + \beta) = \frac{4+7\sqrt{2}}{30}$
(b) $\sin(\alpha + \beta) = \frac{28-\sqrt{2}}{30}$ | (c) $\tan(\alpha + \beta) = \frac{-28+\sqrt{2}}{4+7\sqrt{2}} = \frac{63-100\sqrt{2}}{41}$
(d) $\cos(\alpha - \beta) = \frac{-4+7\sqrt{2}}{30}$ | (e) $\sin(\alpha - \beta) = \frac{28+\sqrt{2}}{30}$
(f) $\tan(\alpha - \beta) = \frac{28+\sqrt{2}}{4-7\sqrt{2}} = \frac{-63+100\sqrt{2}}{41}$ |
|--|---|--|

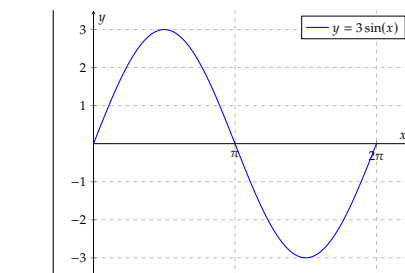
- | | | |
|--|--|--|
| 97. (a) $\sin(\alpha + \beta) = \frac{16}{65}$ | (b) $\cos(\alpha - \beta) = \frac{33}{65}$ | (c) $\tan(\alpha - \beta) = \frac{56}{33}$ |
|--|--|--|

- | | | |
|---|--|---|
| 98. (a) $\csc(\alpha - \beta) = -\frac{5}{4}$ | (b) $\sec(\alpha + \beta) = \frac{125}{117}$ | (c) $\cot(\alpha + \beta) = \frac{117}{44}$ |
|---|--|---|

- | | | |
|--|---|--|
| 111. $\cos(75^\circ) = \frac{\sqrt{6}-\sqrt{2}}{4}$
112. $\sin(105^\circ) = \frac{\sqrt{6}+\sqrt{2}}{4}$
113. $\cos(67.5^\circ) = \frac{\sqrt{2}-\sqrt{6}}{4}$ | 114. $\sin(157.5^\circ) = \frac{\sqrt{2}-\sqrt{6}}{4}$
115. $\tan(112.5^\circ) = -1 - \sqrt{2}$
116. $\cos\left(\frac{7\pi}{12}\right) = \frac{\sqrt{6}-\sqrt{2}}{4}$
117. $\sin\left(\frac{\pi}{12}\right) = \frac{\sqrt{6}-\sqrt{2}}{4}$ | 118. $\cos\left(\frac{\pi}{8}\right) = \frac{\sqrt{2}+\sqrt{2}}{2}$
119. $\sin\left(\frac{5\pi}{8}\right) = \frac{\sqrt{2}+\sqrt{2}}{2}$
120. $\tan\left(\frac{7\pi}{8}\right) = 1 - \sqrt{2}$ |
|--|---|--|

135. $y = 3 \sin(x)$

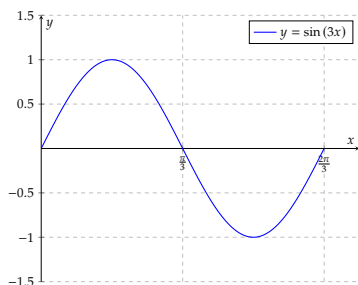
- Period: 2π
- Amplitude: 3
- Domain: $[0, 2\pi]$
- Vertical Shift: None



136. $y = \sin(3x)$

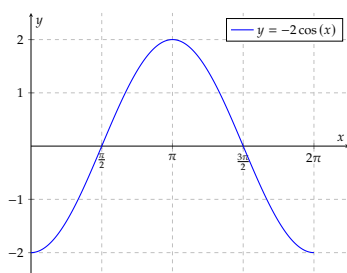
- Period: $\frac{2\pi}{3}$
- Amplitude: 1

- Domain: $[0, \frac{2\pi}{3}]$
- Vertical Shift: None



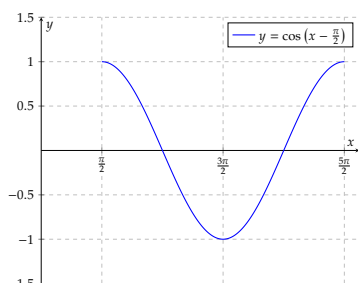
137. $y = -2\cos(x)$

- Period: 2π
- Amplitude: 2 (inverted)
- Domain: $[0, 2\pi]$
- Vertical Shift: None



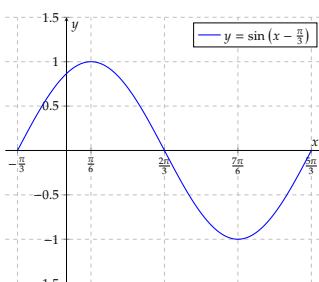
138. $y = \cos(x - \frac{\pi}{2})$

- Period: 2π
- Amplitude: 1
- Domain: $[\frac{\pi}{4}, \frac{5\pi}{4}]$
- Vertical Shift: None



139. $y = \sin(x + \frac{\pi}{3})$

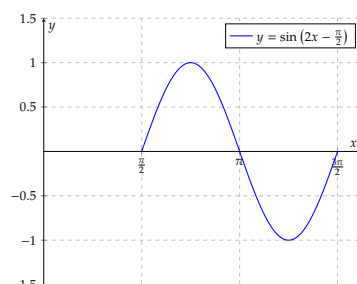
- Period: 2π
- Amplitude: 1
- Domain: $[-\frac{\pi}{3}, \frac{5\pi}{3}]$
- Vertical Shift: None



140. $y = \sin(2x - \pi)$

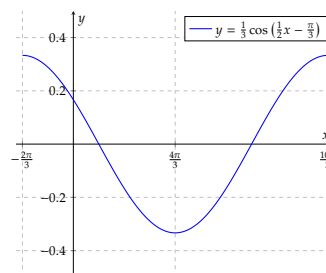
- Period: π

- Amplitude: 1
- Domain: $[\frac{\pi}{2}, \frac{3\pi}{2}]$
- Vertical Shift: None



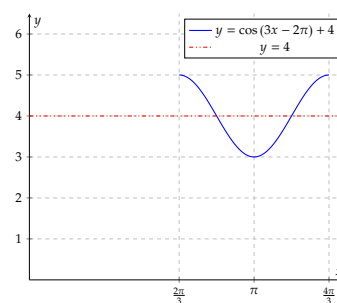
141. $y = -\frac{1}{3}\cos(\frac{1}{2}x + \frac{\pi}{3})$

- Period: 4π
- Amplitude: $\frac{1}{3}$ (inverted)
- Domain: $[-\frac{2\pi}{3}, \frac{10\pi}{3}]$
- Vertical Shift: None



142. $\cos(3x - 2\pi) + 4$

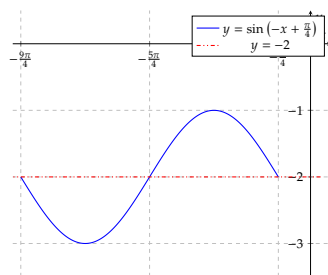
- Period: $\frac{2\pi}{3}$
- Amplitude: 1
- Domain: $[\frac{2\pi}{3}, \frac{4\pi}{3}]$
- Vertical Shift: 4



143. $y = \sin(-x - \frac{\pi}{4}) - 2$

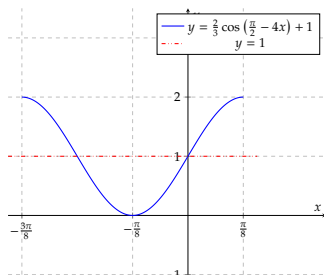
- Period: 2π (to the left)
- Amplitude: 1
- Domain (one cycle): $[-\frac{9\pi}{4}, -\frac{\pi}{4}]$
- Vertical Shift: -2 Resting at $y = -2$

NOTE: When graphing this electronically, use its equivalence $y = -\sin(x + \frac{\pi}{3})$ not the original function.



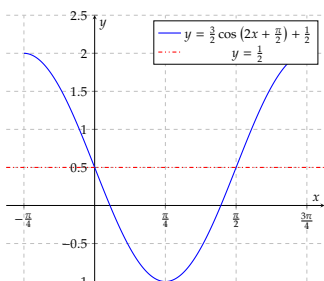
144. $y = \frac{2}{3} \cos(\frac{\pi}{2} - 4x) + 1$

- Period: $\frac{\pi}{2}$
- Amplitude: $\frac{2}{3}$
- Domain (one cycle): $[-\frac{3\pi}{8}, \frac{\pi}{8}]$
- Vertical Shift: 1 (rests at $y = 1$)



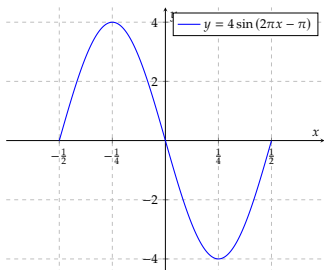
145. $y = -\frac{3}{2} \cos(2x + \frac{\pi}{3}) - \frac{1}{2}$

- Period: π
- Amplitude: $\frac{3}{2}$
- Domain (one cycle): $[-\frac{\pi}{4}, \frac{3\pi}{4}]$
- Vertical Shift: $\frac{1}{2}$ Resting at $y = \frac{1}{2}$



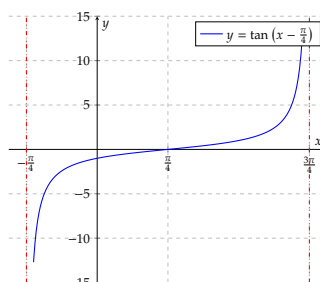
146. $y = 4 \sin(2\pi x + \pi)$

- Period: 1
- Amplitude: 4
- Domain (one cycle): $[-\frac{1}{2}, \frac{1}{2}]$
- Vertical Shift: none



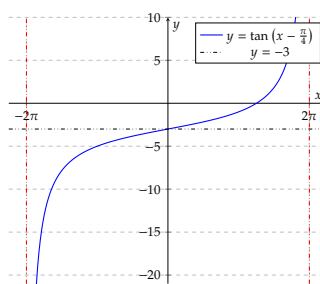
147. $y = \tan(x - \frac{\pi}{4})$

- Period: π
- Domain (one cycle): $(-\frac{\pi}{4}, \frac{3\pi}{4})$
- x - intercept = $\frac{\pi}{4}$



148. $y = 2 \tan(\frac{1}{4}x) - 3$

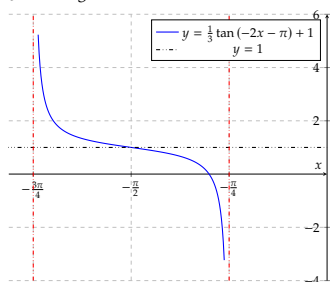
- Period: 2π
- Domain (one cycle): $(-2\pi, 2\pi)$
- Vertical Shift: -3 axis: $y = -3$
- Axis intercept: 0



149. $y = \frac{1}{3} \tan(-2x - \pi) + 1$

- Period: $\frac{\pi}{2}$
- Domain (one cycle): $(-\frac{3\pi}{4}, -\frac{\pi}{4})$
- Vertical Shift: 1 axis: $y = 1$
- Axis-Intercept: $-\frac{1}{2}$

NOTE: When graphing this use the equivalence $y = -\frac{1}{3} \tan(2x + \pi) + 1$



NOTE: All the equivalences used in graphing different functions were found using the negative angle properties from Subsection 1.4.1

Appendix

B.1 List of Compound Angle Formulae

Sum and Difference Equations

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\cos 2\alpha = 1 - 2 \sin^2 \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

Double and Half Angle Formulae

$$\cos \frac{\alpha}{2} = \sqrt{\frac{\cos \alpha + 1}{2}}$$

$$\sin \frac{\alpha}{2} = \sqrt{\frac{1 - \cos \alpha}{2}}$$

$$\tan \frac{\alpha}{2} = \sqrt{\frac{1 + \cos \alpha}{1 - \cos \alpha}}$$

Product and Factor Formulae

$$\sin \alpha \cos \beta = \frac{\sin(\alpha + \beta) + \sin(\alpha - \beta)}{2}$$

$$\cos \alpha \sin \alpha = \frac{\sin(\alpha + \beta) - \sin(\alpha - \beta)}{2}$$

$$\cos \alpha \cos \beta = \frac{\cos(\alpha + \beta) + \cos(\alpha - \beta)}{2}$$

$$\sin \alpha \sin \beta = \frac{\cos(\alpha - \beta) - \cos(\alpha + \beta)}{2}$$

$$\sin u + \sin v = 2 \sin \left(\frac{u + v}{2} \right) \cos \left(\frac{u - v}{2} \right)$$

$$\sin u - \sin v = 2 \cos \left(\frac{u + v}{2} \right) \sin \left(\frac{u - v}{2} \right)$$

$$\cos u + \cos v = 2 \cos \left(\frac{u + v}{2} \right) \cos \left(\frac{u - v}{2} \right)$$

$$\cos u - \cos v = -2 \sin \left(\frac{u + v}{2} \right) \sin \left(\frac{u - v}{2} \right)$$

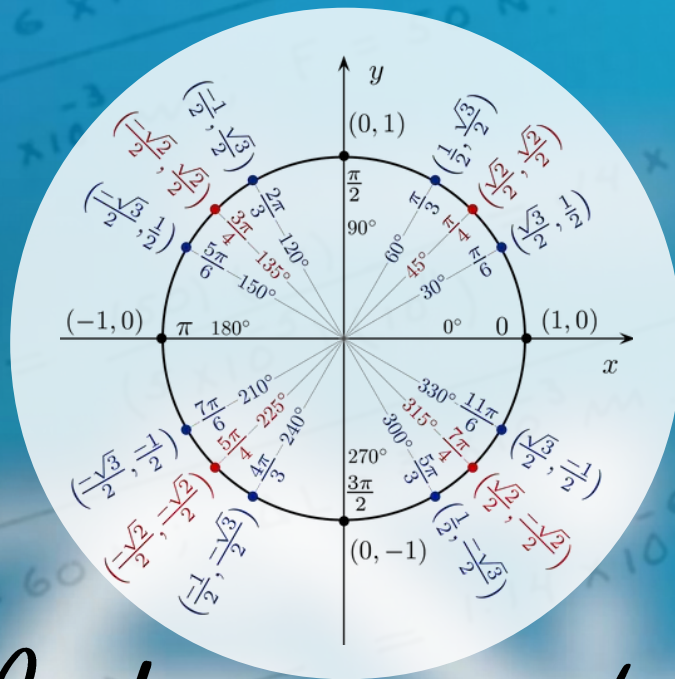
Greek Letters with Pronunciations

Character	Name	Character	Name
α	alpha <i>AL-fuh</i>	ν	nu <i>NEW</i>
β	beta <i>BAY-tuh</i>	ξ, Ξ	xi <i>KSIGH</i>
γ, Γ	gamma <i>GAM-muh</i>	$\omicron$	omicron <i>OM-uh-CRON</i>
δ, Δ	delta <i>DEL-tuh</i>	π, Π	pi <i>PIE</i>
ϵ	epsilon <i>EP-suh-lon</i>	ρ	rho <i>ROW</i>
ζ	zeta <i>ZAY-tuh</i>	σ, Σ	sigma <i>SIG-muh</i>
η	eta <i>AY-tuh</i>	τ	tau <i>TOW (as in cow)</i>
θ, Θ	theta <i>THAY-tuh</i>	υ, Υ	upsilon <i>OOP-suh-LON</i>
ι	iota <i>eye-OH-tuh</i>	ϕ, Φ	phi <i>FEE, or FI (as in hi)</i>
κ	kappa <i>KAP-uh</i>	χ	chi <i>KI (as in hi)</i>
λ, Λ	lambda <i>LAM-duh</i>	ψ, Ψ	psi <i>SIGH, or PSIGH</i>
μ	mu <i>MEW</i>	ω, Ω	omega <i>oh-MAY-guh</i>

Capitals shown are the ones that differ from Roman capitals.

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Worked Examples For

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